JUNIOR IOAA

Problems by Topic

2022–2025

A collection of individual problems and available official solutions from the junior competition.

Source-paper edition
Original pages are reproduced as vector crops, preserving the diagrams, equations, and typography of the supplied papers.

This independent educational compilation is not affiliated with the competition organisers. Source-paper rights remain with their authors and organisers. Consult the repository notice for attribution and reuse information.

Contents

1. The Celestial Sphere 3
Theory problems 3
Solutions to The Celestial Sphere 7
2. Time and Calendars 15
Theory problems 15
Solutions to Time and Calendars 17
3. The Solar System 19
Theory problems 19
Solutions to The Solar System 23
4. Orbital Mechanics 27
Theory problems 27
Solutions to Orbital Mechanics 40
5. Light and Telescopes 67
Theory problems 67
Solutions to Light and Telescopes 71
6. The Sun and Stars 77
Theory problems 77
Solutions to The Sun and Stars 87
7. Stellar Evolution 101
Theory problems 101
Solutions to Stellar Evolution 103
8. Galaxies and Cosmology 104
Theory problems 104
Solutions to Galaxies and Cosmology 107
9. Observational Problems 110
Sky maps and charts 110
Telescope 136
Night observation 143
Solutions to Observational Problems 145
Index by year 169

1. The Celestial Sphere

Theory problems

Problem 1.1 The Sun on the Horizon

2022 | Câmpulung Moldovenesc, Romania | Theory | Q2

Solution p. 7

I. Multiple choice problems – 100 points

For these problems, you have to choose only one correct answer. Please write your answer on answer-sheet. **Don't forget to write the number of the problem, i.e Grid 1. (c).**

Grid 2. The Sun on the Horizon

The astronomical refraction correction, ρ_r, has the minimum value, $\rho_{r,min} = 0$, when the star is at the zenith of the Earth's surface observer $(z = 0; h = 90^0)$, and has the maximum value, $\rho_{r,max}$, when the star, whose light passes through the Earth's atmosphere, in order to reach the Earth's surface observer, is at the horizon $(z = 90^0; h = 0)$, that is, when the star rises or sets.

It is known, that for an observer, atmospheric refraction raises the Sun in the moments of sunrise and sunset, that is, when it is below the horizon, approximately at its edge, by an apparent disk**.**

It is given: the radius of the Sun, $R_S = 6.96 \cdot 10^5\ km$ the distance between the centre of the Earth and the centre of the Sun, $d = 15 \cdot 10^7\ km$.

The value of $\rho_{r,max}$,is:

a) $\rho_{r,max} \approx 23.7'$ b) $\rho_{r,max} \approx 43.7'$ c) $\rho_{r,max} \approx 53.7'$ d) $\rho_{r,max} \approx 33.7'$

Problem 1.2 Horizon crossing

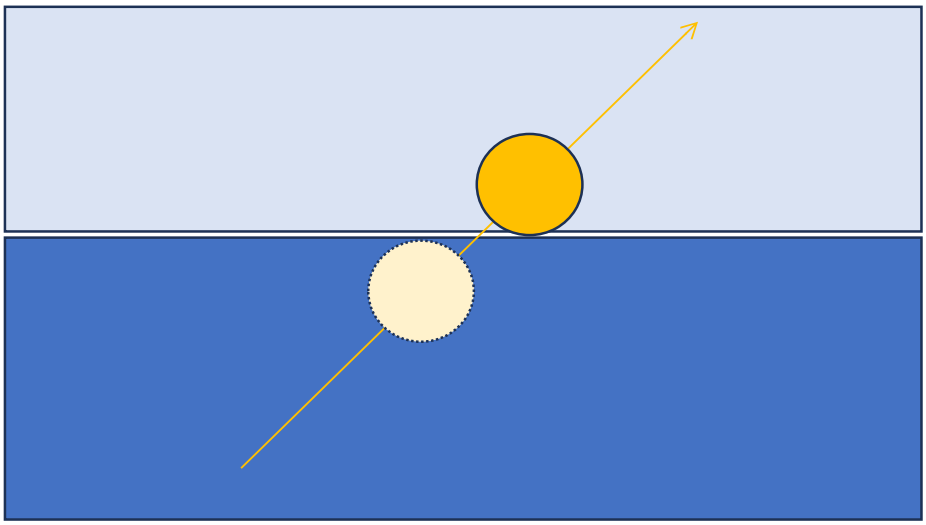
2023 | Volos, Greece | Theory | Q1 | 10 points

Solution p. 9

Question 1: Horizon crossing.

The Sun, at some location, crosses the horizon at sunrise on autumnal equinox day at an angle of 45°. The angular diameter of the Sun is 32 arcminutes. How long does it approximately take for the entire solar disk to cross the horizon?

You may neglect atmospheric effects.



Problem 1.3 Rising Sagarmatha and angular separation

2024 | Kathmandu, Nepal | Theory | T8 | 15 points

Solution p. 9

T8 (15 points)

As we saw in the group competition, Nepal got the chance to name a star in the constellation of Leo. The name given to that star is Sagarmatha. The coordinates of this star are $\alpha = 11^h35^m52^s$, $\delta = -4^o45'21''$. Given that the Autumnal equinox for the year 2024 happened on Sep 22 at 6:28 pm Nepal time, and neglecting the effects of atmospheric refraction,

1. Find the rise time of Sagarmatha as seen from Kathmandu on 6th October 2024. (Note: The rise time here refers to local time in Kathmandu)
2. We want to find Sagarmatha through a telescope. Since Sagarmatha is too faint, we first point to ϕ Leo ($\alpha = 11^h16^m38^s$, $\delta = -3^o38'58''$). Find angular separation between ϕ Leo and Sagarmatha.

Information about the host city

Latitude of Kathmandu City, $\phi_{kathmandu} = 27.7172^{o}N$
Time Zone for Nepal = UTC + 5:45:00

Problem 1.4 Where is Theodor?

2025 | Piatra Neamț, Romania | Theory | Th.3 | 10 points

Solution p. 10

Th.3 Where is Theodor? (10 Points)

Theodor is in a random city somewhere in the world. Using his knowledge of geography, he narrows down the possible locations to the four cities listed below. Then, on October 20th, he measures the time between sunrise and sunset and finds it to be $t = 10^{h}42^{m}$.
Can you help him figure out which city he is in? Ignore atmospheric refraction and Sun's apparent size.

a) Compute the Sun's equatorial coordinates on that date. (Consider that the sun moves with constant velocity on the ecliptic from the autumn equinox onward)

b) Based on the measured daytime duration t, determine which of the four cities Theodor is in.

c) On this day, exactly at the moment when the Sun sets for Theodor, determine the geographical coordinates of the point on Earth where the Sun is directly overhead.



1)Kutaisi (φ=42.27° N, L=42.66° E)



2)Vassouras (φ=22.41° S,L=43.66° W)



3)Mumbai (φ=18.96° N, L=72.83° E)



4)Chorzów (φ= 50.31° N, L=18.97° E)

Problem 1.5 Ceahlau Mountain

2025 | Piatra Neamț, Romania | Theory | Th.5 | 15 points

Solution p. 12

Th.5 Ceahlău Mountain (15 Points)

Ceahlău Mountain is located in Neamț County, not far from the city of Piatra Neamț. An observer on the top of the mountain ($\varphi = 46°53'57'' N, L = 26°24'41'' E$, altitude $h = 1907\,\text{m}$ above sea level), looks at the starry sky, making observations of the star Almach in the constellation Andromeda.

a) Determine whether, from the top of Mount Ceahlău, the star Almach (γ AND, $\alpha = 02^h03^m54^s$, $\delta = 42°19'47''$) is circumpolar. The radius of the Earth is known to be $R_\oplus = 6378\,\text{km}$.

b) In 1778, it was discovered that γ AND is a double star. The two stars are separated by an angle of about $10''$. What should be the diameter of the objective lens of a telescope to see the two components distinctly (the wavelength is known to be $\lambda = 550\,\text{nm}$)?

c) It was later discovered that the Almach star is a multiple star system consisting of the primary component γ_1 (A) with apparent magnitude $m_1 = 2.27$ and a subsystem γ_2 with apparent magnitude $m_2 = 4.84$. Calculate the apparent magnitude of the system consisting of the star γ_1 and the subsystem γ_2.

d) What is the absolute magnitude of the star, knowing that it is approximately 355 light-years away?

Problem 1.6 The Flying Aquila

2025 | Piatra Neamț, Romania | Theory | Th.6 | 15 points

Solution p. 13

Th.6 The Flying Aquila (15 Points)

Radu is observing the night sky from Aveiro ($\varphi = 40.67°N$; $L = 8.65°W$ in the year 2024.
During his observations, he sees Altair ($\alpha_{Altair} - 19^h51^m$; $\delta_{Altair} = 8°55'50''$) rising at local solar time $\text{T}_{\text{Local,1}}$=22:39:43. Later, at local solar time $\text{T}_{\text{Local,2}}$=02:42:58, he observes that Altair and Arcturus ($\delta_{Arcturus} = 19°33'6''$) are at the same altitude. The observations are made from sea level. Ignore atmospheric effects.

a) Determine the sidereal time at the moment Altair rises.

b) Determine the current date (at the moment Altair rises). Ignore the equation of time, and consider that the right ascension of the Sun is increasing uniformly through the year, at a rate of 3^m56^s per day. The spring equinox occurs on March 21st.

c) Determine the right ascension of Arcturus. Take into account that Arcturus is in the Western hemisphere of the sky at $\text{T}_{\text{Local,2}}$.

Note: In your solution, please use the following notation: a - altitude; H - hour angle; δ- declination; α- right ascension

Solutions to The Celestial Sphere

Solution 1.1 The Sun on the Horizon

2022 | Câmpulung Moldovenesc, Romania | Theory | Q2

Problem p. 3

Solution

Atmospheric refraction raises the Sun above its horizon, at its limit, at sunrise and sunset, approximately, with an apparent disk, as shown in the drawing in Figure 1.

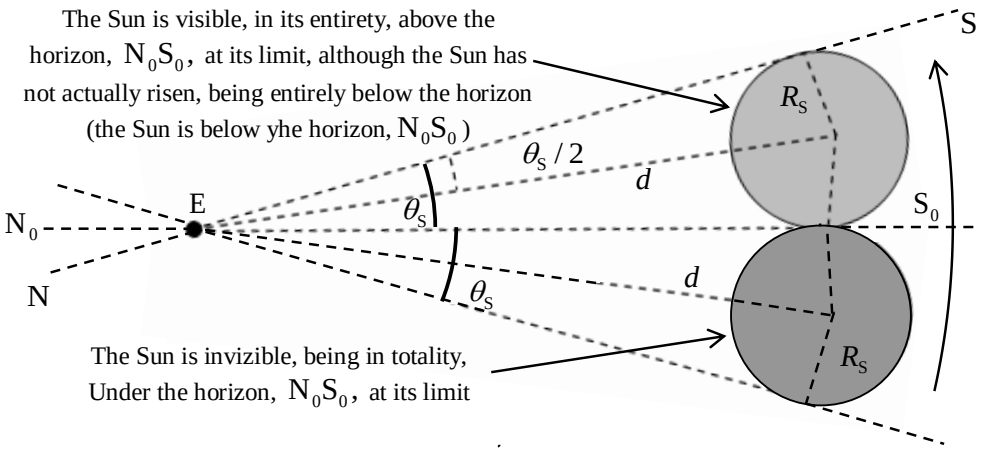


Fig. 1

Therefore, in reality, the rising of the upper arc of the Sun's disk begins after the moment when we already see the entire solar disk above the horizon. We see the sunrise and sunset, later than they actually occur, which makes the day a little longer.

I mean, I see the disk of the whole Sun above the horizon, at its limit, but the Sun has not yet risen, it is still below the horizon !

I mean, the sun went down, it went down below the horizon, at the edge of it, but I still see the whole disk of it above the horizon !

Under these conditions, the astronomical refraction has the maximum value :

$$\rho_{r,max} = \theta_S;$$

$$\sin\frac{\theta_S}{2} = \frac{R_S}{d} \approx \frac{\theta_S}{2};$$

$$\theta_S = \frac{2R_S}{d} = \frac{2 \cdot 6{,}96 \cdot 10^5 \text{ km}}{15 \cdot 10^7 \text{ km}} = 0{,}0098 \text{ radiani};$$

$$1\,\text{rad} = \frac{180 \cdot 60'}{3{,}14};$$

$$\theta_S = 0{,}0098 \cdot \frac{180 \cdot 60'}{3{,}14} \approx 33{,}7' = \rho_{r,max},$$

which proves that the value of the maximum astronomical refraction correction, $\rho_{r,max}$, is approximately equal to the apparent angular diameter of the Sun, θ_S.

The official key prints 0.0098 rad and 33.7 arcmin in the same calculation. Those numbers and the source wording about sunrise are inconsistent; they are reproduced with source comments.

Solution 1.2 Horizon crossing

2023 | Volos, Greece | Theory | Q1 | 10 points

Problem p. 4

Solution: The sun covers ~361 degrees on the celestial sphere in 24 hours, thus the velocity will be: v=361/(24x3600) degrees/sec. **[5 points]**

The Sun will need to move s=(32/60)/cos(π/4) degrees to cross the horizon. Therefore, the time it will need is: t =s/v=[(32/60)/cos(π/4)]/[361/(24x3600)]= 3 minutes. **[5 points]**

Solution 1.3 Rising Sagarmatha and angular separation

2024 | Kathmandu, Nepal | Theory | T8 | 15 points

Problem p. 4

T8

1. (3 points) We know:

$$\sin(a) = \sin(\phi)\sin(\delta) + \cos(\phi)\cos(\delta)\cos(H)$$

(1 point) At rise, a = 0.

$$\cos(H) = -\tan(\phi)\tan(\delta)$$

(3 points) Solve for H to get $H = 5$ hours 49 minutes 59 seconds.
(2 points)

$$t_{rise} = 12\text{pm} - H = 6{:}10 \text{ am}$$

2. (1 point) Since the stars are very close, you can use planar geometry to solve this problem.

$$d = \sqrt{(\Delta\alpha)^2 + (\Delta\delta)^2}$$

(2 point) Substitute $\Delta\alpha = 0.08376$ rad
(2 points) $\Delta\delta = 0.01927$ rad
(1 points) Solving in the original equation, obtain $d = 0.08595$ rad or in degrees, $4.92\,\text{deg}$. Answers that round to 4.9 degrees are acceptable. Answers in degrees or subunits there of for any part are acceptable. Correct solutions with spherical trigonometry will also receive full points.

The official rise-time calculation assumes local-noon culmination and does not use the supplied equinox epoch, date or right ascension. The official answer is retained.

Solution 1.4 Where is Theodor?

2025 | Piatra Neamț, Romania | Theory | Th.3 | 10 points

Problem p. 5

Marking Scheme

a) Compute the Sun's equatorial coordinates on that date. (We consider that the Sun moves with constant velocity on the ecliptic).
On 22nd September, we can consider $\lambda_S = 180°$ (also 23rd September)

$$\lambda_S = 0°/180° + \frac{D}{365.25} \times 360° \qquad \textbf{0.5pt}$$

where D is the number of days between the chosen point and 20th of October.

$\lambda_S = 207.6°$	$206.6°$	**0.5pt**
$\sin\delta = \sin\beta\cos\varepsilon + \cos\beta\sin\varepsilon\cos(90° - \lambda)$		
$\sin\delta = \sin\varepsilon\sin\lambda$	(Since $\beta = 0°$)	**1pt**
$\delta = -10°38'$	$-10°16'$	**0.5pt**
$\cos\alpha = \frac{\cos\lambda}{\cos\delta}$		**0.5pt**
$\alpha = 13^h42^m$	13^h38^m	**1pt**

Note: We would give 75% for students computing α directly, considering constant velocity on the equator.
Note: We would give 25% for students using Euclidian geometry instead of spherical trigonometry.

b) Based on the measured daytime duration t, determine which of the four cities Theodor is in.

$$\cos(90^\circ - h) = \cos(90^\circ - \varphi)\cos(90^\circ - \delta) + \sin(90^\circ - \varphi)\sin(90^\circ - \delta)\cos H$$
$$\therefore 0 = \sin\varphi \sin\delta + \cos\varphi\cos\delta\cos H$$
$$\cos H = -\tan\varphi\tan\delta \qquad \textbf{2pt}$$

Note: If students write the last formula directly, they receive the maximum.

$$t_l = H + 12h + n - L + \eta + D.S.T.$$

but all are constants (we can consider η constant for a day), so

$$\Delta t_l \approx \Delta H \qquad \textbf{0.5pt}$$

Considering that the two solutions for $\cos x = a \text{ in } [0,\, 2\pi]$ are $x = \arccos a$ and $x = 2\pi - \arccos a$, that means $t = 2\arccos(-\tan\varphi\tan\delta)$ (after conversion to hours)

$$t_1 = 10^h 42^m,\ t_2 = 12^h 35^m,\ t_3 = 11^h 31^m,\ t_4 = 10^h 16^m, \qquad \textbf{1pt}$$

Theodor is in city 1: *Kutaisi* **0.5pt**

Note: We award full points for the calculation of both solutions for H for each city and for calculating the time as their difference.

c) For the sun to be viewed as directly above ($h_S = 90^\circ$), the condition is $\varphi = \delta$ **0.5pt**

$$\therefore \varphi = -10^\circ 38' \qquad \textbf{0.5pt}$$

$$L = 12h - T_{GMT} \qquad \textbf{0.5pt}$$

$$T_{GMT} = H + 12h - L$$

$L = 37^\circ 47' W$ *or* $37^\circ 1' W$ if using 23 sept. **0.5pt**

Solution 1.5 Ceahlau Mountain

2025 | Piatra Neamț, Romania | Theory | Th.5 | 15 points

Problem p. 6

Marking Scheme

a) A circumpolar star is a star that, viewed from a certain latitude on Earth, never sets below the horizon, so when the minimum altitude is *0*. It follows that to be circumpolar, the minimum declination

$$\delta_m = 90^\circ - \varphi \Rightarrow \delta_m = 90^\circ - 46^\circ 53' 57'' = 43^\circ 6' 3'' \text{ at sea level.}$$ **1pt**

Considering atmospheric refraction $\alpha = 34'$, $\delta_m = 42^\circ 32' 3''$, therefore, at the given latitude, the Almach star is not circumpolar. **1pt**

At an altitude of 1907 meters, the horizon appears to drop.

$$\cos\theta = \frac{R}{R+h} \Rightarrow \arccos\frac{R}{R+h} \approx 1.4^\circ$$ **2pt**

$$\delta_m = 90^\circ - \varphi - \alpha = 41^\circ 8'$$

so from Mount Ceahlău, the star Almach is circumpolar. **1pt**

b)

$$\theta = \frac{1.22\lambda}{D} \Rightarrow D = \frac{1.22\lambda}{\theta}$$ **2pt**

$$\theta = 10'' = 4.84 \cdot 10^{-5}\, rad$$

$$D \approx 14\, mm$$ **1pt**

c) From Pogson's law $\frac{E_1}{E_2} = 10^{0.4\cdot(m_2 - m_1)}$

$$E_1 = 10.537 \cdot E_2$$ **2pt**

Total brightness (amount of energy per unit time per unit area): $E = E_1 + E_2$

$$E = 11.537 \cdot E_2$$ **1pt**

$$\frac{E_1}{E} = 2.5^{m - m_1}$$

$$m = 2.17$$ **1pt**

d)

$$M = m + 5 - 5 \cdot log_{10}(d)$$ **1pt**

$$d = 355\, ly = 108.896\, Pc$$ **1pt**

$$M = -3.04$$ **1pt**

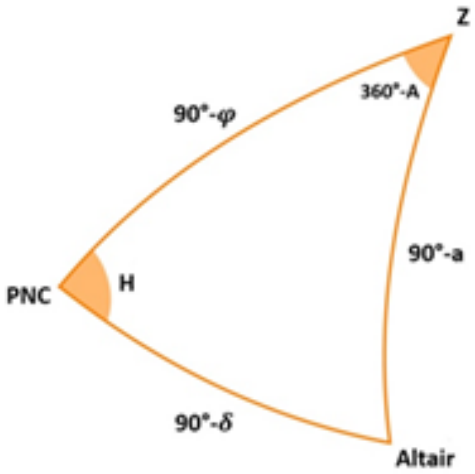
Solution 1.6 The Flying Aquila

2025 | Piatra Neamț, Romania | Theory | Th.6 | 15 points

Problem p. 6

Marking Scheme

a) Drawing of the spherical triangle: North Celestial Pole – Zenith – Star



Using the spherical law of cosines in the previous triangle:

$$\sin a_{Altair} = \sin\varphi \sin\delta_{Altair} + \cos\varphi \cos\delta_{Altair} \cos H_{Altair}$$

At its rising, neglecting horizon effects and refraction, the altitude of Altair is $a_{Altair} = 0°$. Therefore $sina_{Altair} = 0$ and

$$\sin\varphi \sin\delta_{Altair} + \cos\varphi \cos\delta_{Altair} \cos H_{Altair} = 0$$

This allows us to calculate Altair's hour angle

$$cosH_{Altair} = -\tan\varphi \tan\delta_{Altair}$$

$$H_{Altair} = \pm 6h31m$$ **2pt**

i.e. it will rise $6h31m$ before crossing meridian When it is at meridian $T_{sideral} = \alpha_{Altair} = 19h51m$ **1pt** **1pt**

$$T_{sideral} = 19h51m - 6h31m = 13h20m$$ **1pt**

When Altair rises, $T_{local} = 22:39:43$, $H_{Sun} = 10:39:43$ **1pt**

As sideral time is known $H_\gamma = 13h20m$ **0.5pt**

$\alpha_{Sun} = 2h40m17s$ **1pt**

40.75 days after γ **0.5pt**

$1st$ May **1pt**

Since the change in α_{Sun} in the span of a few hours is negligible, we can claim that

$$\Delta T_{Local} \approx \Delta T_{Sideral}$$

Therefore, when the stars are at the same altitude:

$$\begin{aligned} T_{Siderasl,2} &\approx T_{Siderasl,1} + T_{Loccal,2} - T_{Loccal,1} \\ &= 13h20m + 2h43m - 22h40m \\ &= 17h23m \end{aligned}$$ **1pt**

The same reasoning as in part a) applied to both Arcturus and Altair allows us to write

$$\sin a_{Altair} = \sin\varphi \sin\delta_{Altair} + \cos\varphi \cos\delta_{Altair} \cos H_{Altair}$$

$$\sin a_{Arcturus} = \sin\varphi \sin\delta_{Arcturus} + \cos\varphi \cos\delta_{Arcturus} \cos H_{Arcturus}$$ **1pt**

Since $a_{Altair} = a_{Arcturus}$, we can equate the two expressions and solve for the hour angle of Arcturus:

$$\cos H_{Arcturus} = \frac{\sin\varphi \sin\delta_{Altair} + \cos\varphi \cos\delta_{Altair} \cos H'_{Altair} - \sin\varphi \sin\delta_{Arcturus}}{\cos\varphi \cos\delta_{Arcturus}}$$

Solving this equation, we obtain:

$$H_{Arcturus} \in \{3h10m36s;\ 20h49m30s\}$$

Since Arcturus is in the western hemisphere of the sky, the valid solution is $H_{Arcturus} = 3h10m36s$ **3pt**

Results between *3h9m* and *3h12m* should receive full marks. Since the sidereal time and the hour angle of Arcturus are both known, we can now calculate its right ascension:

$$\alpha_{Arcturus} = T_{Sideral,2} - H_{Arcturus}$$

$$\alpha_{Arcturus} \approx 14h12m$$ **1pt**

Results between *14h10m* and 14h18m should receive full marks.

2. Time and Calendars

Theory problems

Problem 2.1 Moon phases

2023 | Volos, Greece | Theory | Q5 | 10 points

Solution p. 17

Question 5: Moon phases

The first full moon for 2024 in Volos is on the 25th of January at 19:54. At what date and time will the last full moon for 2024 (visible from Volos) occur?

Problem 2.2 Longitude and local time in Nepal

2024 | Kathmandu, Nepal | Theory | T3 | 15 points

Solution p. 17

T3 (15 points)

Nepal has a unique timezone of UTC +5:45, with the meridian for the country passing through the peak of Mt. Gaurishankar in the Himalayas.

1. Knowing that Mt. Gaurishankar is about 105km due East from Kathmandu, estimate Kathmandu's longitude.
2. Binod, who stays in Kathmandu, sees a bright star at meridian at exactly 8:35 pm as per his watch. Calculate Kathmandu's local time for this star's meridian crossing.
3. Binod wants to tell about this bright star to his friend Ravi who lives in Ranchi, a city in India that has the same longitude as Kathmandu, but a timezone of UTC +5:30. What is the difference in local time and time in the watch for Ravi? What time will the same star cross the meridian as per Ravi's watch?

Object	Mean radius (km)	Mass (kg)	Semi-major axis (AU)	Eccentricity	Albedo
Earth	$6\,378$	5.972×10^{24}	1.000	0.016710	0.31

Information about the host city

Latitude of Kathmandu City, $\phi_{kathmandu} = 27.7172^{o}N$
Time Zone for Nepal = UTC + 5:45:00

Problem 2.3 The lunar calendar and Dashain

2024 | Kathmandu, Nepal | Theory | T6 | 15 points

Solution p. 18

T6 (15 points)

The festival of Dashain is one of the most important celebrations in the Nepali calendar. The final day of the festival coincides with a Waxing Gibbous Moon that is exactly 10 days old. The festival always happens in a fixed month of lunar calendar, which has 12 months. Answer the following questions regarding the estimated timing of the Dashain festival.

1. In 2024, the final day is scheduled to fall on the 12th of October. Assuming that the Moon is exactly 10 days old at 11 am, predict on which date the final day of Dashain occurred in 2023.
2. Predict on which day *Kojagrat Purnima*, which is the Full Moon immediately after Dashain, will fall in 2025.

1 Lunar mean synodic month	29.53059 days

Solutions to Time and Calendars

Solution 2.1 Moon phases

2023 | Volos, Greece | Theory | Q5 | 10 points

Problem p. 15

The lunar month (from full moon to full moon) is 29.53 days. (29.5 days is acceptable).

[3 points]

12 lunar months correspond to 354.36 days, which gets us to 7/1/2025. Thus, we need to account for 11 lunar months: 324.83, which corresponds to the 15:49 on 15 December 2024.

If they use 29.5 they obtain 7:54 am on 15 December 2024.

[7 points]

Solution 2.2 Longitude and local time in Nepal

2024 | Kathmandu, Nepal | Theory | T3 | 15 points

Problem p. 15

T3

3. (a) (3 points)

$$lon_g = \frac{5.75}{12} \times 180 = 86.25 \deg E$$

(4 points)

$$lon_k = lon_g - \frac{105 \times \cos\phi}{6378} \times \frac{180}{\pi} = 85.415 \deg E$$

(b) (3 points)

$$t_{diff} = \frac{105 \times \cos\phi}{6378} \times \frac{12 \times 3600}{\pi} \text{ s } = 200.4 \text{ s } = 3 \text{ min } 20.4\text{s}$$

$$t_{ktm} = 8{:}35 \text{ pm} - t_{diff} = 8{:}31{:}40$$

(1 point) Answer can be written as 8:32pm, or 8:31:40pm.

(c) (2 points) Ravi has the same local time as Kathmandu. On his watch, however, his time zone is UTC +5:30. Thus, the difference will be 15 - t_{diff} mins, which is 11 mins 40 (39.6) seconds.

(2 points) As per his watch, therefore, the occultation will occur at 8:16:40 pm, or 8:17 pm.

The official longitude calculation multiplies by cos(latitude), and its Ranchi clock time is inconsistent with the stated watch offsets. The published expressions and answers are retained.

Solution 2.3 The lunar calendar and Dashain

2024 | Kathmandu, Nepal | Theory | T6 | 15 points

Problem p. 16

T6

1. (1 point) The mean synodic period of the Moon is 29.53059 days. Dashain happened 12 synodic periods earlier, so

   (3 points) $12 \times 29.53059 = 354.36708$ days $= 354$ days 8 hours 48 minutes.

   (1 point) Because 2024 is a leap year, there are 366 days between the two 12 Octobers.

   (3 points) Dashain in 2023 will be 354 days before the 12th of October.

   (2 point) In other words, it was 12 days after the 12 October, 2023, which was 24 October, 2023.

2. (1 point) Dashain in 2025 will be 354 days 8 hours 48 minutes after 11am on 12 October 2024,

   (2 points) which is 11 days before 12 October 2025 at around 7:48pm on 1 October 2025.

   (2 points) 4 days and 12 hours after this is 7:48am on 6 October 2025. Therefore, *Kojagrat Purnima* will fall on 6 October 2025.

3. The Solar System

Theory problems

Problem 3.1 The Sun as Seen from Saturn

2022 | Câmpulung Moldovenesc, Romania | Theory | Q6

Solution p. 23

I. Multiple choice problems – 100 points

For these problems, you have to choose only one correct answer. Please write your answer on answer-sheet. **Don't forget to write the number of the problem, i.e Grid 1. (c).**

Grid 6. The Sun as seen from Saturn
It is known that: the distance Saturn - Sun is 9.54 times greater than the distance Earth - Sun; the angular diameter of the Sun's disk, seen from Earth, is 32'.
The angular diameter of the Sun seen from Saturn, is:

a) $\alpha_{\text{Sun–Saturn}} \approx 0.001$ radian b) $\alpha_{\text{Sun–Saturn}} \approx 0.002$ radian c) $\alpha_{\text{Sun–Saturn}} \approx 0.003$ radian d) $\alpha_{\text{Sun–Saturn}} \approx 0.004$ radian

Problem 3.2 Solar eclipse

2023 | Volos, Greece | Theory | Q6 | 10 points

Solution p. 24

Question 6: Solar Eclipse.

Assume that one day the Moon's apogee coincides with the Earth's perihelion. An observer, at an appropriate location on the Earth, observes the eclipse at zenith at its maximum. What fraction of the solar disk area is covered?

The maximum distance between the centers of the Earth and the Moon is 405,000 km, while the minimum distance between the Earth and the Sun is 147,100,000 km.

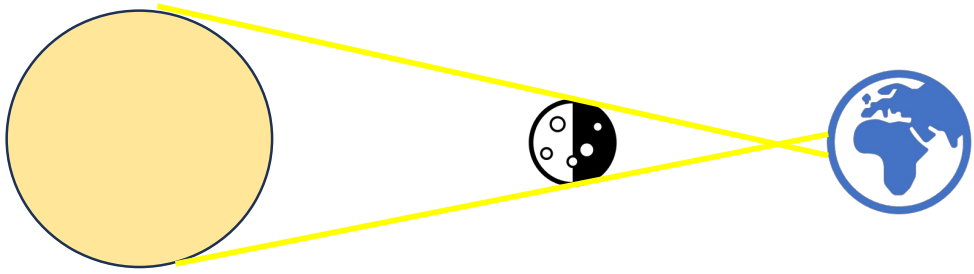


Figure not to scale.

Object	Mean radius (km)	Mass (kg)	Orbital Radius (au)
Sun	695 500	1.988×10^{30}	---
Mercury	2 440	3.301×10^{23}	0.387
Venus	6 052	4.867×10^{24}	0.723
Earth	6 378	5.972×10^{24}	1.000 000
Moon	1 737	7.346×10^{22}	0.002 572

Problem 3.3 Lunar Eclipse

2025 | Piatra Neamț, Romania | Theory | Th.7 | 30 points

Solution p. 24

Th.7 Lunar Eclipse (30 Points)

On September 7th 2025, there was a total lunar eclipse, visible from Australia, Asia, Africa and Europe. A detailed lunar eclipse figure, obtained from the website eclipsewise.com of eclipse predictions by Fred Espenak, is shown below. Based on the figure, respond to the following items:

a) Measure the inclination between the orbit of the Moon around the Earth and the Ecliptic plane. The orbit of the Moon is assumed to be circular.

b) During the total eclipse phase, the moon appears red in colour. Which of the following factors are necessary to explain this red colour? Write their item letters in the aswersheet.

   A) Density of atmosphere
   B) Albedo of Moon
   C) Orbital radius of Moon
   D) Albedo of Earth
   E) Effective temperature of the Sun

c) Measure the angular diameter of the Moon. Knowing the value of the radius of the Moon, calculate the Earth - Moon distance during the eclipse, in kilometres.

d) Measure the angular diameter of Earth's umbra and the angular diameter of Earth's penumbra.

e) What is the instantaneous angular velocity of the Moon relative to the observer, during the total eclipse? Give the answer in arcminutes per hour ('/h).

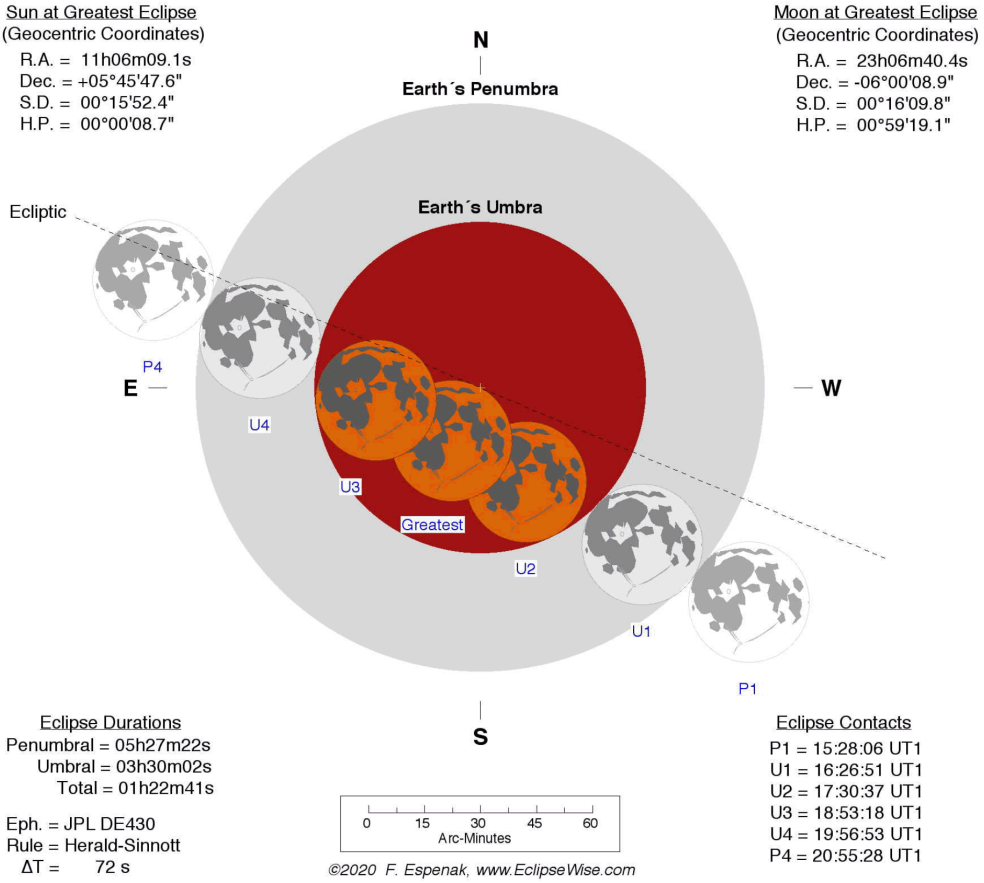
f) On 26 June 2029, a central total lunar eclipse will take place. This means that the centre of the disk of the Moon will pass through the axis of Earth's umbral shadow (see figure below). If the angular velocity is assumed to be the same for the two eclipses, by what percentage will the 2029 total lunar eclipse last longer than the one in 2025 (percentage to be calculated of the totality duration in 2025)?

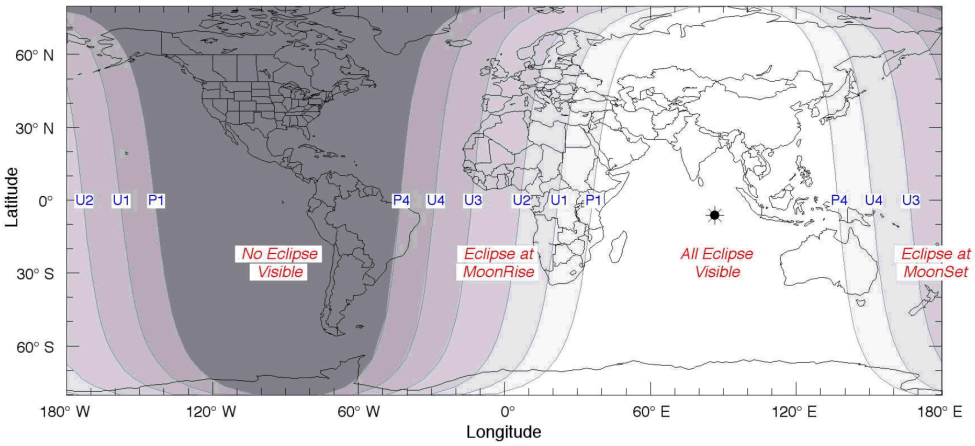
g) Calculate the total duration of the eclipse in 2029 (from Penumbral first contact to Penumbral last contact).

Total Lunar Eclipse of 2025 Sep 07

Greatest Eclipse = 18:12:58.0 TD (= 18:11:46.1 UT1)

Penumbral Magnitude = 2.3459 Gamma = -0.2752 Saros Series = 128
Umbral Magnitude = 1.3638 Axis = 0.2721° Saros Member = 41 of 71



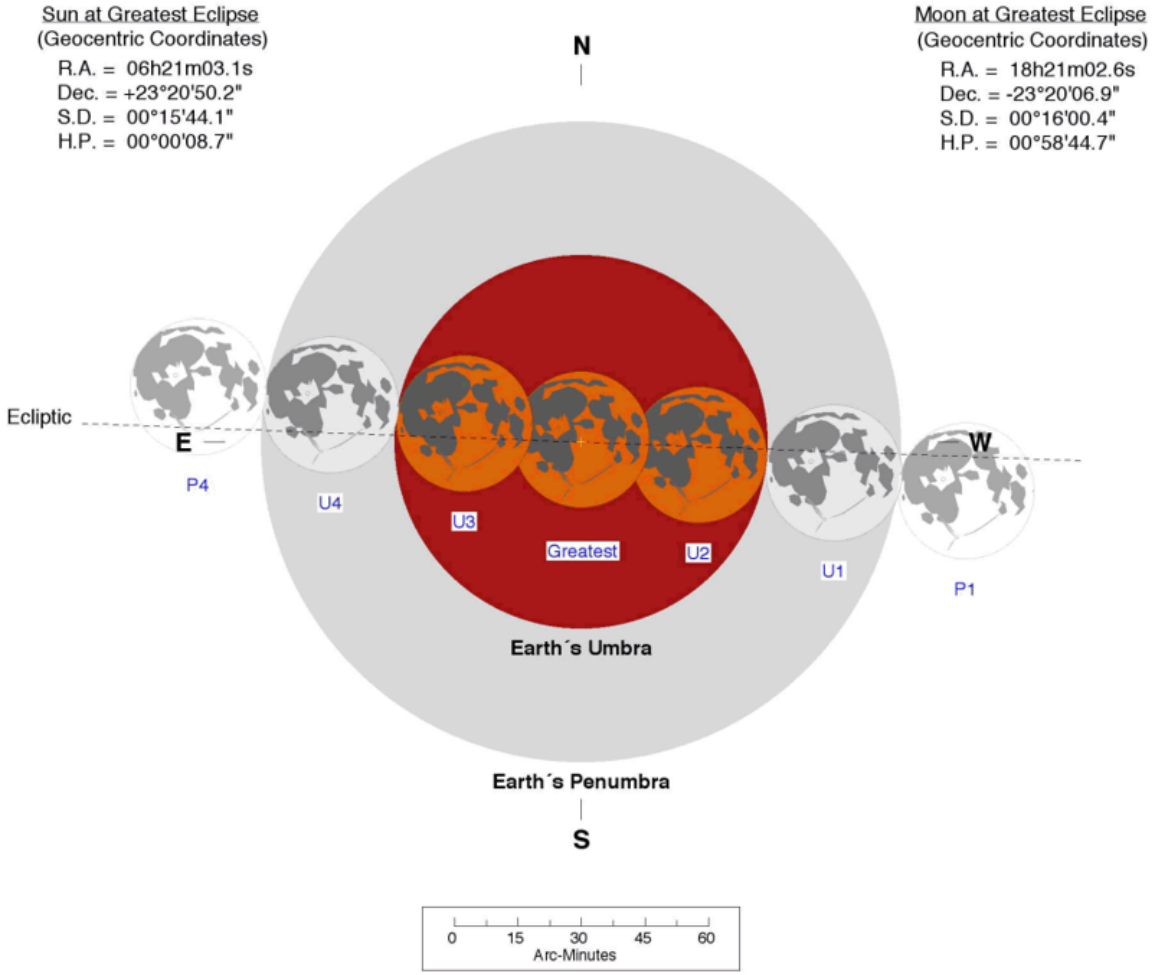


COURTESY OF *21ST CENTURY CANON OF LUNAR ECLIPSES*, FRED ESPENAK, ASTROPIXELS PUBLISHING, 2020

Total Lunar Eclipse of 2029 Jun 26

Greatest Eclipse = 03:23:22.5 TD (= 03:22:08.8 UT1)

Penumbral Magnitude = 2.8282 Gamma = 0.0124 Saros Series = 130
Umbral Magnitude = 1.8452 Axis = 0.0121° Saros Member = 35 of 71



Solutions to The Solar System

Solution 3.1 The Sun as Seen from Saturn

2022 | Câmpulung Moldovenesc, Romania | Theory | Q6

Problem p. 19

Solution

Since the distance between Saturn and the Sun is 9.54 times greater than the distance from Earth to the Sun, it turns out that the angular diameter of the Sun's disk, observed from Saturn, is 9.54 times smaller than the angular diameter of the Sun, observed from Earth. so that, using the drawing in Figure 1, it results:

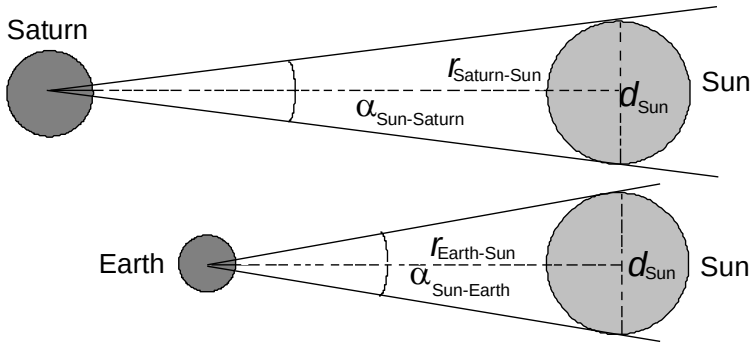


Fig. 1

representing the angular diameter of the Sun seen from Saturn.

Solution 3.2 Solar eclipse

2023 | Volos, Greece | Theory | Q6 | 10 points

Problem p. 19

The angular diameter of the Sun during perihelion is

$$d_1 = 2 \times \frac{695{,}500}{147{,}100{,}00 - 6{,}378} = 2 \times\ 0.004731 = 0.009463 \text{ rad},$$

while the angular diameter of the moon during apogee is

$$d_2 = 2 \times \frac{1{,}737}{405{,}000 - 6{,}378} = 2 \times 0.004358 = 0.008715 \text{ rad}.$$

[6 points]

The fraction of the area that is covered is the ratio of the squares of these values: $r = \left(\frac{d_2}{d_1}\right)^2 = 0.8482 = 84.82\%$. **[4 points]**

Solution 3.3 Lunar Eclipse

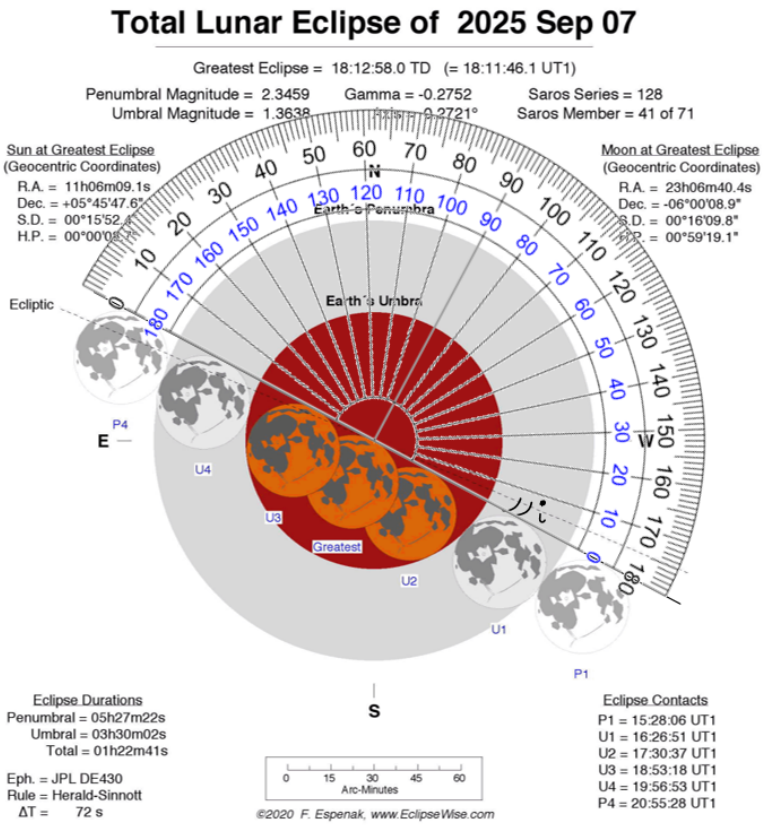
2025 | Piatra Neamț, Romania | Theory | Th.7 | 30 points

Problem p. 20

Marking Scheme

a) The angle of inclination measured is about $5° - 6°$. **4pt**

b) A) and C), each correct option **+2pt**, each wrong option **-1pt**
Note for graders: the minimum score on this part is 0, no negative score can be obtained. **4pt**



c) The scale of the image is given at the bottom of the page. From there, we find the correspondence 2.5 cm – 60 arc-minutes.

Note for graders: **Distances in cm may change due to scaling, but answers in arcmin should remain the same.**

Measuring the diameter of the Moon on the figure, we get $1.6cm$.

Applying a simple proportion, we determine the angular diameter of the Moon to be:

$$d_{Moon} = \frac{1.3\ cm \times 60'}{2.5\ cm} = 31.2' \qquad \textbf{4pt}$$

Answers between $31'$ and $32'$ are accepted (lack of precision due to ruler markings).
The angular radius of the moon will be:

$$r_{Moon} = \frac{d_{Moon}}{2} = 15.6' \approx 16' \qquad \textbf{1pt}$$

The linear value will be:

$$R_{Moon} = D_{Earth-Moon} \times \sin(r_{Moon}) = 1737km$$

$$D_{Earth-Moon} = \frac{R_{Moon}}{\sin(r_{Moon})} = \frac{1737}{\sin(16')} \approx 373200\ km \qquad \textbf{2pt}$$

Answers between $1730km$ and $1790km$ are accepted (corresponding to $15.5'$ and $16'$).

d) Similar to point c, we measure the diameter of Earth's umbra – 3.7 cm – and the diameter of Earth's penumbra – 6.3 cm.

The angular diameters will be:

$$d_{umbra} = \frac{3.7cm \times 60'}{2.5cm} = 89' = 1°29'$$

(88' - 90')

$$d_{penumbra} = \frac{6.3cm \times 60'}{2.5cm} = 151' = 2°31' \quad \textbf{6pt}$$

(150' - 153')

e) To determine the velocity, we choose a time interval (for example, between U2 and U3) and we measure on the figure what angular distance the moon traveled along its trajectory during that time.

$$\Delta t_{U3-U2} = t_{U3} - t_{U2} = 18^h53^m18^s - 17^h30^m37^s$$

$$\Delta t_{U3-U2} = 1^h22^m41^s \quad \textbf{0.5pt}$$

The linear distance travelled by the Moon between these two times is $\Delta l_{U3-U2} = 1.9cm$, which corresponds to:

$$\frac{1.9cm \times 60'}{2.5cm} = 45.6' \quad \textbf{0.5pt}$$

(45' - 47') The angular velocity will then be:

$$\omega_{Moon} = \frac{\Delta l_{U3-U2}}{\Delta t_{U3-U2}} = \frac{45.6'}{1^h22^m41^s} \approx 33.1'/h \quad \textbf{1pt}$$

(32.5'/h - 34.1'/h)

f) Since for this point we have the angular velocity and we want to determine the duration of the total eclipse, we need to measure the distance that the Moon travelled during the totality. First, we need to establish the scale of the new image. Here, 2.7 cm correspond to the $60'$.

Now, by measuring the path of the Moon, we get 2.5 cm.

The angular path will then be:

$$\Delta l_{totality} = \frac{2.5cm \times 60}{2.7cm} = 55.56' \quad \textbf{1.5pt}$$

(55' - 57')

The duration will be:

$$\Delta t_{totality} = \frac{\Delta l_{totality}}{\omega_{Moon}} = \frac{55.56'}{33.1'/h} = 1.69\ hours \approx 1^h41^m \quad \textbf{0.5pt}$$

(1.6 - 1.75 hours), (1h36m - 1h75m) The difference between the totality durations will be:

$$\Delta t_{totality2029} - \Delta t_{totality2025} = 1^h41^m - 1^h22^m41^s = 18^m19^s \quad \textbf{0.5pt}$$

Expressing it as a percentage of the 2025 duration:

$$\frac{\Delta t_{totality2029} - \Delta t_{totality2025}}{\Delta t_{totality2025}} = \frac{18^m19^s}{1^h22^m41^s} \approx 22.15\% \quad \textbf{0.5pt}$$

(15.7% - 26.6%)

g) To calculate the duration of the eclipse, we need to first measure the distance the Moon travels between P_1 and P_4: 8.2 cm.

$$\Delta l_{eclipse} = \frac{8.2cm \times 60'}{2.7cm} = 3.037° \approx 3°2'13'' \quad \textbf{2pt}$$

(3 - 3.1)

$$\Delta t_{eclipse} = \frac{\Delta l_{eclipse}}{\omega_{Moon}} = \frac{3.037 \times 60'}{33.1'/h} \approx 5.5\ hours \approx 5^h30^m \quad \textbf{2pt}$$

(5h17m - 5h42m)

4. Orbital Mechanics

Theory problems

Problem 4.1 The Escape Velocity

2022 | Câmpulung Moldovenesc, Romania | Theory | Q3

Solution p. 40

I. Multiple choice problems – 100 points

For these problems, you have to choose only one correct answer. Please write your answer on answer-sheet. **Don't forget to write the number of the problem, i.e Grid 1. (c).**

Grid 3. The escape velocity

It is given: $V_0 \approx 30\,\frac{\text{km}}{\text{s}}$, the speed of the Earth in its circular orbit around the Sun; $v_0 \approx 7.9\ \frac{\text{km}}{s}$, the speed of a terrestrial satellite orbiting the Earth in a very low circular orbit (the first cosmic speed).

It is known that: $\frac{M_{\text{Earth}}}{R_{\text{Earth}}} << \frac{M_{\text{Sun}}}{R_{\text{Earth-Sun}}}$.

The variation of the kinetic energy of the body, in relation to the Sun, is neglected during the motion of the body from the surface of the Earth to the limit of the gravitational attraction of the Earth.
The approximate minimum escape velocity of a body B, relative to the Sun, launched at a distance from the Sun equal to *1 AU*, so that it leaves the Solar System forever, is:

a) $v_B \approx 45.52\ \frac{\text{km}}{s}$ b) $v_B \approx 32.32\ \frac{\text{km}}{s}$ c) $v_B \approx 22.22\ \frac{\text{km}}{s}$ d) $v_B \approx 42.42\ \frac{\text{km}}{s}$.

Problem 4.2 Balls Suspended inside a Terrestrial Satellite

2022 | Câmpulung Moldovenesc, Romania | Theory | Q7

Solution p. 42

I. Multiple choice problems – 100 points

For these problems, you have to choose only one correct answer. Please write your answer on answer-sheet. **Don't forget to write the number of the problem, i.e Grid 1. (c).**

Grid 7. Balls Suspended inside a Terrestrial Satellite

An artificial satellite moves around the Earth in a circular orbit with radius *r*. We observe the satellite at the position and orientation shown in figure 1. Inside the satellite are suspended, by very light wires, four identical spherical balls, each with mass *m*, such that the balls (b) and (c) are symmetrical to the ball (a), the balls (a) and (d) are symetrical to the ball (c), and the difference of their distances to the center of the Earth is $\Delta r = 2d$.

It is given: M – the mass of the Earth; G– gravitational attraction constant.

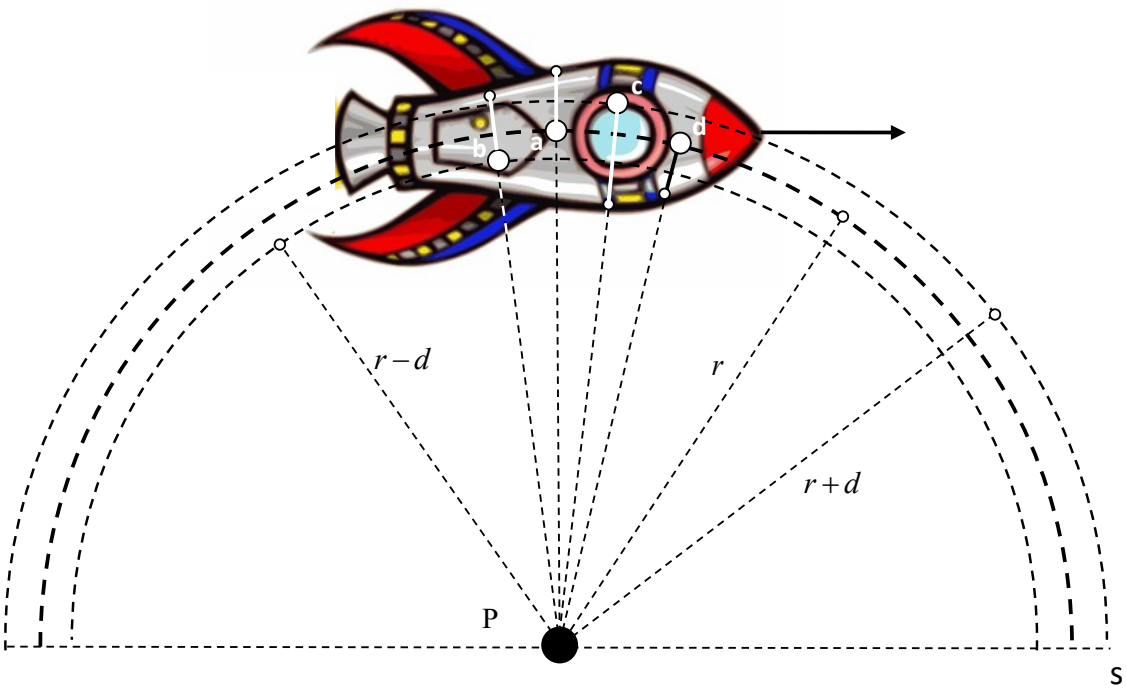


Fig. 1

The tension in each suspension wire, is:

a) $T_{(a)} = 0; T_{(b)} = 3G\frac{mMd}{r^3} > 0; T_{(c)} = 3G\frac{mMd}{r^3} > 0;\ T_{(d)} = 0$

b) $T_{(a)} = G\frac{mMd}{r^3};\ T_{(b)} = G\frac{mMd}{r^3} > 0;\ T_{(c)} = G\frac{mMd}{r^3} > 0;\ T_{(d)} = G\frac{mMd}{r^3}$

c) $T_{(a)} = 3G\frac{mMd}{r^3} > 0;\ T_{(b)} = 0;\ T_{(c)} = 0;\ T_{(d)} = 3G\frac{mMd}{r^3} > 0$

d) $T_{(a)} = 2G\frac{mMd}{r^3} > 0; T_{(b)} = 2G\frac{mMd}{r^3} > 0;\ T_{(c)} = 2G\frac{mMd}{r^3} > 0;\ T_{(d)} = 2G\frac{mMd}{r^3} > 0.$

Problem 4.3 Spaceship to the Sun

2022 | Câmpulung Moldovenesc, Romania | Theory | Q8

Solution p. 44

I. Multiple choice problems – 100 points

For these problems, you have to choose only one correct answer. Please write your answer on answer-sheet. **Don't forget to write the number of the problem, i.e Grid 1. (c).**

Grid 8. Spaceship to the Sun

In a future program, NASA plans to launch a spacecraft, aimed directly at the Sun, without a human crew, to gather information, on its way to the Sun, both about all the inner planets and, in particular, about the Sun.

They are known: the distance Earth - Sun, $r_{ES} = 1.5 \cdot 10^{11}$ m; the period of the Earth's rotation around the Sun, $T_E = 3.15 \cdot 10^7$ s; the radius of the Sun, $R_S \approx \frac{r_{ES}}{200}$.

If the launch of the spacecraft will be done in such a way that its motion relative to the Sun is a free fall, then the approximate duration of the Earth-Sun flight is:

a) $\boldsymbol{t \approx 3.6 \cdot 10^5}\ s$ b) $\boldsymbol{t \approx 6.6 \cdot 10^6}\ s$ c) $\boldsymbol{t \approx 5.6 \cdot 10^6}\ s$ d) $\boldsymbol{t \approx 4.6 \cdot 10^6}\ s$

Problem 4.4 Zodiac IOAA-J 2022

2022 | Câmpulung Moldovenesc, Romania | Theory | Q11 | 100 points

Solution p. 46

II. First long problem - Zodiac IOAA – J – 2022 – 100 points

In their apparent motions, the Moon and the other large planets in our solar system do not stray far from the plane of the ecliptic. Their apparent trajectories described on the celestial sphere remain contained in a region that extends symmetrically on both sides of the ecliptic, having a total width of about 18º. In the drawing in figure 1, in the plane of the ecliptic, the heliocentric orbit of the Earth is shown, and in the drawing in figure 2, the equivalent apparent geocentric orbit of the Sun, in relation to the Earth, is shown.

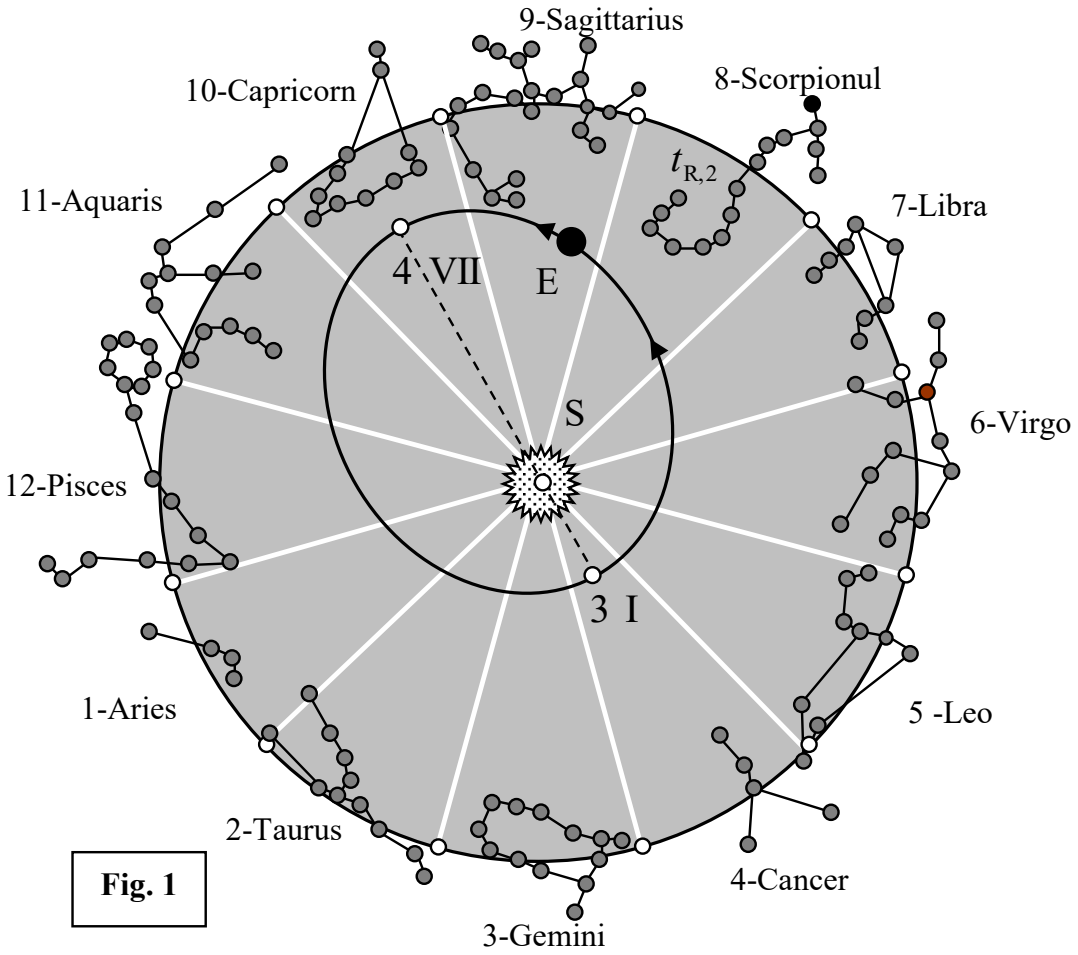

9-Sagittarius
10-Capricorn
8-Scorpionul
$t_{R,2}$
11-Aquaris
7-Libra
4 VII
E
S
6-Virgo
12-Pisces
3 I
1-Aries
5 -Leo
2-Taurus
4-Cancer
3-Gemini

Fig. 1

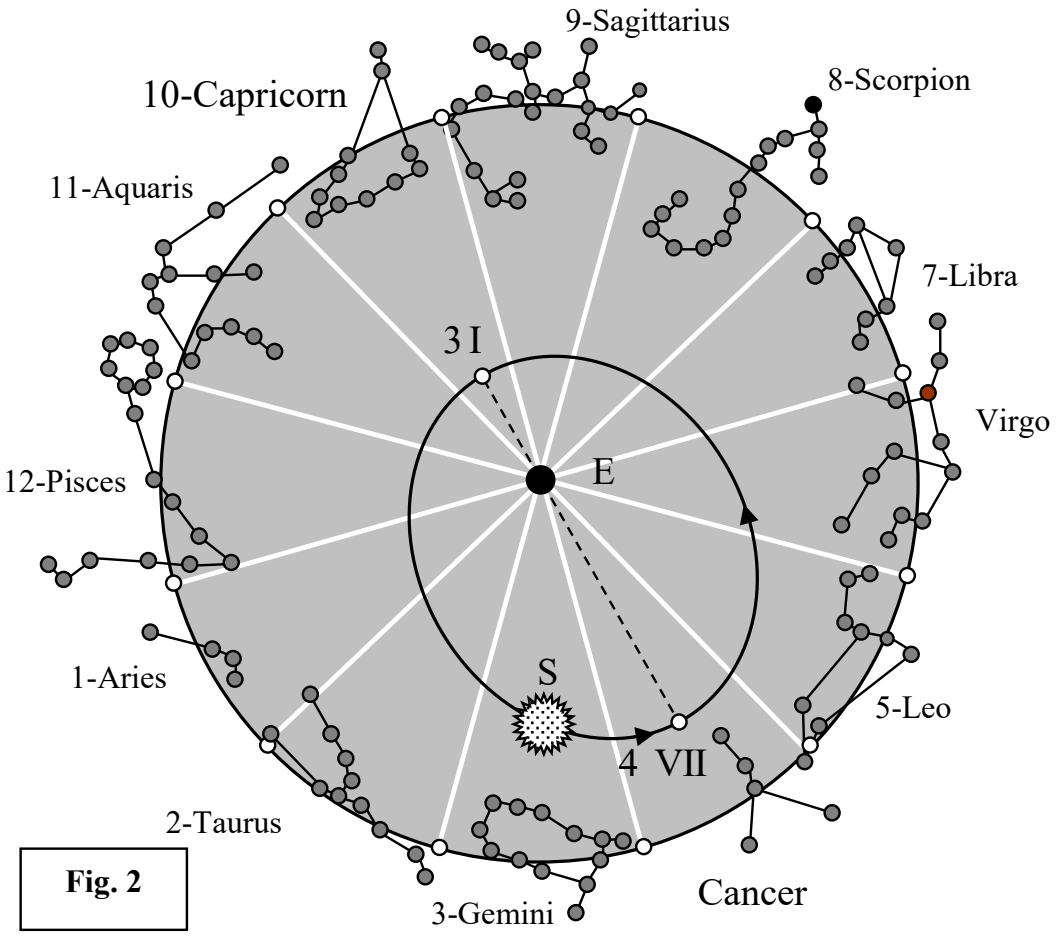

9-Sagittarius
10-Capricorn
8-Scorpion
11-Aquaris
7-Libra
3 I
Virgo
E
12-Pisces
1-Aries
S
5-Leo
4 VII
2-Taurus
3-Gemini
Cancer

Fig. 2

The time intervals of the evolution of the true Sun in each of the 12 constellations of the Zodiac are as follows: **1)** Aries, 21 III - 20 IV; **2)** Taurus, 21 IV - 20 V; **3)** Gemini, 21 V - 20 VI; **4)** Cancer, 21 VI - 22 VII; **5)** Leo, 23 VII - 22 VIII; **6)** Virgo, 23 VIII - 22 IX; **7)** Libra, 23 IX - 22 X; **8)** Scorpio, 23 X - 21 XI; **9)** Sagittarius, 22 XI - 21 XII; **10)** Capricorn, 22 XII - 21 I; **11)** Aquarius, 22 I - 19 II; **12)** Pisces, 20 II - 21 III.

The International Olympiad on Astronomy and Astrophysics for Juniors, Edition I, was to take place in SUCEAVA, in ROMANIA, on March 28 - April 3, 2020. The event, proposed by ROMANIA, was established at the International Olympiad of Astronomy and Astrophysics, 13th Edition, held in Hungary, on August 2-10, 2019.

But, given the known conditions, IOAA-jr, Edition I, is taking place in Romania, only now, in November 2022!

Hint: *All the tasks which require direct measurements on drawings will be performed on the pictures in page 5 (figure 1) and page 6 (figure 2). Both pages are part of answer-sheets*

A) (30 points) *Identify*:

1. the sign of the OIAA development, 13th Edition, Hungary, August 2 - 10, 2019;

2. the sign when the OIAA - J was to take place - Edition I, March 28 - April 3, 2020, Romania (Edition I, proposal);

3. the sign of the OIAA - J - Edition I, November, 2022.

B) (20 points) Each of the 3 specified Olympic events can be considered to have taken place on the day when the Sun, viewed from Earth, was in the middle of the angular range corresponding to each of the 3 signs.

1. *Determine* the time interval between any two of these three events:

$$\Delta t_{\mathrm{H-R,1}};\ \Delta t_{\mathrm{R,1-R,2}};\ \Delta t_{\mathrm{H-R,2}}.$$

2. *Estimate*, by direct measurements on the drawing, by means of a protractor, the value of the angle between the position vector of the Earth, in relation to the Sun, corresponding to each of the three specified moments, ie the directions of the vectors $\vec{r}_{\mathrm{H}}$, $\vec{r}_{\mathrm{R,1}}$ and respectively $\vec{r}_{\mathrm{R,2}}$, and the direction of the apse line, Aph – Ph, ie the angles α_{H}, $\alpha_{\mathrm{R,1}}$ and respectively $\alpha_{\mathrm{R,2}}$.

C) (20 points) *Determine* the area of the surface described by the position vector of the center of the Earth, $\vec{r}$, in relation to the center of the Sun:

1. from position $\vec{r}_{\mathrm{Hungry}}$ to position $\vec{r}_{\mathrm{Romania,\,1}}$;

2. from position $\vec{r}_{\mathrm{Romania,\,1}}$ to position $\vec{r}_{\mathrm{Romania,\,2}}$;

3. from position $\vec{r}_{\mathrm{Hungry}}$ to position $\vec{r}_{\mathrm{Romania,\,2}}$.

D) (30 points) *Determine*:

1. the distance between the Earth and the Sun, on the day when in Suceava, in 2020, the IOAA for Juniors, Edition I should have taken place, $r_{R,1}$, if, for the ellipse representing the Earth's orbit around the Sun, the following are known: semimajor axis, $a = 149\,597\,500$ km and semiminor axis, $b = 149\,580\,670$ km;

2. the acceleration of the center of the Earth, on the day when in Suceava, in 2020, the IOAA for Juniors should have taken place, $a_{E,1}$, and compared with the gravitational acceleration in the gravitational field of the Sun, corresponding to the distance $r_{R,1}$ from the center of the Sun, $g_{S,1}$;

3. the components of the speed of the center of the Earth, $\vec{v}_{R,1}$, parallel to the major axis of the ellipse and respectively perpendicular to the major axis of the ellipse, in the days of the International Olympiad in Romania, $v_{//}$ and respectively $v_{\perp}$.

It is known that the angular momentum of the Earth in relation to the center of the Sun and the total mechanical energy of the Earth-Sun system are given by the expressions:

$$L = M_E b \cdot \sqrt{\frac{G M_S}{a}}; \qquad E = -G\frac{M_E M_S}{2a}.$$

It is given: gravitational attraction constant, $G = 6.67 \cdot 10^{-11}\ Nm^2\,kg^{-2}$ mass of the Sun, $M_S = 1.989 \cdot 10^{30}\ kg$, and period of Earth's revolution is *T=365.256 days.*

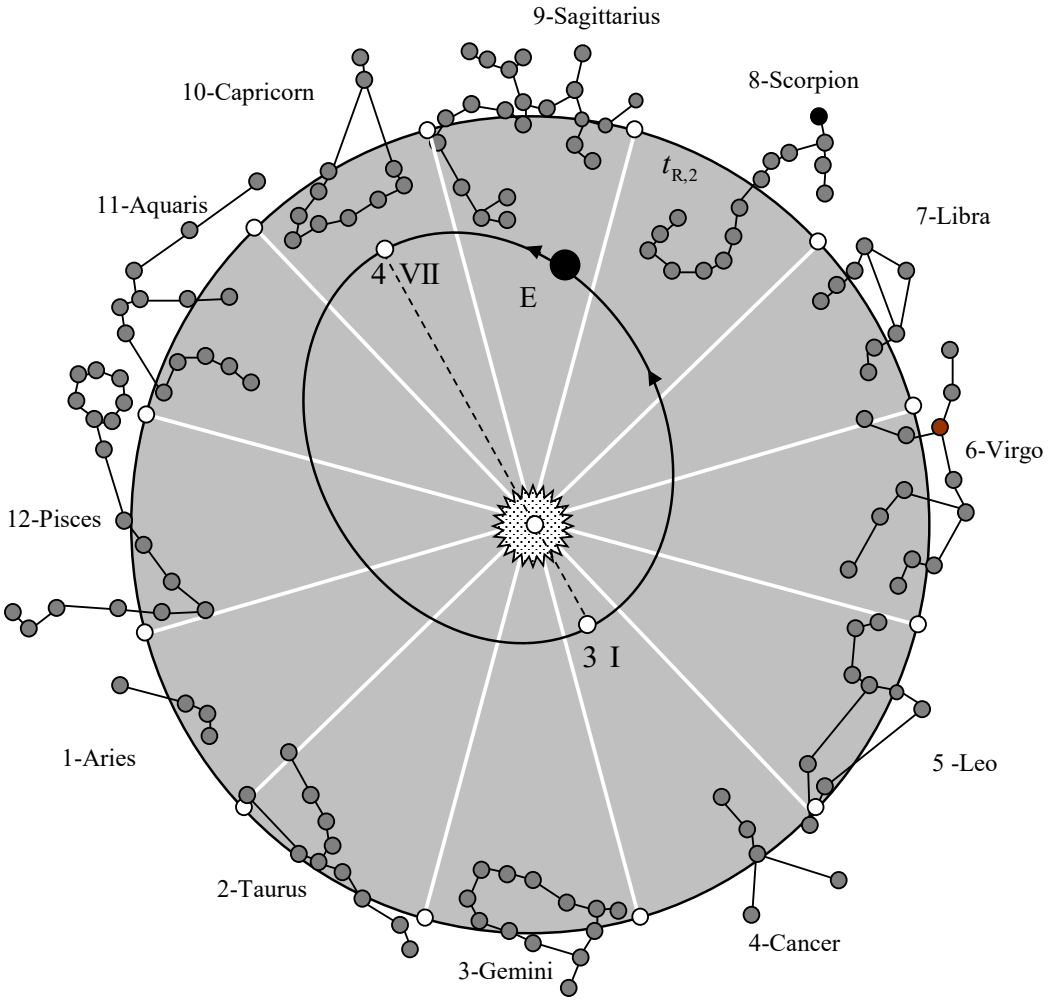


Figure 1, reproduced in enlarged format!
This image will be worked on and will be handed in with the competition sheets!

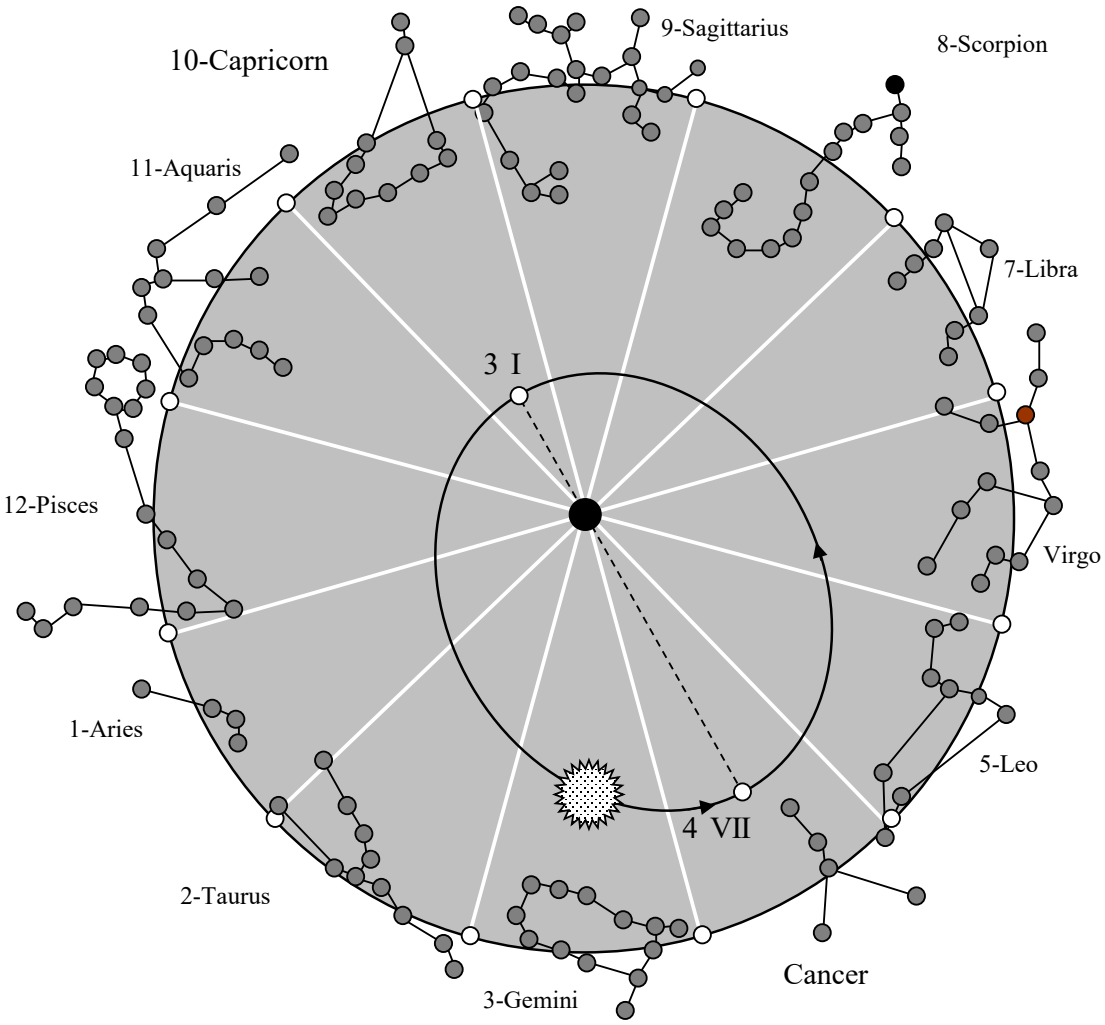


Figure 2, reproduced in enlarged format!
This image will be worked on and will be handed in with the competition sheets!

Problem 4.5 The Rocket that Destroys the Threatening Asteroid

2022 | Câmpulung Moldovenesc, Romania | Theory | Q12 | 100 points

Solution p. 52

III. The second long problem - The rocket that destroys the threatening asteroid – 100 points

With the help of a rocket, R, launched from a spaceship, N, an asteroid, A, must be destroyed, approaching the Earth threateningly, coming right in the direction of the center of the Earth, as shown in the drawing in Figure 1. There are positions of the ship spacecraft, N and the asteroid A, respectively, at the time of the launch of the rocket R, to meet the asteroid A, when the speed of the spacecraft, N, relative to Earth, is constant,, oriented along the spacecraft line - asteroid, and the speed of asteroid A, relative to Earth, is considered constant, $\vec{v}_N$, oriented along the spacecraft line - asteroid, and the speed of asteroid A, relative to Earth, is considered constant, $\vec{v}_A$, its direction forming an angle α with the direction of the spacecraft, N.
During the flight of the rocket, from its launch to the impact with the asteroid, it is considered that the Earth, relative to the Sun, is at rest, and the movements of the rocket, R, the spacecraft, N, and the asteroid, A, are rectilinear and uniform movements.

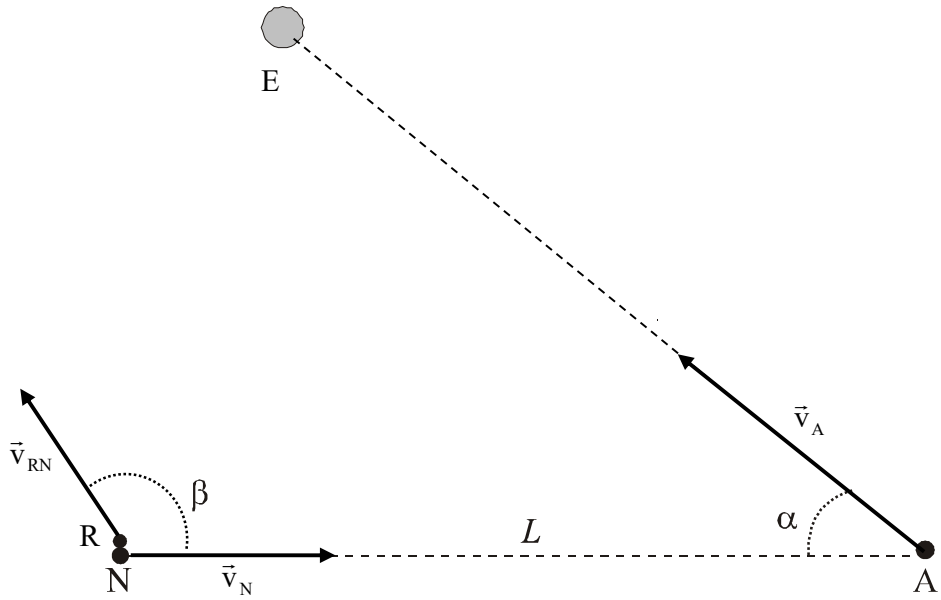


Fig. 1

We assume, that the direction of absolute displacement of the rocket R is perpendicular to the direction of absolute movement of asteroid A.
Determine the angle β, at which the rocket R was launched, relative to the direction of movement spacecraft, N.

Problem 4.6 Planet orbits

2023 | Volos, Greece | Theory | Q2 | 10 points

Solution p. 54

Question 2: Planet orbits

Two planets, A and B, orbit a star S in circular, coplanar orbits, in the same direction. The orbital radius of planet A is 9 times smaller than the orbital radius of planet B. The orbital period of planet A is 10 Earth days. An astronomer on Earth realizes that at 10:00 am on 26/9/2023, an observer on planet B would observe a transit of planet A in front of star S. When is the next transit due to planet A to be seen by an observer on planet B?

(All dates/times are given on Earth times).

The masses of planets A and B are very small compared to the mass of star S.

Problem 4.7 Satellites

2023 | Volos, Greece | Theory | Q12 | 40 points

Solution p. 55

Question 12: SATELLITES

The term geostationary (GEO) refers to a satellite whose orbit is on the equatorial plane, its orbit is circular, and its orbital period is equal to Earth's period of revolution around its axis.

For this problem, all orbits around Earth are considered circular. An important characteristic of the orbit of a satellite is the elevation angle of the satellite, which is defined as the angle at which the satellite is visible above the local horizon for an observer on Earth.

A) What is the maximum latitude the geostationary satellite is visible at? **[12 Points]**

Geostationary satellites, because of their distance from the center of the Earth, observe only a fraction of the Earth's surface. If a satellite needs to view a larger fraction of the Earth, it must change its orbit using rockets.

B) The satellite moves from its location and observes the Earth having an angular diameter of 10°. What is the distance of the satellite from the center of the Earth? **[10 Points]**

C) How much does the total (kinetic and potential) energy of the satellite need to change to move from the geostationary orbit to the one described in question (B)? The satellite has a mass of 1000 kg. **[10 Points]**

D) What is the ratio of the gravitational acceleration g on the Earth's surface compared to the gravitational acceleration at the location described in question (B). **[8 Points]**

Universal gravitational constant	G	=	$6.674\times10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$
1 sidereal day		=	$23^{h}56^{m}04^{s}$

Object	Mean radius (km)	Mass (kg)	Orbital Radius (au)
Sun	695 500	1.988×10^{30}	---
Mercury	2 440	3.301×10^{23}	0.387
Venus	6 052	4.867×10^{24}	0.723
Earth	6 378	5.972×10^{24}	1.000 000

Problem 4.8 Energy and speed in an elliptical orbit

2024 | Kathmandu, Nepal | Theory | T2 | 15 points

Solution p. 57

T2 (15 points)

Starting from the law of conservation of energy, derive an expression for the total mechanical energy of a body of mass m on an elliptical orbit, in terms of G, the mass of the object being orbited (M) and the semi-major axis of the orbit a. Use your derivation to then find the ratio of the speeds at perihelion and aphelion, for an object that has an orbit with eccentricity e.

Problem 4.9 Altitude of a geostationary satellite

2024 | Kathmandu, Nepal | Theory | T5 | 15 points

Solution p. 58

T5 (15 points)

Nepal has ambitions to launch a geostationary satellite of its own. If the satellite is to maintain a geostationary orbit such that, when viewed from Kathmandu, it is at the prime meridian. What would be its altitude when observed from Kathmandu? You can neglect refraction due to the atmosphere and other smaller effects.

Universal Gravitational constant	$G = 6.674 \times 10^{-11}\ \mathrm{N\ m^2\ kg^{-2}}$

Object	Mean radius (km)	Mass (kg)	Semi-major axis (AU)	Eccentricity	Albedo
Earth	6 378	5.972×10^{24}	1.000	0.016710	0.31

Information about the host city

Latitude of Kathmandu City, $\phi_{kathmandu} = 27.7172^{o}N$
Time Zone for Nepal = UTC + 5:45:00

1 sidereal day	$23^{\mathrm{h}}56^{\mathrm{m}}04^{\mathrm{s}}$

Problem 4.10 Free fall inside a gas cloud

2024 | Kathmandu, Nepal | Theory | T9 | 15 points

Solution p. 59

T9 (15 points)

Suppose you are travelling on a spacecraft that encounters a uniformly spherically distributed gas cloud with radius R, and volume density ρ. At your moment of approach, you decelarate till you are at rest with respect to the gas cloud and then turn off your engines. You may neglect gas friction in this problem.

*Note: Gravitational Potential inside the uniform solid sphere is given by

$$V\left(r\right) = -\frac{GM}{R^3}\left(\frac{3R^2}{2} - \frac{r^2}{2}\right)$$

where r is the distance from the center of the gas cloud. You may also find the following fact useful. For an equation of the form $a = -k^2 r$, where k is some constant and a is acceleration, the time period of the oscillation is given by: $T = \frac{2\pi}{k}$.

1. Once inside the gas cloud, how long will it take you to return to your original position if you are in free fall inside the cloud?
2. Start from the law of conservation of energy. Suppose you are tired of being inside the gas cloud after investigating it for a while and want to escape it (to infinity). What is your required speed if you are initially located in the center of the cloud and are at rest?

Problem 4.11 Speed of Asteroid

2025 | Piatra Neamț, Romania | Theory | Th.2 | 10 points

Solution p. 60

Th.2 Speed of Asteroid (10 Points)

An asteroid moves in an elliptical orbit around the Sun. Its maximum orbital speed is 19.00 km/s and its minimum orbital speed is 14.00 km/s.

a) Compute the eccentricity of the orbit.

b) What is the speed of the asteroid when it is exactly at either of the end points of the minor axis of its orbit?

c) A comet from near the outer edge of our Solar System reaches perihelion with an orbital speed of 90.00 km/s. At this instant, the asteroid is at aphelion and the comet, Sun, and asteroid are aligned (comet-Sun-asteroid). Compute the distance between the comet and the asteroid.

Problem 4.12 Roche Limit, Hill Sphere, Lagrange Points

2025 | Piatra Neamț, Romania | Theory | Th.8 | 40 points

Solution p. 61

Th.8 Roche Limit, Hill Sphere, Lagrange Points (40 points)

The following astronomical data are given:

- The masses of *Mars* M_M and the *Sun* $M_\odot$ are expressed relative to *Earth's* mass $M_\oplus$: $M_M = 0.107 M_\oplus$ and $M_\odot = 333054 M_\oplus$.
- The mean *Mars-Sun* distance is $r_{MS} = 1.524\,\text{au}$ and we consider that *Mars* has a circular orbit.
- The mean *Earth-Sun* distance is $r_{ES} = 1\,\text{au} = 1.496 \times 10^{11}\,\text{m}$ and we consider that *Earth* has a circular orbit.
- *Mars's* moon *Phobos* has a mass equal to $M_{Phobos} = 1.5 \times 10^{-8} M_M$;
- *Earth's* sidereal period is $P_{Sid\oplus} = 1\,\text{year} = 365.2564\,\text{days}$;
- The average radius of the satellite Phobos is approximately $R_{Phobos} = 12.5\,\text{km}$;
- The radius of its orbit around *Mars* is $d_{PhobosM} = 9380\,\text{km}$;
- Phobos is spiraling inward towards *Mars* at the constant rate of $1.8\,\text{cm/yr}$.
- You can use the following approximation:

$$(1+x)^n \approx 1 + nx, \text{when } x << 1.$$

- Roche limit is the boundary inside which the tidal force of the planet on a small mass is more than its self-gravity.

a) Derive the formula for the L_1 Lagrange point (distance between secondary body and the L_1 point) and calculate its numerical value for the *Sun-Mars* system. Find the ratio of Phobos's orbital radius to this distance.

b) Derive the formula for the L_2 Lagrange point (distance between secondary body and the L_2 point) and calculate its numerical value for the the *Mars-Phobos* system.

c) Derive the formula for the *Roche* limit of the *Mars-Phobos* system. How long would it take Phobos to reach this limit? It is known that it approaches the surface of the planet during its rotation and will either collide with it or begin to disintegrate, forming a ring. The satellite is considered to be solid.

d) Calculate the distance l between the *Lagrange* points L_4 and L_5 for Sun - Mars system.

Solutions to Orbital Mechanics

Solution 4.1 The Escape Velocity

2022 | Câmpulung Moldovenesc, Romania | Theory | Q3

Problem p. 27

Solution

Let $\vec{v}_B$ the speed of body B at the time of its launch from Earth, in relation to the Sun, so that the body reaches the limit of the gravitational attraction of the Sun and there it is at rest in relation to the Sun. Using the details in Figure 1, in accordance with the law of conservation of mechanical energy, it follows:

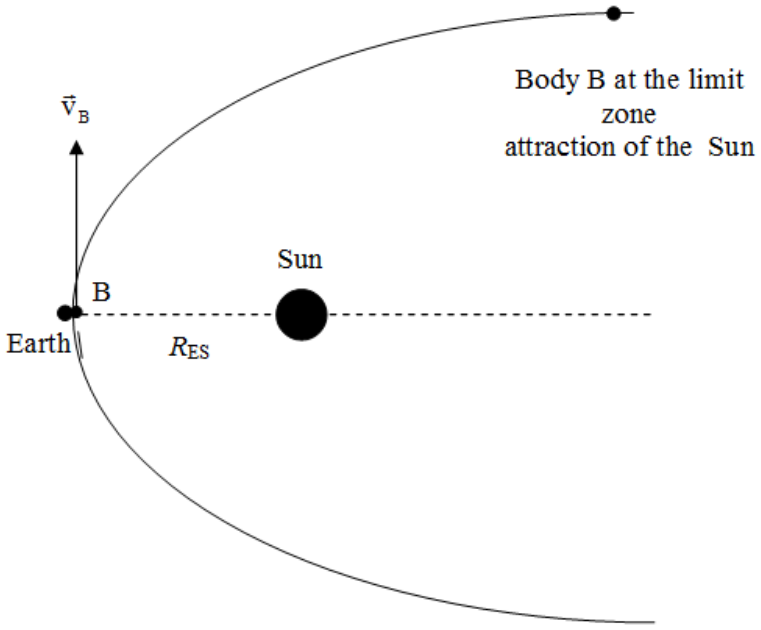


Fig. 1

$$R_E << R_{ES};$$

$$\frac{m\mathrm{v}_\mathrm{B}^2}{2} - K\frac{mM_\mathrm{E}}{R_\mathrm{E}} - K\frac{mM_\mathrm{S}}{R_\mathrm{ES}} = 0;$$

$$\frac{M_\mathrm{E}}{R_\mathrm{E}} << \frac{M_\mathrm{S}}{R_\mathrm{ES}};$$

$$\frac{m\mathrm{v}_\mathrm{B}^2}{2} - K\frac{mM_\mathrm{S}}{R_\mathrm{ES}} = 0;$$

$$\mathrm{v}_\mathrm{B} = \sqrt{2}\sqrt{K\frac{M_\mathrm{S}}{R_\mathrm{ES}}};$$

$$\sqrt{K\frac{M_\mathrm{S}}{R_\mathrm{ES}}} = \mathrm{v}_\mathrm{ES} = V_\mathrm{orbital} = V_0,$$

representing the orbital speed of the Earth on the circular trajectory around the Sun;

$$\mathrm{v}_\mathrm{B} = \sqrt{2}\ V_0,$$

representing the second cosmic speed in relation to the Sun (parabolic speed);

$$R_\mathrm{ES} \approx 1.5\cdot 10^8 \text{ km};\ T_\mathrm{ES} = 1 \text{ year};$$

$$V_0 = \frac{2\pi R_\mathrm{ES}}{T_\mathrm{S}} \approx 30\,\frac{\text{km}}{\text{s}};\ \mathrm{v}_\mathrm{B} \approx 42.42\,\frac{\text{km}}{\text{s}}.$$

Conclusion: the body launched from the Earth, reaches the limit of the gravitational attraction of the Sun, in relation to the Sun, on a parabolic trajectory, with the Sun in its focus.

The S3 file restates the problem as an Earth-relative third cosmic speed, while its derivation and final 42.42 km/s result use the Sun-relative speed asked for in the question paper.

Solution 4.2 Balls Suspended inside a Terrestrial Satellite

2022 | Câmpulung Moldovenesc, Romania | Theory | Q7

Problem p. 27

Solution

The orientation of the spacecraft being constant, and it is, as a whole, a rigid solid body, means that all points belonging to the spacecraft move around the Earth at the same angular velocity, ω, whose expression is obtained as follows:

$$F_{\text{ag(NC)}} = K\frac{m_{\text{(NC)}}M}{r^2} = m_{\text{(NC)}}\omega^2 r;\ \ \omega^2 = K\frac{M}{r^3}.$$

The centripetal accelerations of the three suspended balls are:

$$a_{\text{cp(a)}} = \omega^2 r;\ \ a_{\text{cp(b)}} = \omega^2(r-d);\ \ a_{\text{cp(c)}} = \omega^2(r+d);\ \ a_{\text{cp(d)}} = \omega^2 r = a_{\text{cp(a)}}.$$

Assuming that all the suspension wires are tensioned, then, for each ball, the resultant of the gravitational pull force with the tension in the respective suspension wire is the centripetal force responsible for the circular motion of each ball.

Results that:

- for *the ball* (a):

$$\vec{F}_{\text{cp(a)}} = \vec{F}_{\text{ag(a)}} + \vec{T}_{\text{(a)}};\ \ F_{\text{cp(a)}} = F_{\text{ag(a)}} - T_{\text{(a)}};\ \ T_{\text{(a)}} = F_{\text{ag(a)}} - F_{\text{cp(a)}};$$

$$F_{\text{ag(a)}} = K\frac{mM}{r^2};\ F_{\text{cp(a)}} = ma_{\text{cp(a)}} = m\omega^2 r;\ \omega^2 = K\frac{M}{r^3};\ F_{\text{cp(a)}} = ma_{\text{cp(a)}} = K\frac{mM}{r^2};$$

$$T_{\text{(a)}} = K\frac{mM}{r^2} - K\frac{mM}{r^2};$$

$$T_{\text{(a)}} = 0,$$

which means that the suspension wire of the ball (a) is not tensioned, so that the ball (a) is in a weightless state;

- *for the ball* (b):

$$\vec{F}_{\text{cp(b)}} = \vec{F}_{\text{ag(b)}} + \vec{T}_{\text{(b)}};\ \ F_{\text{cp(b)}} = F_{\text{ag(b)}} - T_{\text{(b)}};\ \ T_{\text{(b)}} = F_{\text{ag(b)}} - F_{\text{cp(b)}};$$

$$F_{\text{ag(b)}} = K\frac{mM}{(r-d)^2} = K\frac{mM}{r^2\left(1-\frac{d}{r}\right)^2} = K\frac{M}{r^2}\left(1-\frac{d}{r}\right)^{-2};$$

$$d << r;\ \left(1-\frac{d}{r}\right)^{-2} \approx 1+2\cdot\frac{d}{r};$$

$$F_{\text{ag(b)}} = K\frac{M}{r^2}\left(1+2\cdot\frac{d}{r}\right);$$

$$F_{\text{cp(b)}} = ma_{\text{cp(b)}};\ \ a_{\text{cp(b)}} = \omega^2(r-d);\ \ F_{\text{cp(b)}} = m\omega^2(r-d);$$

$$\omega^2 = K\frac{M}{r^3};\ F_{\text{cp(b)}} = K\frac{mM}{r^3}(r-d);$$

$$T_{\text{(b)}} = K\frac{M}{r^2}\left(1+2\cdot\frac{d}{r}\right) - K\frac{mM}{r^3}(r-d) = K\frac{mM}{r^2} + 2K\frac{mMd}{r^3} - K\frac{mM}{r^2} + K\frac{mMd}{r^3};$$

$$T_{\text{(b)}} = 3K\frac{mMd}{r^3} \neq 0;\ \ T_{\text{(b)}} = 3K\frac{mMd}{r^3} > 0,$$

which means that the ball suspension wire (b) is tensioned so that the ball (b) is not weightless;

- *for the ball* (c)

$$\vec{F}_{cp(c)} = \vec{F}_{ag(c)} + \vec{T}_{(c)};\ \ F_{cp(c)} = F_{ag(c)} + T_{(c)};\ \ T_{(c)} = F_{cp(c)} - F_{ag(c)};$$

$$F_{ag(c)} = K\frac{mM}{(r+d)^2} = K\frac{mM}{r^2\left(1+\frac{d}{r}\right)^2} = K\frac{mM}{r^2}\left(1+\frac{d}{r}\right)^{-2};$$

$$d << r;\ \left(1+\frac{d}{r}\right)^{-2} \approx 1-2\cdot\frac{d}{r};$$

$$F_{ag(c)} = K\frac{mM}{r^2}\left(1-2\cdot\frac{d}{r}\right);$$

$$F_{cp(c)} = ma_{cp(c)};\ \ a_{cp(c)} = \omega^2(r+d);\ \ F_{cp(c)} = m\omega^2(r+d);$$

$$\omega^2 = K\frac{M}{r^3};\ F_{cp(c)} = K\frac{mM}{r^3}(r+d);$$

$$T_{(c)} = K\frac{mM}{r^3}(r+d) - K\frac{mM}{r^2}\left(1-2\cdot\frac{d}{r}\right) = K\frac{mM}{r^2} + K\frac{mMd}{r^3} - K\frac{mM}{r^2} + 2K\frac{mMd}{r^3};$$

$$T_{(c)} = -3K\frac{mMd}{r^3} \neq 0;\ \ T_{(c)} = 3K\frac{mMd}{r^3} > 0,$$

which means that the ball suspension wire (c) is tensioned so that the ball (c) is not weightless;

- *for the ball* (d)

$$\vec{F}_{cp(d)} = \vec{F}_{ag(d)} + \vec{T}_{(d)};\ \ F_{cp(d)} = F_{ag(d)} + T_{(d)};\ \ T_{(d)} = F_{cp(d)} - F_{ag(d)};$$

$$F_{ag(d)} = K\frac{mM}{r^2};\ F_{cp(d)} = ma_{cp(d)} = m\omega^2 r;\ \omega^2 = K\frac{M}{r^3};\ F_{cp(d)} = ma_{cp(d)} = K\frac{mM}{r^2};$$

$$T_{(d)} = K\frac{mM}{r^2} - K\frac{mM}{r^2};$$

$$T_{(d)} = 0,$$

which means that the suspension wire of the ball (d) is not tensioned, so that the ball (d) is in a weightless state.

Solution 4.3 Spaceship to the Sun

2022 | Câmpulung Moldovenesc, Romania | Theory | Q8

Problem p. 29

Solution

Imagine that a spacecraft is launched from the Earth whose speed in relation to the Sun is slightly lower than the orbital speed of the Earth in relation to the Sun. The ship will evolve around the Sun in an elliptical trajectory, with the Sun in the focus opposite the launch point. The lower this initial speed of the ship, the longer the ellipse on which the spacecraft will evolve around the Sun will be longer. For a certain value of this speed, the spacecraft, in its elliptical motion around the Sun, will touch the surface of the Sun, as shown in Figure 1. In this case, the height of the ship's orbit is equal to the Sun's radius. which we know is:

$$r_{\min} = R_S \approx \frac{r_{ES}}{200}.$$

As a result, the semi-axes of the ship's elliptical orbit, tangent to the Sun's surface, are a and b << a, respectively, so that the small half-axis of this elliptical orbit can be neglected ($b \approx 0$). Under these conditions, the trajectory of the ship in relation to the Sun can be estimated to be rectilinear, joining the launching point of the ship with the Sun, so that the ship will reach the surface of the Sun, in point B, after a free fall towards it.

Conclusion: In order for the ship to fall freely on the surface of the Sun, in point B, after a rectilinear course in relation to it, at the initial moment of its launch from the surface of the Earth, the speed of the ship in relation to the Sun must have been zero.

To meet this condition, it is necessary that at the time of launching the ship from the Earth's surface, a speed relative to the Earth be imprinted on the ship, so that the speed of the ship relative to the Sun is zero.

The Earth's launch velocity relative to Earth must be inversely related to the orientation of the Earth's orbital velocity relative to the Sun, as shown in Figure b in Figure 1, and their modulus must be equal:

$$v_{ES} = \frac{2\pi r_{ES}}{T_E} = \frac{2 \cdot 3{,}14 \cdot 1{,}5 \cdot 10^{11}\ \text{m}}{3{,}15 \cdot 10^{7}\ \text{s}} \approx 3 \cdot 10^{4}\ \text{m/s} = 30\ \text{km/s}.$$

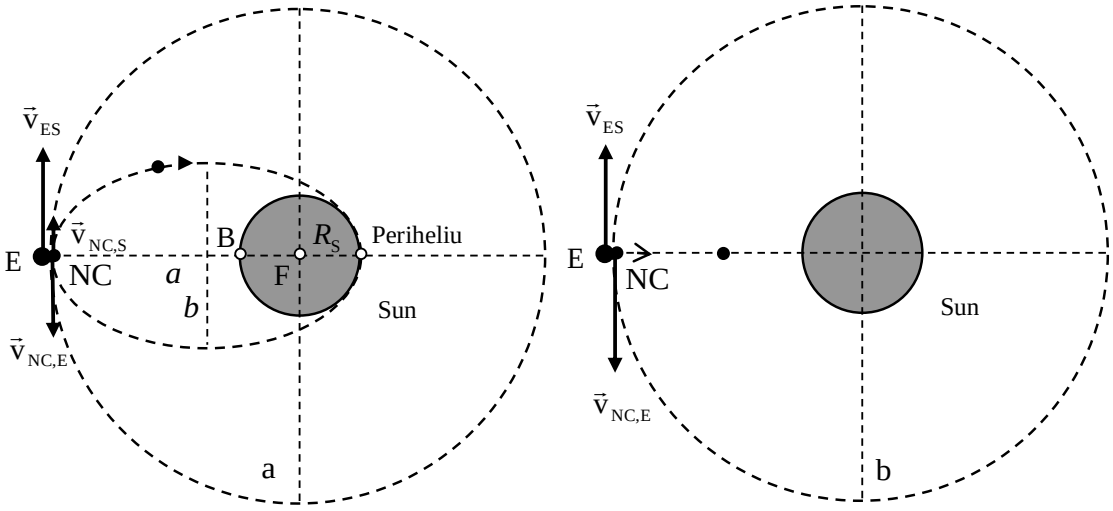


Fig. 1

According to Kepler's third law, it can be written that:

$$\frac{T_E^2}{r_{ES}^3} = \frac{T_{NC}^2}{\left(\frac{r_{ES}}{2}\right)^3},$$

where T_{NC} is the period of the spacecraft's motion on the elongated elliptical trajectory around the Sun;

$$T_{NC} = \frac{T_E}{2\sqrt{2}},$$

so that the duration of the free fall of the ship on the surface of the Sun is:

$$t = \frac{T_{NC}}{2} = \frac{T_E}{4\sqrt{2}} \approx 5.6 \cdot 10^6 \text{ s}.$$

Solution 4.4 Zodiac IOAA-J 2022

2022 | Câmpulung Moldovenesc, Romania | Theory | Q11 | 100 points

Problem p. 29

Solving

a)

1) In the drawing in figure 3 are represented the positions of the Earth on its elliptical orbit around the Sun, as well as the positions of the projections of the Sun in the zodiacal belt, corresponding to the moments of the three events, specified in the statement of the problem.

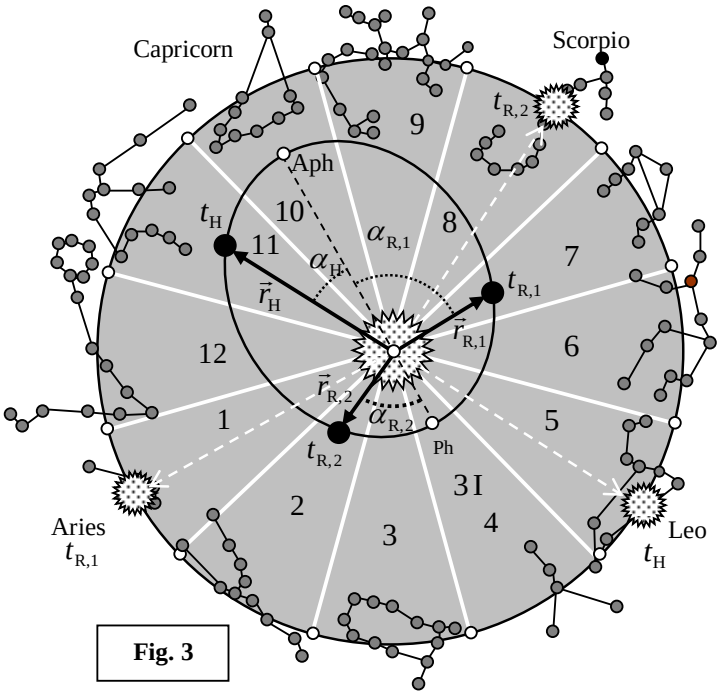


Fig. 3

In the drawing in figure 4, the positions of the Sun on its apparent elliptical orbit around the Earth are represented, as well as the positions of the projections of the Sun in the zodiac belt, corresponding to the moments of the three events, specified in the statement of the problem.

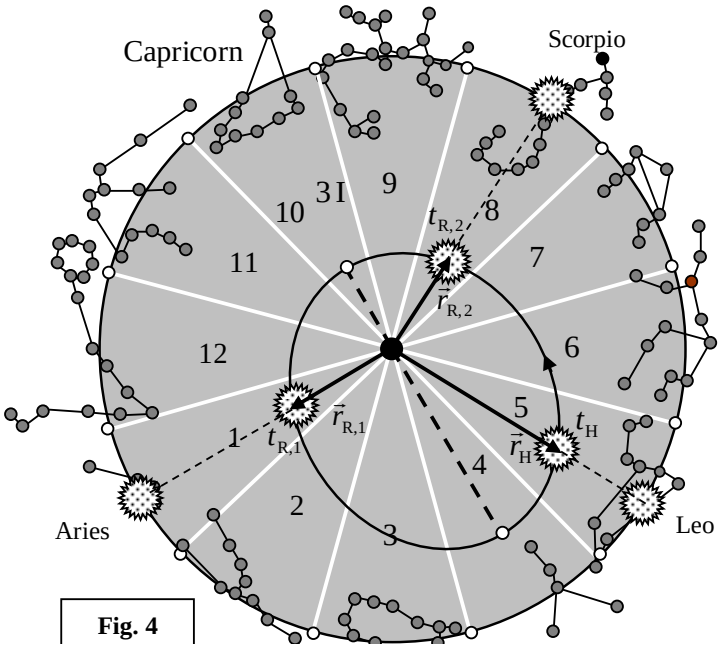


Fig. 4

In these conditions, we identify:

1. the announcement from Hungary was made in the sign of LEO, in the middle of it, on August 6, 2019;

2. the event proposed to take place in Romania was to take place in the sign of ARIES, in the middle of it, on April 4, 2020;

3. the International Olympics of Astronomy and Astrophysics, for Juniors, Edition I, takes place in Romania, in November, 2022, in the sign of SCORPIO.

b) The position vectors of the Earth, in relation to the Sun, as well as the moments corresponding to the three events, are:

1. for IOAA – 2019, Hungary: $\vec{r}_{\mathrm{H}}$; $t_{\mathrm{H}} =$ 6 August, 2019;

2. for IOAA – J – 2020, Romania - proposal: $\vec{r}_{\mathrm{R},1}$; $t_{\mathrm{R},1} = 4$ April, 2020,

3. for IOAA – J – 2022, Romania: $\vec{r}_{\mathrm{R},2}$; $t_{\mathrm{R},2} =$

so that the time intervals between any two of these three events are:

$$\Delta t_{\mathrm{H-R},1} = t_{\mathrm{R},1} - t_{\mathrm{H}} = 223 \text{ days};$$

$$\Delta t_{\mathrm{R},1-\mathrm{R},2} = t_{\mathrm{R},2} - t_{\mathrm{R},1} = \text{ days};$$

$$\Delta t_{\mathrm{H-R},2} = t_{\mathrm{R},2} - t_{\mathrm{H}} = \Delta t_{\mathrm{H-R},1} + \Delta t_{\mathrm{R},1-\mathrm{R},2} = 223 \text{ days} + 886 \text{ days} = 1109 \text{ days}.$$

From direct measurements, performed on the drawing in Figure 3, it results:

$$\alpha_{\mathrm{H}} \approx 30^0;$$

$$\alpha_{\mathrm{R},1} \approx 90^0;\ \vec{r}_{\mathrm{R},1} \perp (\mathrm{Aph-Ph});$$

$$\alpha_{\mathrm{R},2} \approx 60^0.$$

c)

1. It is known that in the evolution of the Earth in the elliptical orbit around the Sun, according to Keppler's third law, the areolar velocity of the Earth is constant, so that:

$$\Omega = \frac{\Delta A}{\Delta t} = \text{constant};$$

$$\frac{A_{\mathrm{ellipse}}}{T} = \frac{\Delta A_{\mathrm{H-R},1}}{\Delta t_{\mathrm{H-R},1}};$$

$$\Delta A_{\mathrm{H-R},1} = \frac{\Delta t_{\mathrm{H-R},1}}{T} A_{\mathrm{ellipse}} = \frac{\Delta t_{\mathrm{H-R},1}}{T} \cdot \pi \cdot a \cdot b;$$

$$\Delta t_{\mathrm{H-R},1} = 223 \text{ days};\ T = 365.256 \text{ days};$$

$$a = 149\,597\,500 \text{ km};\ b = 149\,580\,670 \text{ km};$$

$$\Delta A_{\mathrm{H-R},1} = \frac{223 \text{ days}}{365.256 \text{ days}} \cdot 3.14 \cdot 149\,597\,500 \text{ km} \cdot 149\,580\,670 \text{ km};$$

$$\Delta A_{\mathrm{H-R},1} = \frac{223}{365.256} \cdot 3.14 \cdot 149\,597\,500 \cdot 149\,580\,670 \text{ km}^2;$$

$$\Delta A_{\mathrm{H-R},1} = 4.2897 \cdot 10^{16} \text{ km}^2.$$

2.

$$\frac{A_{\text{ellipse}}}{T}=\frac{\Delta A_{\text{R,1–R,2}}}{\Delta t_{\text{R,1–R,2}}};$$

$$\Delta A_{\text{R,1–R,2}}=\frac{\Delta t_{\text{R,1–R,2}}}{T}A_{\text{ellipse}}=\frac{\Delta t_{\text{R,1–R,2}}}{T}\cdot\pi\cdot a\cdot b;$$

$$\Delta A_{\text{R,1–R,2}}=\frac{886\text{ days}}{365.256\text{ days}}\cdot 3.14\cdot 149\,597\,500\text{ km}\cdot 149\,580\,670\text{ km};$$

$$\Delta A_{\text{R,1–R,2}}=\frac{886}{365.256}\cdot 3.14\cdot 149\,597\,500\cdot 149\,580\,670\text{ km}^2;$$

$$\Delta A_{\text{R,1–R,2}}=1.704\cdot 10^{17}\text{ km}^2;$$

3.

$$\frac{A_{\text{ellipse}}}{T}=\frac{\Delta A_{\text{H–R,2}}}{\Delta t_{\text{H–R,2}}};\quad \frac{A_{\text{ellipse}}}{T}=\frac{\Delta A_{\text{H–R,1}}}{\Delta t_{\text{H–R,1}}};$$

$$\Delta A_{\text{H–R,2}}=\frac{\Delta t_{\text{H–R,2}}}{T}A_{\text{ellipse}}=\frac{\Delta t_{\text{H–R,2}}}{T}\cdot\pi\cdot a\cdot b;$$

$$\Delta t_{\text{H–R,2}}=1109\text{ days};\ T=365.256\text{ days};$$

$$a=149\,597\,500\text{ km};\ b=149\,580\,670\text{ km};$$

$$\Delta A_{\text{H–R,2}}=\frac{1109\text{ days}}{365.256\text{ days}}\cdot 3.14\cdot 149\,597\,500\text{ km}\cdot 149\,580\,670\text{ km};$$

$$\Delta A_{\text{H–R,2}}=\frac{1109}{365.256}\cdot 3.14\cdot 149\,597\,500\cdot 149\,580\,670\text{ km}^2;$$

$$\Delta A_{\text{H–R,2}}=2.133\cdot 10^{17}\text{ km}^2.$$

d)

1. According to the notations in figure 5, representing the position of the Earth, in relation to the Sun, on the day of, when, as we have shown, $\vec{r}_{\text{R,1}}\perp(\text{Aph}-\text{Ph})$, it results:

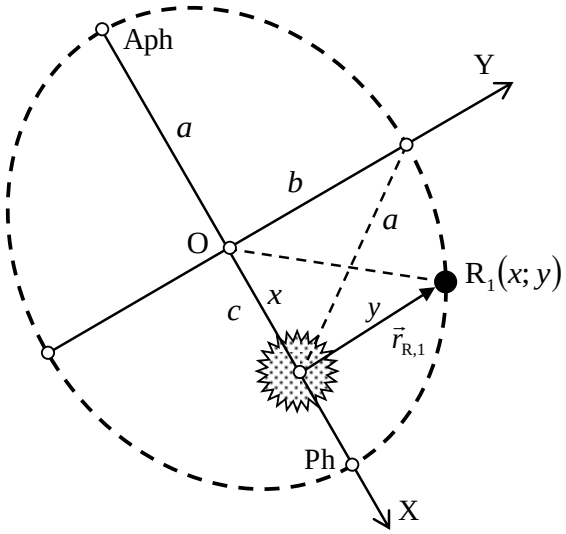


Fig. 5

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1;\ \ x=c;\ y=r_{\mathrm{R},1};$$

$$c^2=a^2-b^2;$$

$$\frac{c^2}{a^2}+\frac{r_{\mathrm{R},1}^2}{b^2}=1;\ \frac{a^2-b^2}{a^2}+\frac{r_{\mathrm{R},1}^2}{b^2}=1;$$

$$1-\frac{b^2}{a^2}+\frac{r_{\mathrm{R},1}^2}{b^2}=1;\ \frac{r_{\mathrm{R},1}^2}{b^2}=\frac{b^2}{a^2};$$

$$r_{\mathrm{R},1}^2=\frac{b^4}{a^2};$$

$$r_{\mathrm{R},1}=\frac{b^2}{a},$$

representing the distance between the center of the Earth and the center of the Sun on the day when the International Olympiad in Astronomy and Astrophysics should have taken place, for Juniorim Edition I, on March 28 - April 3, 2020, in Romania;

$$a=149\,597\,500\ \text{km};\ \ b=149\,580\,670\ \text{km};$$

$$r_{\mathrm{R},1}=149\,563\,841.9\ \text{km}.$$

2.

$$K\frac{M_{\mathrm{E}}M_{\mathrm{S}}}{r_{\mathrm{R},1}^2}=M_{\mathrm{E}}a_{\mathrm{E},1};\ a_{\mathrm{E},1}=K\frac{M_{\mathrm{S}}}{r_{\mathrm{R},1}^2};$$

$$K=6.67\cdot10^{-11}\ \mathrm{Nm^2kg^{-2}};\ M_{\mathrm{S}}=1.989\cdot10^{30}\ \mathrm{kg};$$

$$a_{\mathrm{E},1}=\frac{6.67\cdot10^{-11}\cdot1.989\cdot10^{30}}{(149\,563\,841.9)^2\cdot10^6}\frac{\mathrm{m}}{\mathrm{s}^2};$$

$$a_{\mathrm{E},1}\approx0.006\frac{\mathrm{m}}{\mathrm{s}^2}.$$

$$g_{S,1} = K \cdot \frac{M_S}{r_{R,1}^2} = a_{E,1}.$$

3. Corresponding to the position R_1 of the Earth, represented in the drawing in Figure 6, when its kinetic moment is:

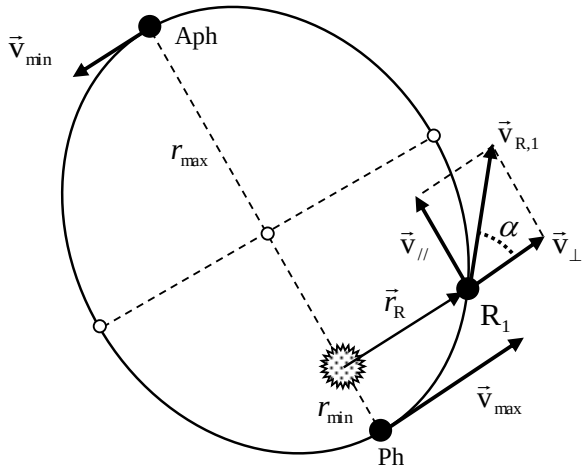


Fig. 6

$$\vec{L}_{R,1} = \vec{r}_{R,1} \times M_E \vec{v}_{R,1};\ L_{R,1} = r_{R,1} M_E v_{R,1} \cdot \sin\left(\vec{r}_{R,1}; \vec{v}_{R,1}\right) = r_{R,1} M_E v_{R,1} \cdot \sin\alpha,$$

according to the law of conservation of kinetic momentum, it follows:

$$v_{R,1} \cdot \sin\alpha = v_{//};$$

$$L_{R,1} = r_{R,1} M_E v_{//};\ L = M_E b \cdot \sqrt{\frac{KM_S}{a}};$$

$$M_E b \cdot \sqrt{\frac{KM_S}{a}} = r_{R,1} M_E v_{//};\ r_R = \frac{b^2}{a};$$

$$v_{//} = \frac{a}{b} \cdot \sqrt{\frac{KM_S}{a}};$$

$$K = 6.67 \cdot 10^{-11}\ \mathrm{Nm^2kg^{-2}};\ M_S = 1.989 \cdot 10^{30}\ \mathrm{kg};$$

$$a = 149\,597\,500\ \mathrm{km};\ b = 149\,580\,670\ \mathrm{km};$$

$$v_{//} = \frac{149\,597\,500\ \mathrm{km}}{149\,580\,670\ \mathrm{km}} \cdot \sqrt{\frac{6.67 \cdot 10^{-11}\ \mathrm{Nm^2kg^{-2}} \cdot 1.989 \cdot 10^{30}\ \mathrm{kg}}{149\,597\,500 \cdot 10^3\ \mathrm{m}}};$$

$$v_{//} = \frac{149\,597\,500}{149\,580\,670} \cdot \sqrt{\frac{6.67 \cdot 10^{-11} \cdot 1.989 \cdot 10^{30}}{149\,597\,500 \cdot 10^3}}\ \frac{\mathrm{m}}{\mathrm{s}};$$

$$v_{//} = \frac{149\,597\,500}{149\,580\,670} \cdot \sqrt{\frac{6.67 \cdot 1.989}{149\,597\,500}} \cdot 10^8\ \frac{\mathrm{m}}{\mathrm{s}};$$

$$v_{//} = \frac{\sqrt{149\,597\,500 \cdot 6.67 \cdot 1.989}}{149\,580\,670} \cdot 10^8\ \frac{\mathrm{m}}{\mathrm{s}};$$

$$v_{//} = 29\,782.9\,\frac{\text{m}}{\text{s}};\ v_{//} = 29\,782.9\cdot 10^{-3}\,\frac{\text{km}}{\text{s}};$$

$$v_{//} = 29.7829\,\frac{\text{km}}{\text{s}}.$$

According to the law of conservation of mechanical energy it results:

$$E = -K\,\frac{M_{\text{E}} M_{\text{S}}}{2a};$$

$$E_{\text{R},1} = \frac{M_{\text{E}}\,v_{\text{R}}^2}{2} - K\,\frac{M_{\text{E}} M_{\text{S}}}{r_{\text{R},1}};$$

$$\frac{M_{\text{E}}\,v_{\text{R},1}^2}{2} - K\,\frac{M_{\text{E}} M_{\text{S}}}{r_{\text{R},1}} = -K\,\frac{M_{\text{E}} M_{\text{S}}}{2a};$$

$$\frac{v_{\text{R},1}^2}{2} - K\,\frac{M_{\text{S}}}{r_{\text{R},1}} = -K\,\frac{M_{\text{S}}}{2a};$$

$$v_{\text{R},1} = \sqrt{KM_{\text{S}}\left(\frac{2}{r_{\text{R},1}} - \frac{1}{a}\right)} = \sqrt{KM_{\text{S}}\cdot\frac{2a - r_{\text{R},1}}{r_{\text{R},1}a}};$$

$K = 6.67\cdot 10^{-11}\ \text{Nm}^2\text{kg}^{-2};\ M_{\text{S}} = 1.989\cdot 10^{30}\ \text{kg};\ r_{\text{R},1} = 149\,563\,841{,}9\ \text{km};\ a = 149\,597\,500\ \text{km};$

$$v_{\text{R},1} = \sqrt{6.67\cdot 10^{-11}\ \text{Nm}^2\text{kg}^{-2}\cdot 1.989\cdot 10^{30}\ \text{kg}\cdot\frac{2\cdot 149\,597\,500\ \text{km} - 149\,563\,841.9\ \text{km}}{149\,563\,841.9\ \text{km}\cdot 149\,597\,500\ \text{km}}};$$

$$v_{\text{R},1} = \sqrt{6.67\cdot 10^{-11}\ \text{Nm}^2\text{kg}^{-2}\cdot 1.989\cdot 10^{30}\ \text{kg}\cdot\frac{2\cdot 149\,597\,500 - 149\,563\,841.9}{149\,563\,841.9\cdot 149\,597\,500\cdot 10^3\ \text{m}}};$$

$$v_{\text{R},1} = \sqrt{6.67\cdot 10^{-11}\cdot 1.989\cdot 10^{30}\cdot\frac{2\cdot 149\,597\,500 - 149\,563\,841.9}{149\,563\,841.9\cdot 149\,597\,500\cdot 10^3}}\cdot 10^8\,\frac{\text{m}}{\text{s}};$$

$$v_{\text{R},1} = 29\,786.25\,\frac{\text{m}}{\text{s}} = 29\,786.25\cdot 10^{-3}\,\frac{\text{km}}{\text{s}};$$

$$v_{\text{R},1} = 29.7862\,\frac{\text{km}}{\text{s}};$$

$$v_{//} = 29.7829\,\frac{\text{km}}{\text{s}};$$

$$v_{\perp} = \sqrt{v_{\text{R},1}^2 - v_{//}^2} = 0.4433\,\frac{\text{km}}{\text{s}}.$$

The supplied official key leaves the 2022 midpoint date and a separate time interval blank. Its printed 223-day interval disagrees with its stated dates; the subsequent swept areas use the source intervals. The numerical values are retained as printed.

Solution 4.5 The Rocket that Destroys the Threatening Asteroid

2022 | Câmpulung Moldovenesc, Romania | Theory | Q12 | 100 points

Problem p. 35

Solving

Reaching the target, asteroid A, by the rocket R, is done in a minimum time, if the direction of absolute displacement of the rocket R, ie the direction of movement NC, is perpendicular to the direction of absolute movement of asteroid A, ie the direction of movement AC, as indicated the drawing in Figure 2, which shows:

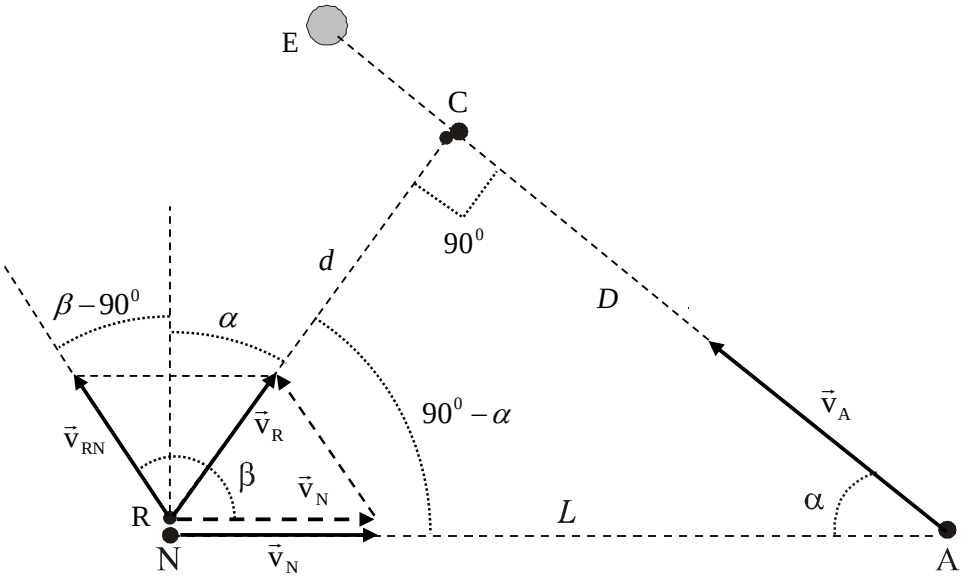


Fig. 2

$$D = v_A \cdot t_{min} = L \cdot \cos\alpha;$$
$$d = v_R \cdot t_{min} = L \cdot \sin\alpha,$$

where v_R is the absolute speed of the R rocket, from the moment of its launch;

$$t_{min} = \frac{L \cdot \cos\alpha}{v_A};\ t_{min} = \frac{L \cdot \sin\alpha}{v_R};$$
$$\frac{L \cdot \cos\alpha}{v_A} = \frac{L \cdot \sin\alpha}{v_R};\ \frac{\cos\alpha}{v_A} = \frac{\sin\alpha}{v_R};$$
$$v_R = v_A \cdot \tan\alpha,$$

where v_R is the absolute speed of the rocket, after launch;

$$\vec{v}_{RN} = \vec{v}_R - \vec{v}_A,$$

where $\vec{v}_{RN}$ is the relative speed of the rocket R, relative to the spacecraft N;

$$v_{RN}^2 = v_R^2 + v_A^2 - 2 \cdot v_R \cdot v_N \cdot \cos(90^0 - \alpha);$$
$$v_{RN}^2 = v_R^2 + v_A^2 - 2 \cdot v_R \cdot v_N \cdot \sin\alpha;$$
$$v_R \cdot \cos\alpha = v_{RN} \cdot \cos(\beta - 90^0),$$

where β is the angle at which the rocket R is launched from the spacecraft, N, relative to the flight direction of the spacecraft, N;

$$v_R \cdot \cos\alpha = v_{RN} \cdot \sin\beta;$$

$$\sin\beta = \frac{v_R}{v_{RN}} \cdot \cos\alpha;$$

$$v_R = v_A \cdot \tan\alpha;$$

$$\sin\beta = \frac{v_A \cdot \tan\alpha}{v_{RN}} \cdot \cos\alpha = \frac{v_A}{v_{RN}} \cdot \frac{\sin\alpha}{\cos\alpha} \cdot \cos\alpha;$$

$$\sin\beta = \frac{v_A}{v_{RN}} \cdot \sin\alpha;$$

$$v_{RN}^2 = v_R^2 + v_A^2 - 2 \cdot v_R \cdot v_N \cdot \sin\alpha;$$

$$v_R = v_A \cdot \tan\alpha;$$

$$v_{RN}^2 = v_A^2 \cdot \frac{\sin^2\alpha}{\cos^2\alpha} + v_A^2 - 2 \cdot v_A \cdot \frac{\sin\alpha}{\cos\alpha} \cdot v_N \cdot \sin\alpha;$$

$$v_{RN}^2 = v_A^2 \cdot \left(\frac{\sin^2\alpha}{\cos^2\alpha} + 1 \right) - 2 \cdot v_A \cdot \frac{\sin^2\alpha}{\cos\alpha} \cdot v_N;$$

$$v_{RN}^2 = v_A^2 \cdot \frac{1}{\cos^2\alpha} - 2 \cdot v_A \cdot v_N \cdot \frac{\sin^2\alpha}{\cos\alpha};$$

$$v_{RN} = v_A \cdot \sqrt{\frac{1}{\cos^2\alpha} - 2 \cdot \frac{v_N}{v_A} \cdot \frac{\sin^2\alpha}{\cos\alpha}};$$

$$v_{RN} = v_A \cdot \sqrt{\frac{1}{\cos^2\alpha} - 2 \cdot \frac{v_N}{v_A} \cdot \frac{\cos\alpha \cdot \sin^2\alpha}{\cos^2\alpha}};$$

$$v_{RN} = \frac{v_A}{\cos\alpha} \cdot \sqrt{1 - 2 \cdot \frac{v_N}{v_A} \cdot \sin^2\alpha \cdot \cos\alpha},$$

representing the relative speed of the rocket R, relative to the spacecraft N;

$$\sin\beta = \frac{v_R}{v_{RN}} \cdot \cos\alpha;$$

$$v_R = v_A \cdot \tan\alpha;$$

$$v_{RN} = \frac{v_A}{\cos\alpha} \cdot \sqrt{1 - 2 \cdot \frac{v_N}{v_A} \cdot \sin^2\alpha \cdot \cos\alpha};$$

$$\sin\beta = \frac{v_A \cdot \frac{\sin\alpha}{\cos\alpha}}{\frac{v_A}{\cos\alpha} \cdot \sqrt{1 - 2 \cdot \frac{v_N}{v_A} \cdot \sin^2\alpha \cdot \cos\alpha}} \cdot \cos\alpha;$$

$$\sin\beta = \frac{\sin\alpha \cdot \cos\alpha}{\sqrt{1 - 2 \cdot \frac{v_N}{v_A} \cdot \sin^2\alpha \cdot \cos\alpha}},$$

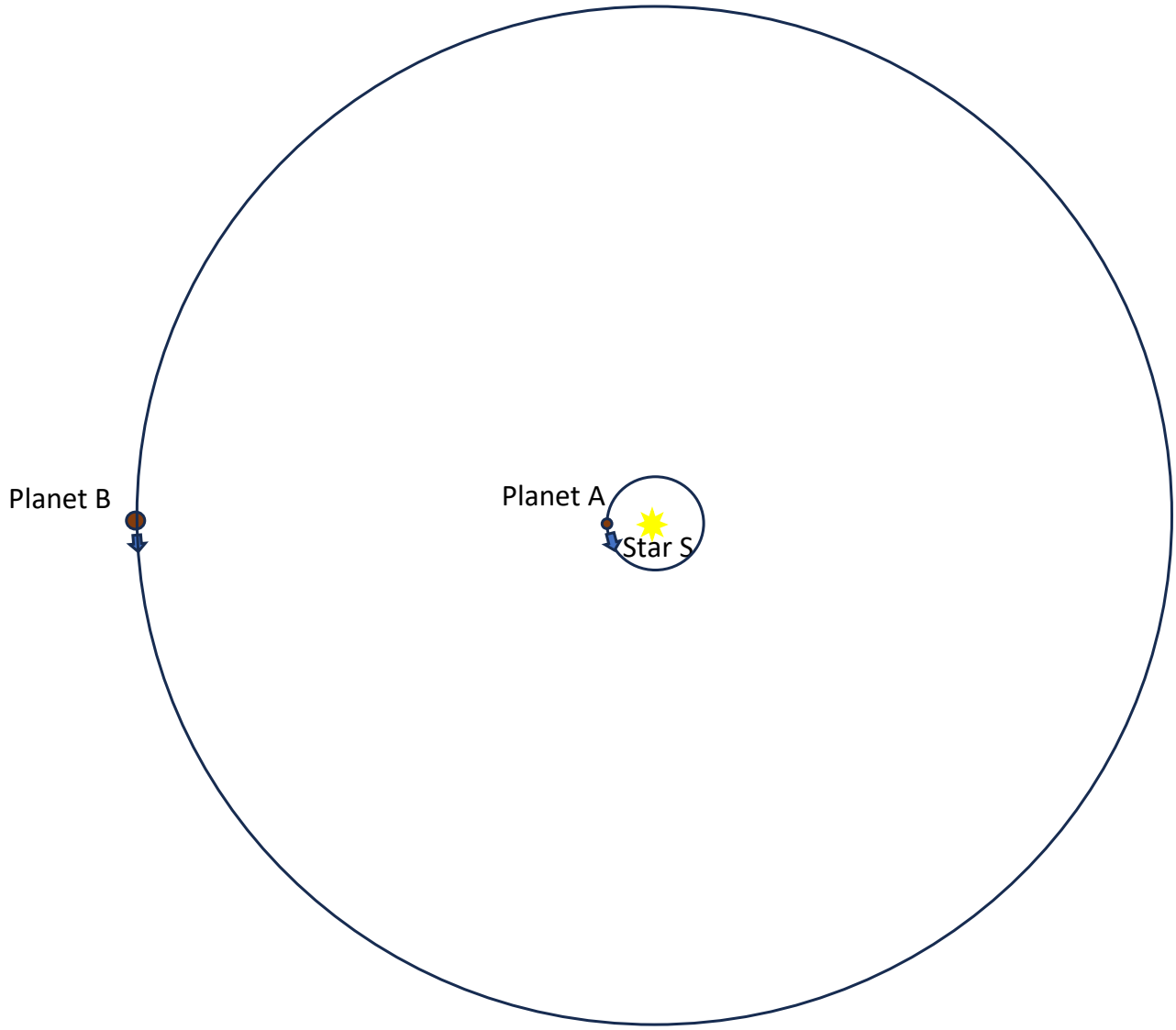
β being the angle at which the rocket R is launched from the spacecraft, N, relative to the direction of flight of the spacecraft, N, to destroy the asteroid.

The supplied solution describes minimum-time interception, whose perpendicular geometry is assumed directly in Q12. Its relative-velocity subtraction and squared-speed expression use v_A where the spacecraft velocity v_N would be expected. The derivation and final expression are reproduced as printed.

Solution 4.6 Planet orbits

2023 | Volos, Greece | Theory | Q2 | 10 points

Problem p. 36



The ratio of the orbits is 9, therefore, according to Kepler's law is the ratio of the squares of the periods is equal to the ratio of the cubes of the radii of the orbits: $(r_1/r_2)^3 = (P_1/P_2)^2$ **[3 points]**

Substituting the numbers, we obtain the period of planet 2:

$P_2=P_1(r_2/r_1)^{3/2} = 10\ 3^3=270$ days. **[2 points]**

The angular velocity is defined as $\omega= 2\pi/P$ and the orbital phase is $\varphi= \omega t$.

A transit will occur when $\varphi_1-\varphi_2=2k\pi$, with k being an integer. The first one will occur for k=1. Therefore:

$2\pi t\ (1/10-1/270) = 2\pi$, t = 10.3846 days = 10 days 9 hours and 14 minutes after the original one, thus on 6 October 2023, 7:14 pm. **[5 points]**

Solution 4.7 Satellites

2023 | Volos, Greece | Theory | Q12 | 40 points

Problem p. 36

A) We need first to evaluate the radius of the geostationary satellite. The radius of the orbit is found through the relation, using the sidereal period $T = 23h\ 56m\ 4s$:

$$T^2 = \frac{4\pi^2 R_S^3}{G M_E} \Rightarrow R_S = 4.2163 \times 10^7\ m = 42163\ km$$

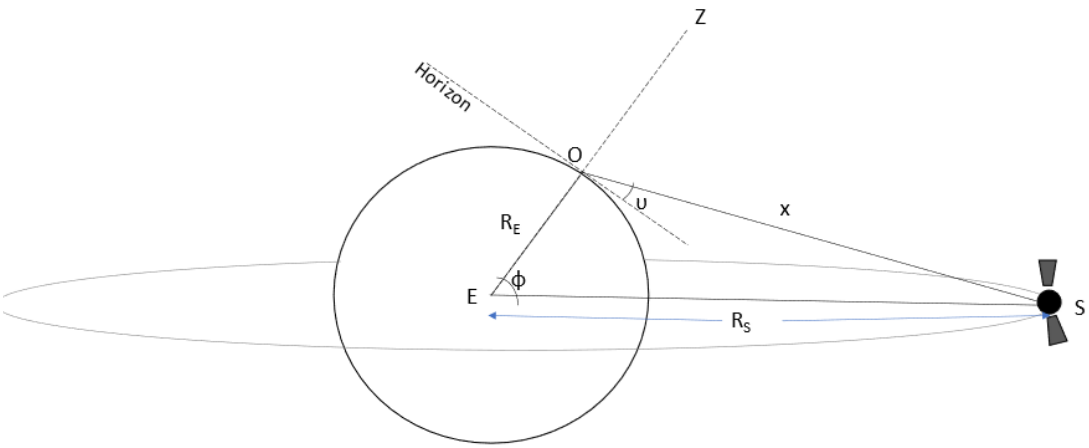


Figure 1.

Naming all the points and lines in Figure 1:

E: center of Earth, S: satellite, O: observer on Earth, Z: observer's zenith

φ: observer's latitude , υ: elevation angle

EO = R_E: Earth's radius, ES = R_S: Satellite's orbital radius, OS = x: observer – satellite distance

From the EOS:

The ends of the range of latitudes the satellite is visible is when υ = 0.

Using the sine law: $\frac{\sin(\widehat{EOS})}{R_S} = \frac{\sin(\widehat{OSE})}{R_E} \Rightarrow \frac{\sin(90°)}{R_S} = \frac{\sin(180°-(90°+\varphi))}{R_E} \Longrightarrow \varphi = \cos^{-1}\left(\frac{R_E}{R_S}\right)$

$$\therefore \varphi \cong 81.3°$$

The maximum latitudes the satellite is visible is 81.3° N and 81.3° S .

B) We need to find the location where angle $O\hat{S}E = 5^o$, while $\upsilon = 0$. This means that the triangle SOE is orthogonal, with angle $E\hat{O}S = 90^o$. From the definition of the $\sin O\hat{S}E = \frac{R_E}{R_S} = \sin 5^o = 0.0872$ we obtain $R_S = 6378/0.0872 = 73179$ km.

C) The total energy of a satellite is given by the following relation:
$E_{tot} = \frac{1}{2}\, m\, v^2 - G\frac{M_E m}{r}$

We obtain the velocity from the centripetal force relation:
$\frac{v^2}{r} = G\frac{M}{r^2}$, therefore the relation for the total energy becomes:

$$E_{tot} = -\frac{1}{2}G\frac{M_E m}{r}.$$

We substitute the physical quantities for the two locations, and we obtain:

$E_{tot,1} = -4.727 \times 10^9$ J, $E_{tot,2} = -2.723 \times 10^9$J, thus energy has increased by 2.00×10^9J.

D) The gravitational acceleration is inversely proportional to the square of the radius: $g = GM/r^2$, thus the ration of the gravitational acceleration is the inverse of the ratio of the squares. Therefore:

$$\frac{g_E}{g_B} = \left(\frac{r_B}{r_E}\right)^2 = \left(\frac{73179000}{6378000}\right)^2 = 131.6.$$

Solution 4.8 Energy and speed in an elliptical orbit

2024 | Kathmandu, Nepal | Theory | T2 | 15 points

Problem p. 37

T2

2. (2 points) $r_p = a(1-e)$ is the perihelion distance. $r_a = a(1+e)$ is the aphelion distance. Use the law of conservation of energy to get:

   (2 points)

$$E = -\frac{GMm}{r_p} + \frac{mv_p^2}{2} = -\frac{GMm}{r_a} + \frac{mv_a^2}{2}$$

   (1 point) From the conservation of angular momentum,

$$mv_a r_a = mv_p r_p$$

$$\implies v_a = v_p \frac{r_p}{r_a}$$

   (3 points) Substitute velocities and rearrange the first equation to get (alternative expression making v_p the subject is also possible and correct):

$$v_a^2 \left(1 - \frac{r_a^2}{r_p^2}\right) = 2GM \left(\frac{1}{r_a} - \frac{1}{r_p}\right)$$

   (3 points) Then, substitute this into the equation for the total mechanical energy:

$$E = \frac{-GMm}{r_a} + \frac{GMmr_p^2}{r_p^2 - r_a^2} \left(\frac{r_p - r_a}{r_a r_p}\right)$$

   Which gives (after simplifying or any equivalent formulation):

$$E = \frac{-GMm}{2a} \left(\frac{2}{1+e} - \frac{1-e}{1+e}\right)$$

   (2 points) Finally giving us:

$$E = -\frac{GMm}{2a}$$

   (2 points) Substitute r_a and r_p in the energy equations to get:

$$\frac{v_a}{v_p} = \frac{1-e}{1+e}$$

   (or use the conservation of angular momentum directly).

The official key displays the reciprocal of the speed ratio requested in the question. Its relation is retained.

Solution 4.9 Altitude of a geostationary satellite

2024 | Kathmandu, Nepal | Theory | T5 | 15 points

Problem p. 37

T5

(3 points) Firstly, calculate the geocentric orbit height by using Newton's law of gravitation and setting T = 23 hours 56 mins 4 secs.

$$\frac{GMm}{d^2} = \frac{mv^2}{d} \text{ and } v = \frac{2\pi d}{T}$$

(1 point) Solve for d to get:

$$d = \left(\frac{GMT^2}{4\pi^2}\right)^{1/3}$$

(1 point) Substitute the correct values to obtain $d = 4.216 \times 10^7$m.

(5 points) Then, construct a triangle, such that from the law of sines we have:

$$\frac{\sin(90+a)}{d} = \frac{\sin(90-(\phi+a))}{r}$$

where r is the radius of the Earth, a is the altitude and ϕ is the latitude of Kathmandu.

(2 points) Simplifying:

$$\cos(a)\frac{r}{d} = \cos(\phi)\cos(a) - \sin(\phi)\sin(a) \implies 1 - \tan(\phi)\tan(a) = \frac{r}{d\cos(\phi)}$$

(2 points) Rearranging and solving for a, you get:

$$a = \tan^{-1}\left(\frac{1 - \dfrac{r}{d\cos(\phi)}}{\tan(\phi)}\right)$$

(1 point)Plugging in the values:

$$a = 57.64\circ$$

The question says prime meridian, while the official solution assumes the satellite is on Kathmandu's local meridian.

Solution 4.10 Free fall inside a gas cloud

2024 | Kathmandu, Nepal | Theory | T9 | 15 points

Problem p. 38

T9

1. (2 points) Notice that the expression for the acceleration due to gravity becomes:

$$a = -\frac{4}{3}G\pi\rho r$$

which can be expressed as:

$$a = -k^2 r, \text{ where: } k = \sqrt{\frac{4}{3}G\pi\rho}$$

(1 point) This is the equation for a harmonic oscillator, as given in the hint. The time period is given by:

$$T = \frac{2\pi}{k}$$

(2 points) which simplifies to:

$$T = \sqrt{\frac{3\pi}{G\rho}}$$

2. (3 points) From the expression of V, the potential difference between r = R and r = 0 is given by $\Delta V_{int} = \dfrac{GM}{2R}$

(2 points) Finally, to escape:

$$\frac{1}{2}mv_{esc}^2 - \frac{4}{3}Gm\rho\pi R^2 - \Delta V_{int} = 0$$

(5 points) Collecting terms and simplifying in terms of v_{esc}:

$$v_{esc} = 2R\sqrt{G\pi\rho}$$

The official escape-energy equation omits a mass factor in its internal-potential term. The published equation and final speed are retained.

Solution 4.11 Speed of Asteroid

2025 | Piatra Neamț, Romania | Theory | Th.2 | 10 points

Problem p. 38

Marking Scheme

a) There are some classical formulas regarding the speed of a celestial object at aphelion and perihelion, respectively (a denotes the semi-major axis of the elliptic orbit, M_S is the mass of the Sun, G is the gravitational constant):

$$v_{max} = \sqrt{\frac{GM_S}{a} \cdot \frac{1+e}{1-e}};\ v_{min} = \sqrt{\frac{GM_S}{a} \cdot \frac{1-e}{1+e}}; \quad \textbf{1.5pt}$$

$$k = \frac{v_{max}}{v_{min}} = \frac{1+e}{1-e};\ k(1-e) = 1+e;$$

$$e = \frac{k-1}{k+1} = \frac{v_{max} - v_{min}}{v_{max} + v_{min}} \quad \textbf{0.5pt}$$

Numerically, the eccentricity of the asteroid's orbit is:

$$e = (19-14)/(19+14) \approx 0.15 \quad \textbf{1pt}$$

b) When being at distance r relative to the Sun, the speed of the asteroid is given by the formula:

$$v(r) = \sqrt{GM_S(\frac{2}{r} - \frac{1}{a})}. \quad \textbf{1pt}$$

When being at one of the co-vertices of the orbit, the asteroid is at the intersection between the ellipse and the semi-minor axis, so r=a. **1pt**
The orbital speed is, in this case:

$$v(a) = \sqrt{\frac{GM_S}{a}} = \sqrt{v_{max} \cdot v_{min}} \approx 16.31\,\text{km/s}. \quad \textbf{1pt}$$

c) Because we are talking about a comet from outer edge of Solar System, we can safely assume its orbit is a parabola. Since the total energy on a parabolic orbit is 0, the speed of the comet is calculated via this formula (d is the Sun-comet distance):

$$v = \sqrt{\frac{2GM_\odot}{d}} \Rightarrow d_{min} = \frac{2GM_\odot}{v_{max}^2} \quad \textbf{1.5pt}$$

$$d_{min} = \frac{2 \times 6.673 \times 10^{-11} \times 1.989 \times 10^{30}}{(9 \times 10^4)^2} \approx 0.22\,\text{au} \quad \textbf{0.5pt}$$

Hence, $a = 3.34\,\text{au}$, $r_{max} = a(1+e) = 3.84\,\text{au}$, this is the distance Sun-asteroid at aphelion. **1.5pt**
Now, the distance comet-asteroid, in the given context, is simply:

$$D = d_{min} + r_{max} = 0.22 + 3.84 = 4.06\,\text{au} \quad \textbf{0.5pt}$$

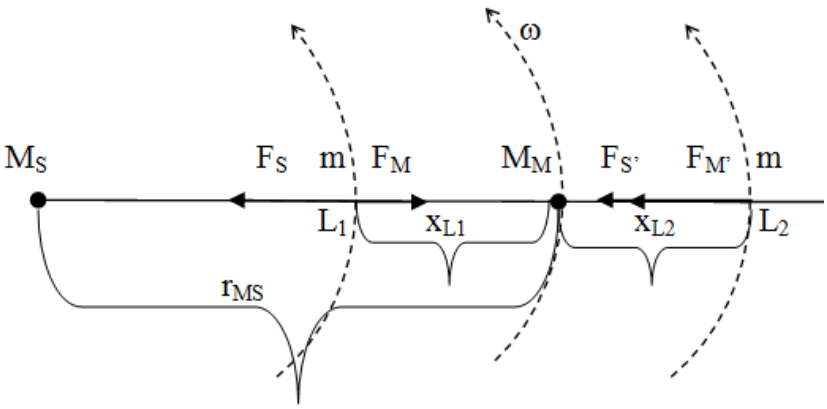
Solution 4.12 Roche Limit, Hill Sphere, Lagrange Points

2025 | Piatra Neamț, Romania | Theory | Th.8 | 40 points

Problem p. 39

Marking Scheme

a) For the Lagrange points L_1 and L_2:



For Mars in orbit around the Sun:

$$F_{cf} = F_{SM}$$

$$M_M \omega^2 r_{MS} = G\frac{M_M M_\odot}{r_{MS}^2} \quad \textbf{2pt}$$

$$\omega = \sqrt{\frac{GM_\odot}{r_{MS}^3}} \quad \textbf{1pt}$$

For a body of mass m at L_1:

$$m\omega^2(r_{MS} - x_{L1}) = G\frac{mM_\odot}{(r_{MS} - x_{L1})^2} - G\frac{mM_M}{x_{L1}^2} \quad \textbf{2pt}$$

$$\frac{M_\odot}{r_{MS}^3}(r_{MS} - x_{L1}) = \frac{M_\odot}{(r_{MS} - x_{L1})^2} - \frac{M_M}{x_{L1}^2}$$

$$\frac{M_\odot}{r_{MS}^3}(r_{MS} - x_{L1})^3 = M_\odot - \frac{M_M(r_{MS} - x_{L1})^2}{x_{L1}^2}$$

$$M_\odot\left(1 - \frac{x_{L1}}{r_{MS}}\right)^3 = M_\odot - M_M\left(\frac{r_{MS}}{x_{L1}} - 1\right)^2$$

Now, $x_{L1} << r_{MS}$, which means that $\frac{x_{L1}}{r_{MS}} << 1$ and $\frac{r_{MS}}{x_{L1}} >> 1$. Thus we can use the following approximations:

$$\left(1 - \frac{x_{L1}}{r_{MS}}\right)^3 \approx 1 - 3\frac{x_{L1}}{r_{MS}} \quad \textbf{2pt}$$

and

$$\frac{r_{MS}}{x_{L1}} - 1 \approx \frac{r_{MS}}{x_{L1}} \quad \textbf{1pt}$$

$$M_\odot\left(1 - 3\frac{x_{L1}}{r_{MS}}\right) = M_\odot - M_M\left(\frac{r_{MS}}{x_{L1}}\right)^2$$

$$3M_\odot\frac{x_{L1}}{r_{MS}} = M_M\left(\frac{r_{MS}}{x_{L1}}\right)^2$$

$$x_{L1}^3 = \frac{M_M}{3M_\odot}r_{MS}^3$$

$$x_{L1} = r_{MS}\sqrt[3]{\frac{M_M}{3M_\odot}} \quad \textbf{2pt}$$

Numerical value for *Sun-Mars* system:

$$x_{L1} = 1.524 \times 1.496 \times 10^{11} \times \sqrt[3]{\frac{0.107M_\oplus}{3 \times 333054M_\oplus}}$$

$$x_{L1} = 1.08 \times 10^9 m = 1.08 \times 10^6 km \quad \textbf{2pt}$$

Comparing this with Phobos's orbit, $d_{PhobosM} = 9380\ km$, we obtain: $\frac{d_{PhobosM}}{x_{L1}} \approx 9 \times 10^{-3}$. **1pt**

b) For a body of mass m at L_2:

$$m\omega^2(r_{MS} + x_{L2}) = G\frac{mM_\odot}{(r_{MS} + x_{L2})^2} + G\frac{mM_M}{x_{L2}^2} \quad \textbf{2pt}$$

$$\frac{M_\odot}{r_{MS}^3}(r_{MS} + x_{L2}) = \frac{M_\odot}{(r_{MS} + x_{L2})^2} + \frac{M_M}{x_{L2}^2}$$

$$\frac{M_\odot}{r_{MS}^3}(r_{MS}+x_{L2})^3 = M_\odot + \frac{M_M(r_{MS}+x_{L2})^2}{x_{L2}^2}$$

$$M_\odot\left(1+\frac{x_{L2}}{r_{MS}}\right)^3 = M_\odot + M_M\left(\frac{r_{MS}}{x_{L2}}+1\right)^2$$

Now, $x_{L2} << r_{MS}$, which means that $\frac{x_{L2}}{r_{MS}} << 1$ and $\frac{r_{MS}}{x_{L2}} >> 1$. Thus we can use the following approximations:

$$\left(1+\frac{x_{L2}}{r_{MS}}\right)^3 \approx 1+3\frac{x_{L2}}{r_{MS}} \quad \textbf{2pt}$$

and

$$\frac{r_{MS}}{x_{L2}}+1 \approx \frac{r_{MS}}{x_{L2}} \quad \textbf{1pt}$$

$$M_\odot\left(1+3\frac{x_{L2}}{r_{MS}}\right) = M_\odot + M_M\left(\frac{r_{MS}}{x_{L2}}\right)^2$$

$$3M_\odot\frac{x_{L2}}{r_{MS}} = M_M\left(\frac{r_{MS}}{x_{L2}}\right)^2$$

$$x_{L2}^3 = \frac{M_M}{3M_\odot}r_{MS}^3$$

$$x_{L2} = r_{MS}\sqrt[3]{\frac{M_M}{3M_\odot}} \quad \textbf{2pt}$$

The position of the L_2 point for the Mars-Phobos system will be at a distance equal to:

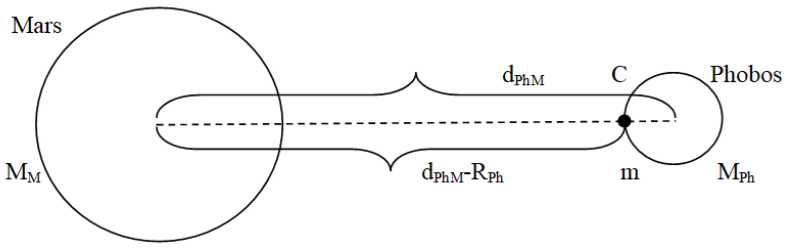
$$x_{L2} = d_{PhobosM}\sqrt[3]{\frac{M_{Phobos}}{3M_M}}$$

$$x_{L2} = 9380 \times \sqrt[3]{\frac{1.5\times10^{-8}M_M}{3M_M}}$$

$$x_{L2} \approx 16km \quad \textbf{2pt}$$

The L_2 point is close to Phobos's surface $R_{Phobos} = 12.5km$.

c) For the Roche limit:



We calculate the tidal force experienced by Phobos due to Mars's gravitational influence. That is, the difference in gravitational attractions exerted by Mars on a material point of mass m, located on the surface of the satellite, and the same mass m located at the center of Phobos:

$$F_{tidal} = G\frac{mM_M}{(d_{PhobosM}-R_{Phobos})^2} - G\frac{mM_M}{d_{PhobosM}^2} \quad \textbf{4pt}$$

$$F_{tidal} = G\frac{mM_M}{d_{PhobosM}^2\left(1-\frac{R_{Phobos}}{d_{PhobosM}}\right)^2} - G\frac{mM_M}{d_{PhobosM}^2}$$

We take into account the relationship $R_{Phobos} << d_{PhobosM}$, which means $\frac{R_{Phobos}}{d_{PhobosM}} << 1$ and we can use the approximation given in the problem:

$$\left(1 - \frac{R_{Phobos}}{d_{PhobosM}}\right)^{-2} \approx 1 + 2\frac{R_{Phobos}}{d_{PhobosM}} \quad \textbf{2pt}$$

$$F_{tidal} = G\frac{mM_M}{d^2_{PhobosM}}\left(1 + 2\frac{R_{Phobos}}{d_{PhobosM}} - 1\right)$$

$$F_{tidal} = 2G\frac{mM_M R_{Phobos}}{d^3_{PhobosM}} \quad \textbf{1pt}$$

The gravitational attraction felt by the point of mass m due to Phobos's influence is:

$$F_{grav} = G\frac{mM_{Phobos}}{R^2_{Phobos}} \quad \textbf{1pt}$$

The Roche limit corresponds to the equality of the two forces:

$$F_{tidal} = F_{grav} \quad \textbf{1pt}$$

$$G\frac{mM_{Phobos}}{R^2_{Phobos}} = 2G\frac{mM_M R_{Phobos}}{d^3_{PhobosM_Roche}}$$

$$d^3_{PhobosM_Roche} = 2\frac{M_M}{M_{Phobos}}R^3_{Phobos}$$

$$d_{PhobosM_Roche} = R_{Phobos}\sqrt[3]{2\frac{M_M}{M_{Phobos}}} \quad \textbf{2pt}$$

$$d_{PhobosM_Roche} = 12.5\sqrt[3]{2\frac{M_M}{1.5 \times 10^{-8} M_M}}$$

$$d_{PhobosM_Roche} = 6386km \quad \textbf{2pt}$$

To calculate the time it would take Phobos to reach that radius around Mars, we need to compute the difference in radii:

$$\Delta d = d_{PhobosM} - d_{PhobosM_Roche} = 9380 - 6386 = 2994km \quad \textbf{1pt}$$

$$\Delta t = \frac{\Delta d}{v} = \frac{2994000}{1.8 \times 10^{-2}}\ years = 1.67 \times 10^8\ years \quad \textbf{1pt}$$

Alternative solution, which will receive the maximum number of points (15 points): Students might calculate the Roche limit by modeling the satellite as two spheres of mass $\frac{M_{Phobos}}{2}$, with density equal to Phobos's density. **3pt**

The radius r of such a sphere will be:

$$2 \times \frac{4\pi r^3}{3} = \frac{4\pi R^3_{Phobos}}{3}$$

$$r = \frac{R_{Phobos}}{\sqrt[3]{2}} \quad \textbf{1pt}$$

The gravitational force between the two bodies will be:

$$F_{grav} = G\frac{M^2_{Phobos}}{4(2r)^2} \quad \textbf{2pt}$$

The difference in the gravitational force due to Mars will be:

$$\Delta F = G\frac{M_M M_{Phobos}}{2(d_{PhobosM} - r)^2} - G\frac{M_M M_{Phobos}}{2(d_{PhobosM} + r)^2} \quad \textbf{2pt}$$

$$\Delta F = G\frac{M_M M_{Phobos}}{2(d_{PhobosM})^2}\left(\frac{1}{\left(1-\frac{r}{d_{PhobosM}}\right)^2}-\frac{1}{\left(1+\frac{r}{d_{PhobosM}}\right)^2}\right)$$

Using the approximation given in the problem:

$$\Delta F \approx G\frac{M_M M_{Phobos}}{2(d_{PhobosM})^2}\left(1+\frac{2r}{d_{PhobosM}}-1+\frac{2r}{d_{PhobosM}}\right)$$

$$\Delta F \approx G\frac{M_M M_{Phobos}}{2(d_{PhobosM})^2}\times\frac{4r}{d_{PhobosM}}$$

$$\Delta F \approx G\frac{M_M M_{Phobos}\times 2r}{(d_{PhobosM})^3} \quad \textbf{2pt}$$

The Roche limit corresponds to the equality of these two forces:

$$G\frac{M_M M_{Phobos}\times 2r}{(d_{PhobosM_Roche})^3} = G\frac{M_{Phobos}^2}{4(2r)^2} \quad \textbf{2pt}$$

$$\frac{32r^3}{(d_{PhobosM\ \ Roche})^3} = \frac{M_{Phobos}}{M_M}$$

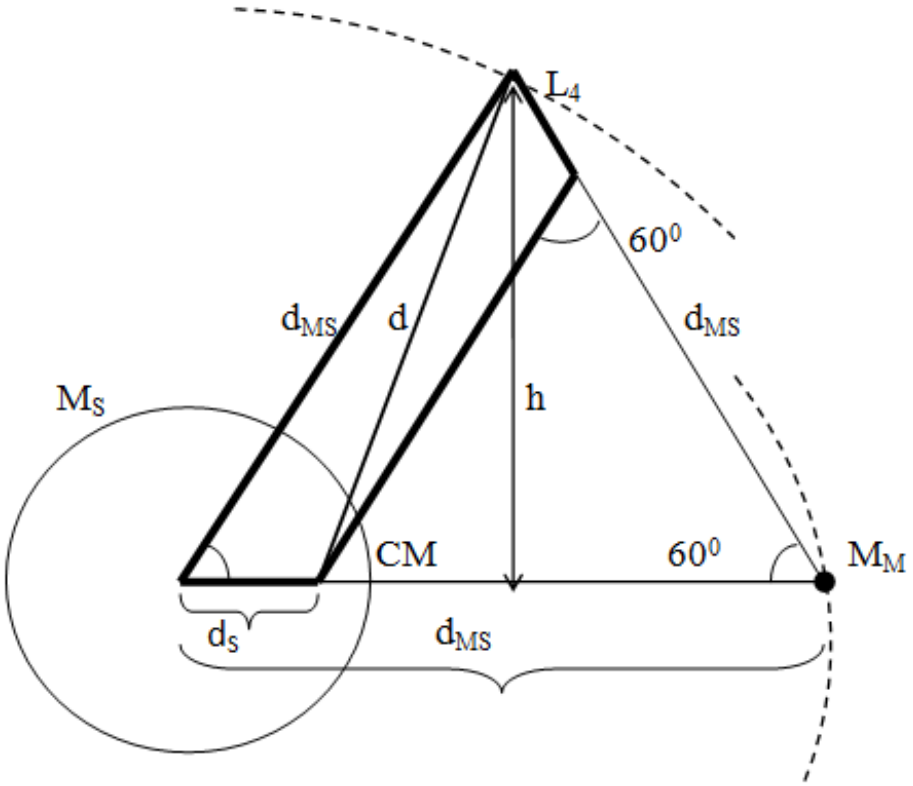
$$\frac{32R_{Phobos}^3}{2(d_{PhobosM\ \ Roche})^3} = \frac{M_{Phobos}}{M_M}$$

$$d_{PhobosM\ \ Roche} = 2R_{Phobos}\sqrt[3]{2\frac{M_M}{M_{Phobos}}} \quad \textbf{1pt}$$

$$d_{PhobosM\ \ Roche} = 12772km \quad \textbf{1pt}$$

This means that Phobos is already closer to Mars than the Roche limit. **1pt**

d) Determining the CM of the Mars-Sun system: Let d_S be the distance from the center of the Sun to the center of mass CM, and d the distance from the CM to the Lagrange point L_4.



Distance l calculation between Lagrange points L_4 and L_5:

$$h^2 = d_{MS}^2 - \frac{d_{MS}^2}{4} \Rightarrow h = 1.32AU$$

$$l = 2h = 2.64AU$$

3pt

5. Light and Telescopes

Theory problems

Problem 5.1 The Stars Observed with the Telescope

2022 | Câmpulung Moldovenesc, Romania | Theory | Q1

Solution p. 71

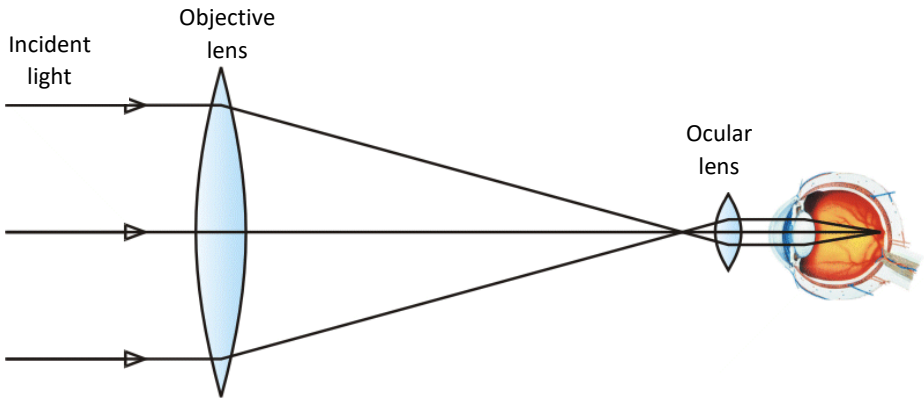
I. Multiple choice problems – 100 points

For these problems, you have to choose only one correct answer. Please write your answer on answer-sheet. **Don't forget to write the number of the problem, i.e Grid 1. (c).**

Grid 1. The stars observed with the telescope

The diameter of the objective lens of a telescope, represented in the drawing in figure 1, is $D_{\text{objective telescope}} = 300$ mm, and the diameter of the pupil of the observer's eye is $D_{\text{pupil eye}} = 6\,\text{mm}$.

The light gathering power, coming from a star, due to this telescope, is:



a) $g_c = 2500;$ b) $g_c = 1500;$ c) $g_c = 1000;$ d) $g_c = 3000.$

Problem 5.2 The Terrestrial Satellite

2022 | Câmpulung Moldovenesc, Romania | Theory | Q4

Solution p. 72

I. Multiple choice problems – 100 points

For these problems, you have to choose only one correct answer. Please write your answer on answer-sheet. **Don't forget to write the number of the problem, i.e Grid 1. (c).**

Grid 4. The Terrestrial Satellite

A small satellite (a luminous sphere) that orbits the Earth, in an equatorial circular orbit at the altitude h, is photographed with a camera on the ground, the lens of which is a converging lens with focal length f. The photography is made when the satellite aligns with the camera and the centre of the Earth.

It is given: the mass of the Earth, M; the gravitational constant, G, the radius of the Earth, R. During the photographic exposure the proper rotation of the Earth is neglected.

The length of the image on the photograph, after developing the photographic plate, placed in the focal plane of the objective lens, as indicated by the drawing in figure 1, if the exposure time of the camera was Δt, is given by the expression:

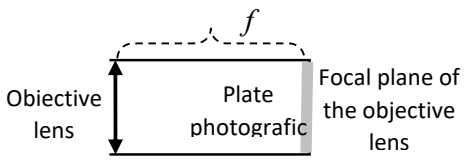


Fig. 1

a) $l = \frac{f}{h-f} \cdot \sqrt{G\frac{M}{R+h}} \cdot \Delta t$ b) $l = \frac{f}{h+f} \cdot \sqrt{G\frac{M}{R+h}} \cdot \Delta t$ c) $l = \frac{f}{h-f} \cdot \sqrt{G\frac{M}{h-R}} \cdot \Delta t$ d) $l = \frac{f}{h-f} \cdot \sqrt{2G\frac{M}{R+h}} \cdot \Delta t$

Problem 5.3 Telescope

2023 | Volos, Greece | Theory | Q8 | 10 points

Solution p. 74

Question 8: Telescope.

The focal length of a telescope is 900 mm and its aperture is 4 inches. We use an eyepiece whose focal length is 25 mm. What is the magnification m of this telescope and what is its focal ratio f?

1 inch is 2.54 cm.

Problem 5.4 Telescope limiting magnitude and interstellar extinction

2024 | Kathmandu, Nepal | Theory | T1 | 15 points

Solution p. 74

T1 (15 marks)

A student used a telescope of diameter (D) 100cm, to observe a distant star whose absolute magnitude is -0.5. If we consider the limiting magnitude of the eye as 6 and the diameter of the pupil (d) to be 7mm, then answer the following questions:

1. If no interstellar medium (ISM) is present, find the maximum distance to which student can detect this star.
2. In reality, an ISM of extinction factor 0.05 mag/Kpc is present between the star and the Earth. Find the maximum distance to which the student can detect the star.

Note: When an equation cannot be solved analytically, we use iterative methods to find an approximate numerical solution. As an example, suppose you want to find out the value of $x = x_{fin}$, for which $x_{fin} = f(x_{fin})$, we start with some value $x = x_0$ and proceed as follows:

- Step 1: Find $f(x_0)$ and call it x_1.
- Step 2: Find $f(x_1)$ and call it x_2.
- Step 3: Find $f(x_2)$ and call it x_3.

Keep repeating this iteration process till x_n is almost same as x_{n-1}. That is your desired x_{fin} value.

Problem 5.5 Exploring NEAs

2025 | Piatra Neamț, Romania | Theory | Th.4 | 10 points

Solution p. 75

Th.4 Exploring NEAs (10 Points)

Near-Earth Asteroids (NEAs) are small rocky bodies orbiting the Sun in the vicinity of Earth's orbit. A science mission designed to study a NEA target is operating a spacecraft that will make its closest approach to the target at a distance of $\Delta_c = 100\,\text{m}$. One of the instruments aboard the spacecraft is a CCD camera called the Asteroid Framer (AF), which can acquire images in the V-band at an effective wavelength of $\lambda = 550\,\text{nm}$. After the mission launched, the operations team conducted initial tests by acquiring images of the Earth using the AF camera with an aperture diameter of $D = 10\,\text{mm}$. At that time, the spacecraft was at a distance of $d = 1.5 \times 10^6\,\text{km}$ from Earth. The Earth was observed on the images within a photometric aperture of $\alpha_\oplus = 60$ pixels in diameter.

a) Compute the pixel scale in arcseconds per pixel.

b) Estimate the linear diameter of the smallest morphological feature that can be observed on the surface of the asteroid at the closest approach.

The saturation test on initial images showed Earth was just over-exposed at 10 s exposure time and just under-exposed at 0.5 s (under-exposed means signal-to-noise ratio < 5). Using this data, the operations team can determine the exposure time interval allowed to observe the NEA target, which has the expected albedo of $A = 0.10$. Ignore phase angle.

Note: Assume that the detected signal is proportional to the object's surface brightness and the exposure time, and that the surface brightness is proportional to the albedo divided by the square of its distance from the Sun.

c) Determine the corresponding exposure time interval for which the planned observations of the NEA target at the closest approach will be successful without any under–exposure or over–exposure. At that time, the target will be located at $r_{NEA} = 1.20\,\text{au}$ from the Sun.

Solutions to Light and Telescopes

Solution 5.1 The Stars Observed with the Telescope

2022 | Câmpulung Moldovenesc, Romania | Theory | Q1

Problem p. 67

Solution

The degree of collection of light from a star, g_c, by a telescop, used as shown in the drawing in Figure 1, is defined as the ratio of the area of the circle's surface, whose diameter is equal to the diameter of the lens of the telescope, and the area of the circle, whose diameter is equal to the diameter of the pupil of the observer's eye:

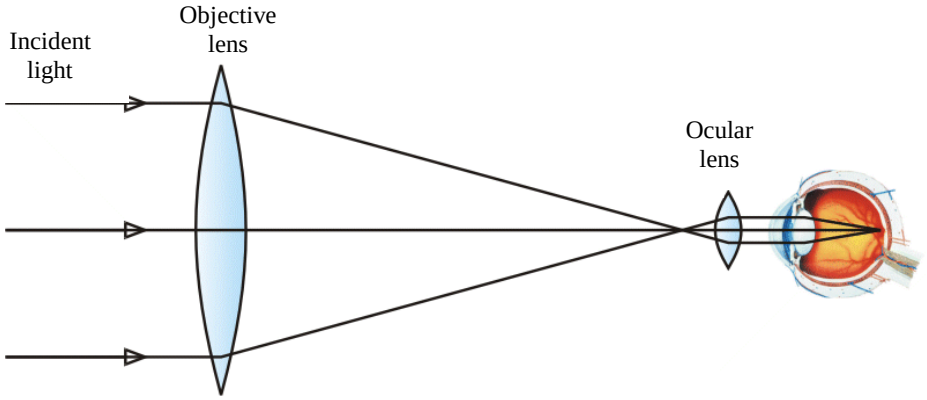


Fig. 1

$$g_c = \frac{\frac{\pi}{4} D_{\text{objectivtelescope}}^2}{\frac{\pi}{4} D_{\text{pupileye}}^2} = \left(\frac{D_{\text{objectivtelescope}}}{D_{\text{pupileye}}} \right)^2 ; \; D_{\text{objectivtelescope}} > D_{\text{pupileye}}; \; g_c > 1.$$

The official S1 solution stops after the light-gathering formula without substituting the values or explicitly giving an answer letter.

Solution 5.2 The Terrestrial Satellite

2022 | Câmpulung Moldovenesc, Romania | Theory | Q4

Problem p. 67

Solution

Sector AB, in the circular orbit of the satellite, represented in the drawing in Figure 2, traveled by the satellite during the time Δt, equal to the exposure time of the camera, has the length:

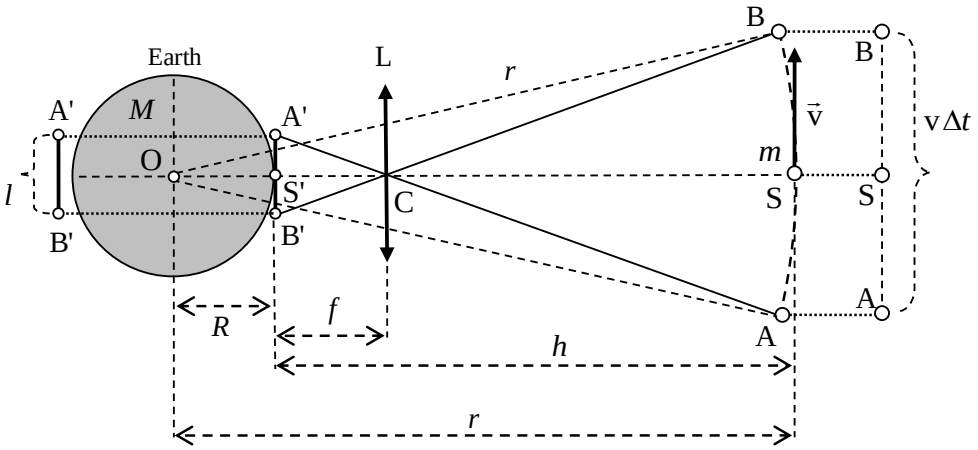


Fig. 1

$$AB = v_{rel} \cdot \Delta t,$$

where v– the speed of the satellite in its circular orbit, relative to the center of the Earth;

$$K\frac{mM}{r^2}=\frac{m\mathrm{v}^2}{r};$$

$$\mathrm{v}=\sqrt{K\frac{M}{r}}=\sqrt{K\frac{M}{R+h}}.$$

Because the distance between the satellite and the camera lens is very large, the image will be formed in the focal plane of the camera lens.

Since the circle sector AB is very short and has a very large radius ($r = R + h$), it can be assimilated with the line segment AB, so that:

$$\Delta(\mathrm{ACS})\sim\Delta(\mathrm{A'CS'});$$

$$\frac{\mathrm{A'S'}}{\mathrm{AS}}=\frac{\mathrm{CS'}}{\mathrm{CS}};\ \frac{\frac{l}{2}}{\frac{\mathrm{v}\cdot\Delta t}{2}}=\frac{f}{h-f},$$

where l – the length of the image on the photo;

$$l=\frac{f}{h-f}\cdot\mathrm{v}\cdot\Delta t;$$

$$l=\frac{f}{h-f}\cdot\sqrt{K\frac{M}{R+h}}\cdot\Delta t.$$

Solution 5.3 Telescope

2023 | Volos, Greece | Theory | Q8 | 10 points

Problem p. 68

The magnification is the ratio of the focal length of the telescope over the focal length of the eyepiece, thus: $m = \frac{900}{25} = 36$. **[5 points]**

The focal ratio f is the ratio of the focal length to the aperture, thus $\frac{900}{40\times2.54} = 8.9$. **[5 points]**

Solution 5.4 Telescope limiting magnitude and interstellar extinction

2024 | Kathmandu, Nepal | Theory | T1 | 15 points

Problem p. 69

T1

1. (a) (1 point) We know $m_t - m_e = -2.5log\left(\frac{F_t}{F_e}\right)$

   (1 point) since,$F_{lim} \propto \frac{1}{D^2}$ so $m_t - m_e = 5log\,(1000/7)$

   $m_t = 10.8 + 6$

   (1 point) $m_t = 16.8$

   Now we know $m = M + 5log\left(\frac{r}{10pc}\right)$

   $\frac{r}{10} = 10^{3.454}$

   (2 points) So $r = 29Kpc$ is the maximum distance till which student can detect the star using telescope.

   (b) (2 points) $m' = m + A$,thus $m' = 16.8 + \alpha r$

   (1 point) Now,$m' = M + 5log\left(\frac{r}{10pc}\right)$ $16.8 + 0.00005 * r = -0.5 + 5log(r) - 5$

   (2 points) $r = 10^{4.46-0.00001r}$
   This equation can't be solve analytically.

   (2 points) So,if we do the iteration for this, taking initial value from previous result of 29Kpc, which is good approximation.

   (3 points) Then, we get r=18.73KPc.

The official extinction solution has a sign inconsistency between its printed intermediate magnitude equation and subsequent iteration formula. Both expressions are retained.

Solution 5.5 Exploring NEAs

2025 | Piatra Neamț, Romania | Theory | Th.4 | 10 points

Problem p. 70

Marking Scheme

a) The pixel scale can be calculated as the ratio between the angular dimension and the corresponding number of pixels within which it is observed,

$$s = \frac{2R_\oplus/d}{\alpha_\oplus} \qquad \textbf{1pt}$$

Numerically,

$$s = \frac{2 \times 6.378 \times 10^3}{1.50 \times 10^6} \times \frac{180 \times 3600}{\pi} \times \frac{1}{60} = 29\,\text{arcsec/pix} \qquad \textbf{1pt}$$

b) First, we apply the Rayleigh criterion to quantify the diffraction effects,

$$\theta = 1.22\frac{\lambda}{D} = \frac{1.22 \times 550 \times 10^{-9}\ \text{m}}{0.01\ \text{m}} \times \frac{180 \times 3600}{\pi}$$
$$= 13.8\,\text{arcsec} \qquad \textbf{0.5pt}$$

The minimum angular separation calculated above is less than the angular resolution of the camera:

$$\theta < s \times (1\ \text{pix}) \qquad \textbf{0.5pt}$$

Thus, the minimum size of a detectable feature should be comparable to the linear projection of a single pixel:

$$l = s \times (1\ \text{pix}) \times \Delta_c \qquad \textbf{1pt}$$

Note for graders: Solutions using 2 or 3 pixels for the estimations will receive full scores.
Numerically,

$$l = 29\ \text{arcsec} \times \frac{\pi}{180 \times 3600} \times 100\ \text{m} = 1.4\ \text{cm} \qquad \textbf{1pt}$$

c) The detected signal per pixel is proportional to the product of the surface brightness of the object, measured in W/(m^2·sr), and the exposure time:

$$S \propto Bt \qquad \textbf{1pt}$$

The surface brightness is due to the sunlight reflected by the object, so it is proportional to the albedo and the inverse of the squared heliocentric distance:

$$B \propto \frac{A}{r^2} \qquad \textbf{1pt}$$

Comparing the surface brightnesses of Earth and the NEA target, one gets:

$$\frac{B_{\mathrm{NEA}}}{B_{\oplus}} = \frac{A_{\mathrm{NEA}} r_{\oplus}^2}{A_{\oplus} r_{\mathrm{NEA}}^2} \qquad \textbf{1pt}$$

Remark. The surface brightness is independent of the distance between the sensor and the extended target. Both Earth and the NEA target are extended objects (not point-like sources).

Both objects should be observed under the same conditions (appropriate exposure times), so one can consider similar signal levels recorded for both objects. Thus, considering the first equation:

$$t_{\mathrm{NEA}} = t_{\mathrm{Earth}} \times \frac{A_{\oplus} r_{\mathrm{NEA}}^2}{A_{\mathrm{NEA}} r_{\oplus}^2} \qquad \textbf{1pt}$$

Numerically,

$$t_{\mathrm{NEA}} = t_{\mathrm{Earth}} \times \frac{0.30}{0.10} \times \left(\frac{1.20}{1.00}\right)^2 = 4.32\, t_{\mathrm{Earth}} \qquad \textbf{0.5pt}$$

The exposure time interval will be:

$$2.16 \text{ s} - 43.2 \text{ s} \qquad \textbf{0.5pt}$$

6. The Sun and Stars

Theory problems

Problem 6.1 The Moon, Seen on the Way to the Moon

2022 | Câmpulung Moldovenesc, Romania | Theory | Q9

Solution p. 87

I. Multiple choice problems – 100 points

For these problems, you have to choose only one correct answer. Please write your answer on answer-sheet. **Don't forget to write the number of the problem, i.e Grid 1. (c).**

Grid 9. The Moon, seen on the way to the Moon

The full Moon, whose linear diameter is $d = 3\,436$ km, at Earth in the distance $r_0 = 384\,000\ km$, has the apparent magnitude $m_0 = -12.7$. The apparent magnitude of the Sun, seen from Earth, is $m_s = -26.84$. The distance, r, from the full Moon, where a cosmonaut is, on the way of his spaceship to the Moon, the magnitude of the full Moon should be 50% of the magnitude of the Sun as seen from Earth, is:

a) $r \approx 198\,243\ km\ ss$ b) $r \approx 320\,732\ km$ c) $r \approx 132\,521\ km$ d) $r \approx 275\,633\ km$

Problem 6.2 The Luminosity of a Star

2022 | Câmpulung Moldovenesc, Romania | Theory | Q10

Solution p. 89

I. Multiple choice problems – 100 points

For these problems, you have to choose only one correct answer. Please write your answer on answer-sheet. **Don't forget to write the number of the problem, i.e Grid 1. (c).**

Grid 10. The luminosity of a star

The luminosity of a star Σ, is $L_\Sigma = 100 \cdot L_S$, where L_S is the luminosity of the Sun, and the star's surface temperature is $T_\Sigma = \frac{1}{2} \cdot T_S$, where T_S is the Sun's surface temperature.

Ratio between the radius of the star and the radius of the Sun is:

a) $\frac{R_\Sigma}{R_S} = 40;$ b) $\frac{R_\Sigma}{R_S} = 30;$ c) $\frac{R_\Sigma}{R_S} = 60;$ d) $\frac{R_\Sigma}{R_S} = 50.$

Problem 6.3 Unresolved binary star

2023 | Volos, Greece | Theory | Q3 | 10 points

Solution p. 90

Question 3: Unresolved binary star

An unresolved binary star consists of two members whose apparent magnitudes are m_1=2.0 mag and m_2=4.5 mag. What is the combined apparent magnitude of this binary system?



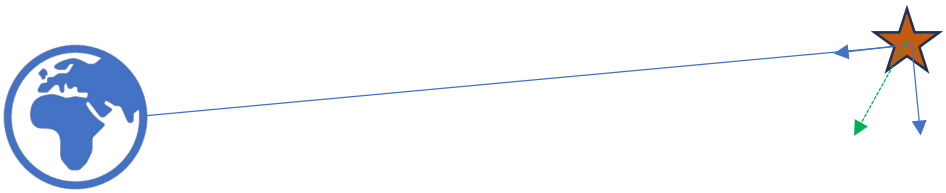
Problem 6.4 Parallax and proper motion

2023 | Volos, Greece | Theory | Q7 | 10 points

Solution p. 90

Question 7: Parallax and proper motion

The parallax of Barnard's star is 0.552 arcseconds and its proper motion is μ=10.31arcseconds/year. From its spectrum, an observer noticed that the absorption line H_α is seen at 656.044 nm instead of 656.281 nm. What is the total speed of the Barnard's star relative to this observer?



Speed of light in vacuum	c	=	2.998×10^{8} m s^{-1}
1 parsec	1 pc	=	3.086×10^{16} m
		=	206 265 au
		=	3.262 ly
1 tropical year		=	365.2422 solar days

Problem 6.5 Sirius A and B

2023 | Volos, Greece | Theory | Q11 | 40 points

Solution p. 91

Question 11: SIRIUS A and B

Sirius is the brightest star of the night sky during the winter of the northern hemisphere and the summer of the southern hemisphere. It has an apparent visual magnitude m_V=-1.47 mag. Sirius B is the binary companion of Sirius A and its magnitude is m_V=8.5 mag. Their orbital period is 50 years and the parallax of the system is 350 milliarcseconds, while their angular separation is 7 arcseconds.

You can assume that Sirius A has a surface temperature of T=10,000 K. The temperature of Sirius B is estimated to be 25,000 K. Both stars have a bolometric correction equal to -0.4. An absolute bolometric magnitude of zero corresponds to a luminosity of 3×10^{28} W. The orbit is assumed to be circular, and the line of sight is perpendicular to the orbital plane of the system.

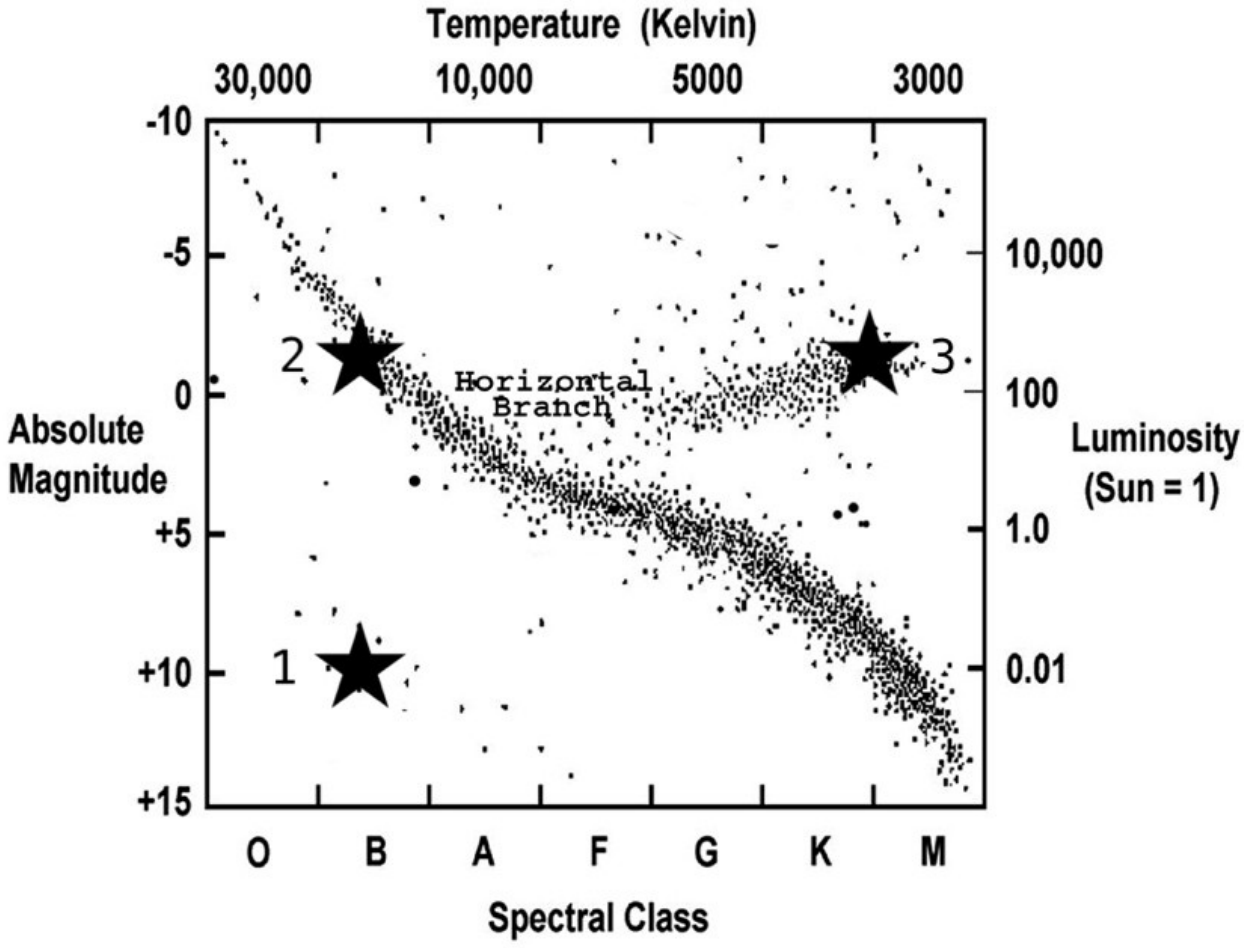
A) Calculate the luminosity and radius of Sirius A and B. **[12 Points]**

B) What is the orbital separation of the system in meters? **[8 Points]**

C) What is the total mass of the system? **[8 Points]**

D) Assuming that the mass of Sirius A is twice the mass of Sirius B, what is the mean density of Sirius B? **[6 Points]**

E) In the H-R diagram shown in question 10, which star number (1 / 2 / 3) will be the closest to the star Sirius B? **[6 Points]**



Stefan-Boltzmann constant	σ	=	5.670×10^{-8} W m^{-2} K^{-4}

1 parsec	1 pc	=	3.086×10^{16} m
		=	206 265 au
		=	3.262 ly
1 astronomical unit (au)	1 au	=	1.496×10^{11} m

Object	Mean radius (km)	Mass (kg)	Orbital Radius (au)
Sun	695 500	1.988×10^{30}	---

Problem 6.6 A Cepheid period-luminosity relation

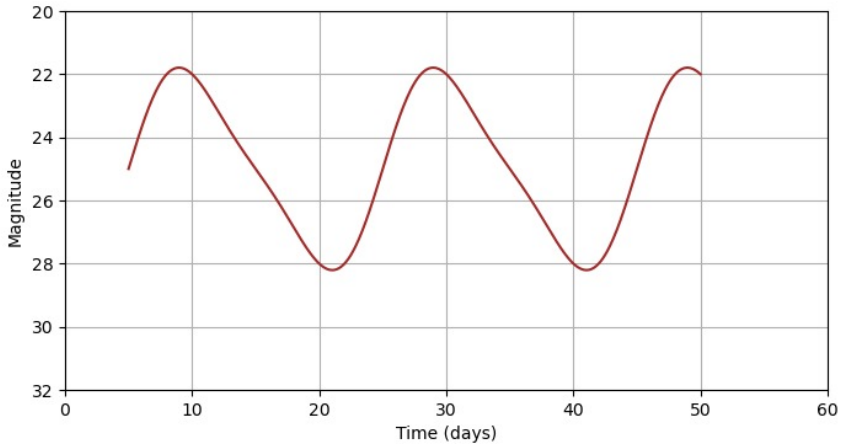
2024 | Kathmandu, Nepal | Theory | T4 | 15 points

Solution p. 92

T4 (15 marks)

In Cosmology, Cepheid variables are called standard candles as they can be used to estimate distances. Edwin Hubble used these standard candles to settle the Shapley-Curtis debate by measuring the distance to the Andromeda Galaxy. Cepheid variables are variable stars whose brightness vary periodically. Henrietta Leavitt discovered the relation between the periodicity and average luminosity of these stars, and this relation can be used to derive the equation linking Absolute magnitude and Period. For Cepheid variables, Luminosity(L) varies with Period(P) as $L \propto P^{2.5}$, where P is in days.

1. Find the relation between Absolute Magnitude (M) and P for this type of variable stars.
2. You might notice that it is a linear equation between M and a function of P. If the unknown constatnt of the given equation is -1.5 mag, then find the distance to the given Cepheid Variable.



Problem 6.7 A spectroscopic binary system

2024 | Kathmandu, Nepal | Theory | T10 | 15 points

Solution p. 93

T10 (15 points)

Spectroscopic binary star systems are those systems in which individual stars of the system can't be resolved visually due to their low angular separation. But we can detect that the system is binary due to shifts in their spectral line caused by Doppler shift. In double lined spectroscopic binaries, we can detect two spectra, one for each star. From the change in wavelength of the spectrum over a period of time, we can find the radial velocity curve.

Suppose an astronomer observed a hypothetical spectroscopic binary system having a circular orbit as shown in Figure 1. The plane of the orbit makes an angle θ w.r.t the plane of the sky. Star 1 (mass m_1) is located on the outer orbit and Star 2 (mass m_2) is located in the inner orbit. The observed radial velocity graph is shown in Figure 2.

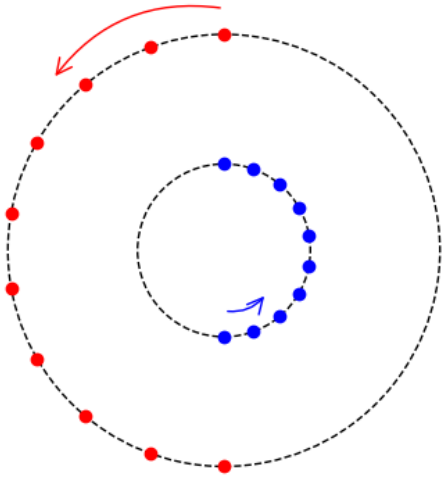


Figure 1. A top-view of the binary system.

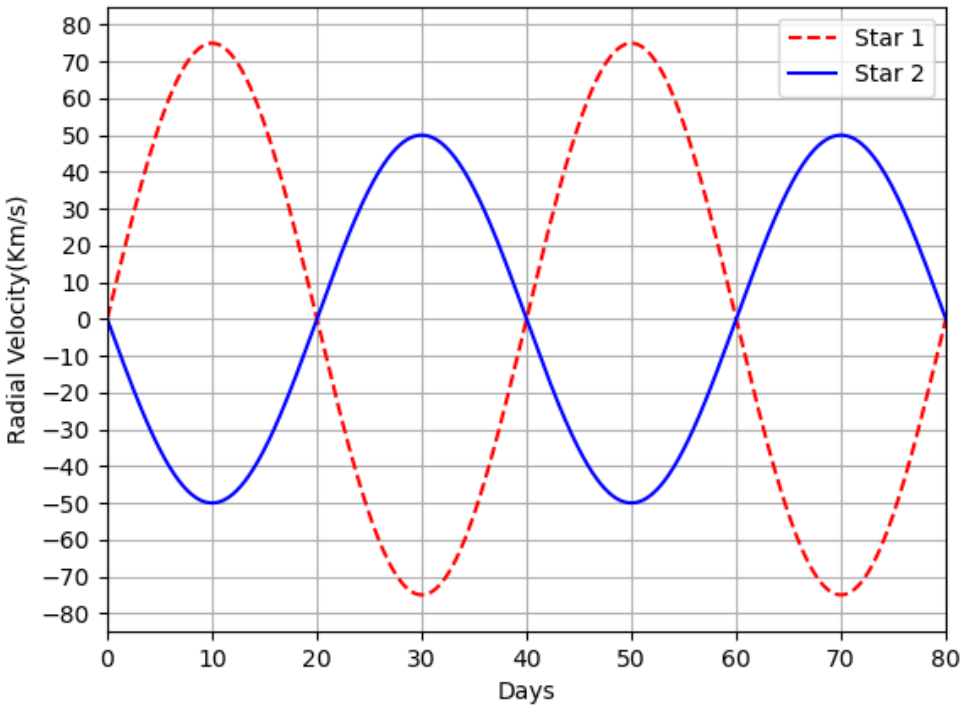


Figure 2. Radial Velocity Plots for the Binary Stars

1. The binary mass function is an expression which depends upon only the observable quantities such as radial velocity and period. Find the binary mass function for both star 1 and star 2.
2. Use your expression in (1) to estimate the masses for the stars. Is your estimate exact, a lower-limit or an upper-limit?

Universal Gravitational constant	$G = 6.674 \times 10^{-11}\ \mathrm{N\ m^2\ kg^{-2}}$

Object	Mean radius (km)	Mass (kg)	Semi-major axis (AU)	Eccentricity	Albedo
Sun	$695\,500$	$1.988\ \times\ 10^{30}$	---	---	---

Problem 6.8 A radiating interstellar dust cloud

2024 | Kathmandu, Nepal | Theory | T12 | 15 points

Solution p. 95

T12 (15 marks)

Interstellar dust is heated up by hot ionizing radiation from young blue stars. They then re-emit the UV light. A dust cloud in a dusty galaxy was recently studied by a researcher. Make the simplifying approximation that you can treat the dust cloud as a blackbody. Help him to answer the following questions:

1. Find the wavelength where the dust emission peaks (λ_{max}) if it has a temperature of $T = 50\ \mathrm{K}$.
2. The researcher determines that the dust has a total luminosity (L_{dust}) = $1.6 \times 10^{11} L_{\odot}$. Estimate the size of the dust cloud, assuming a spherical distribution.
3. If the cosmological redshift of the galaxy (z) is 0.01, estimate the angular size (θ) of the dust cloud.
4. What aperture telescope is minimally needed if we want to resolve the dust cloud?

Speed of light in a vacuum	$\mathrm{c} = 2.998 \times 10^{8}\ \mathrm{m\ s^{-1}}$
Hubble constant	$H_0 = 70\ \mathrm{km\ s^{-1}\ Mpc^{-1}}$
1 parsec	$1\ \mathrm{pc}\ =\ 3.086 \times 10^{16}\ \mathrm{m}\ =\ 206\,265\ \mathrm{AU}$ $=\ 3.262\ \mathrm{ly}$

Object	Mean radius (km)	Mass (kg)	Semi-major axis (AU)	Eccentricity	Albedo
Sun	$695\,500$	$1.988\ \times\ 10^{30}$	---	---	---

Solar luminosity	$L_{\odot} = 3.826\ \times\ 10^{26}\ \mathrm{W}$

Effective temperature of Sun	$T_{\text{eff},\odot} = 5778\ \text{K}$

Wien's displacement law	$\lambda_m T = 2.898 \times 10^{-3}\ \text{m K}$

Problem 6.9 Binary System

2025 | Piatra Neamț, Romania | Theory | Th.1 | 10 points

Solution p. 96

Th.1 Binary System (10 Points)

Two stars A and B, that form a non-eclipsing binary system, have peaks of their blackbody radiation spectra at 650 nm and 400 nm, respectively. We also know that the radius of star A is twice the radius of star B.

a) Compute the difference between the absolute magnitudes of the stars A and B.

b) If the combined absolute magnitude of the binary system is 0.00, find the absolute magnitude of each star.

Problem 6.10 Lightcurves

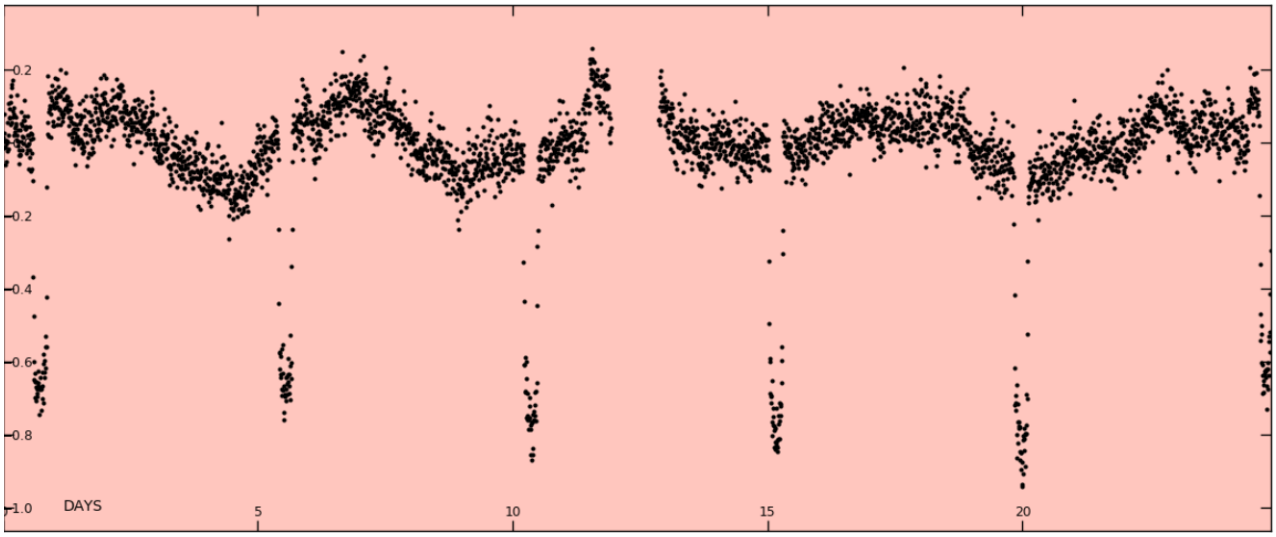
2025 | Piatra Neamț, Romania | Theory | Th.9 | 40 points

Solution p. 97

Th.9 Lightcurves (40 points)

(A) Transiting Exoplanet

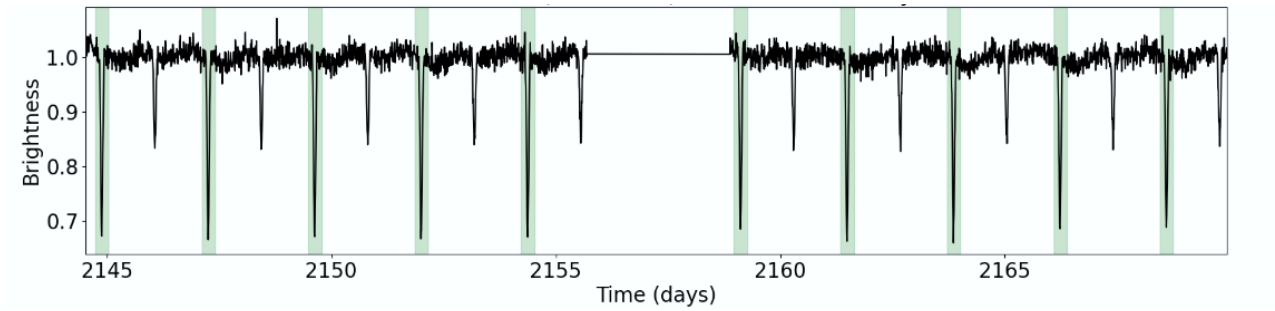
The figure below presents the light curve of an exoplanet transit. On the X-axis are represented the days elapsed since the first observation, and on the Y-axis is shown the percentage of the average luminosity by which the brightness of the system increases or decreases relative to the average brightness of the star, that is: $100 \times \frac{L(t)-L_{average}}{L_{average}}$. Determine the period P of the planet and the ratio $\frac{Planet\ Radius}{Star\ Radius}$.



Note: On Y-axis *0.2* means *0.2%* and *1.0* means *1%*

(B) Eclipsing Binary Star System

Another type of object with an interesting light curve is an eclipsing binary star system. The lightcurve of such an object is presented below, with the X-axis representing days since the start of the observing mission, and the Y-axis representing the relative brightness of the system, that is: $\frac{L(t)}{L_{average}}$. We know that the smaller star is hotter than the larger star. Calculate the period P of the system and the ratio $\frac{Small\ Star\ Radius}{Big\ Star\ Radius}$.



Note: On Y-axis *0.7* means *70%* and *1.0* means *100%*

Solutions to The Sun and Stars

Solution 6.1 The Moon, Seen on the Way to the Moon

2022 | Câmpulung Moldovenesc, Romania | Theory | Q9

Problem p. 77

Solution

The approach of the cosmonaut to the Moon must be such that it determines a variation of the apparent magnitude of the Moon:

$$\Delta m = m - m_0 = m_S - m_0 = -26.84 - (-12.7) = -14.14,$$

so that the apparent magnitude of the Moon, viewed from this new distance, is:

$$m = m_0 + \Delta m = m_S = -26.84;$$

$$\Delta m = m_S - m_0 = -26.84 + 12.7 = -14.14.$$

Knowing that the luminous intensity, *I*, of a star, in particular of the Moon, is:

$$I = \frac{\phi}{\Omega} = \frac{L}{\Omega} = \frac{W}{t \cdot \Omega},$$

that is, the energy flow emitted by the star, the Moon, in the unit of solid angle, then:

$$\Omega = \frac{A}{\Delta^2};\ A = \Omega \cdot \Delta^2;\ \phi = I \cdot \Omega;\ E = \frac{\phi}{A} = \frac{I \cdot \Omega}{\Omega \cdot \Delta^2} = \frac{I}{\Delta^2};\ E = \frac{L}{4\pi\Delta^2}.$$

According to Pogson's formula, if E_0 and respectively *E* are the brightnesses of the Moon, corresponding to the initial (r_0) and respectively the final distances (*r*), shown in the drawing in figure 1, it results:

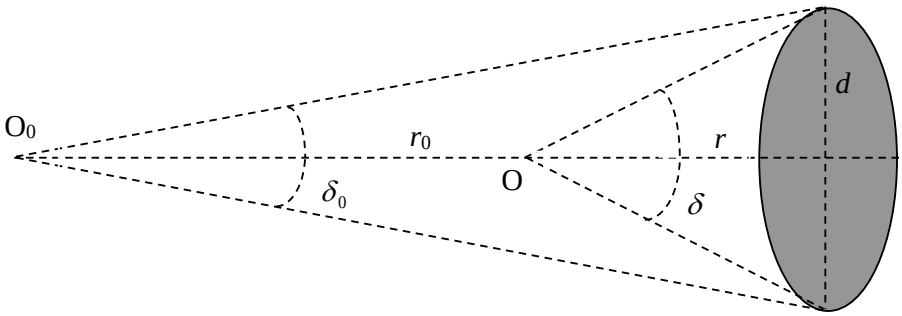


Fig. 1

$$\log \frac{E}{E_0} = -0.4 \cdot (m - m_0) = \log 10^{-0.4(m-m_0)};\ E_0 = \frac{I}{r_0^2};\ E = \frac{I}{r^2},$$

where *I* is the luminous intensity of the Moon;

$$\log \frac{r_0^2}{r^2} = \log 10^{-0.4(m-m_0)};$$

$$2 \log \frac{r_0}{r} = -0.4 \cdot \Delta m;\ \log \frac{r_0}{r} = 0.2 \cdot 14.14 = 2.828 = \log 10^{2.828};$$

$$\frac{r_0}{r} = 10^{2.828};\ r = \frac{r_0}{10^{2.828}}$$

$$x = 10^{2.828};\ \log x = 2.828;\ \ x \approx 675;$$

$$r = \frac{r_0}{10^{2.828}};\ 10^{2.828} \approx 675;$$

$$r = \frac{r_0}{675} = \frac{384000\,\text{km}}{675} \approx 568.88\,\text{km},$$

this is the distance from which the Moon should be viewed, so that its brightness is the same as the brightness of the Sun, seen from Earth.

The supplied S9 file solves a different version requiring the Moon's brightness to equal the Sun's brightness, giving 568.88 km. Q9 instead asks for 50% of the Sun's magnitude. The solution also initially prints 348000 km for the lunar distance, then substitutes 384000 km.

Solution 6.2 The Luminosity of a Star

2022 | Câmpulung Moldovenesc, Romania | Theory | Q10

Problem p. 77

Solution

The luminosity of a star, meaning the energy of the total radiation emitted by that star, in the unit of time, through its entire surface, at all wavelengths, in accordance with the Stefan-Boltzmann law, results:

$$L_\Sigma = 100 \cdot L_S;$$

$$L_\Sigma = 4\pi \cdot R_\Sigma^2 \cdot \sigma \cdot T_\Sigma^4;\ L_S = 4\pi \cdot R_S^2 \cdot \sigma \cdot T_S^4;$$

$$4\pi \cdot R_\Sigma^2 \cdot \sigma \cdot T_\Sigma^4 = 10^2 \cdot 4\pi \cdot R_S^2 \cdot \sigma \cdot T_S^4;\ R_\Sigma^2 \cdot T_\Sigma^4 = 10^2 \cdot R_S^2 \cdot T_S^4;$$

$$\frac{R_\Sigma^2}{R_S^2} = 10^2 \cdot \frac{T_S^4}{T_\Sigma^4} = 10^2 \cdot \left(\frac{T_S}{T_\Sigma}\right)^4 = 10^2 \cdot 2^4;$$

$$\frac{R_\Sigma}{R_S} = 40.$$

Solution 6.3 Unresolved binary star

2023 | Volos, Greece | Theory | Q3 | 10 points

Problem p. 78

The apparent magnitude of a star is defined as: $m = -2.5 \log_{10}\left(\frac{F_x}{F_{x,0}}\right)$. **[4 points]**

We solve for the flux F_x and we add the fluxes corresponding to each star, obtaining the relation: $m_b = -2.5 \log_{10}(10^{-0.4\, m_1} + 10^{-0.4\, m_2}) = 1.89 \approx 1.9$ mag. **[6 points]**

Solution 6.4 Parallax and proper motion

2023 | Volos, Greece | Theory | Q7 | 10 points

Problem p. 78

Based on the information given we can extract the radial and the transverse velocity of Barnard's star. The radial velocity is evaluated through the doppler shift, due to the low values, we can use the non-relativistic formula: $v_r = \frac{\Delta\lambda}{\lambda} c = \frac{656.044-656.281}{656.281} \times 2.998 \times 10^5 = -108.27$ km/s.

[3 points]

For the tangential velocity, we need first to evaluate the radial distance. Starting from the parallax, we find that the distance is $d = \frac{1}{0.552} = 1.81$pc , as it moves 10.31 arcseconds/year, this corresponds to a tangential velocity of:

$v_t = \mu\, d = \frac{10.31}{206265} \times 1.81 \times 3.086 \times 10^{13} \times \frac{1}{365.2422 \times 24 \times 3600} = 88.55$ km/s. **[4 points]**

The speed can be evaluated through Pythagorean theorem:
$v = \sqrt{(108.27^2 + 88.55^2)} = 139.87$ km/s . **[3 points]**

Solution 6.5 Sirius A and B

2023 | Volos, Greece | Theory | Q11 | 40 points

Problem p. 79

A) We will use the relation between the apparent and absolute magnitude of the star:
$m - M = 5\log(d) - 5$.
The distance can be evaluated using the parallax: $d = \frac{1}{0.350} = 2.86$ pc. Combining this with the above relation we obtain the absolute visual magnitude of Sirius A:
$M_V = -1.47 - 5\log 2.86 + 5 = 1.25$mag. Taking into account the bolometric correction, the absolute bolometric magnitude is: $M_B = 1.25 - 0.4 = 0.85$mag.
Regarding Sirius B, we repeat the same process, we obtain: $M_B = 0.85 + (8.5 + 1.47) = 10.82$.
To obtain the luminosity we use the following relation:
$M_V = -2.5\log\frac{L}{L_0}$
$L = L_0 \times 10^{-0.4\,M_V}$

Substituting for Sirius A and B, we obtain, for Sirius A $L_A = 1.37 \times 10^{28}$ W,
Sirius B $L_B = 1.41 \times 10^{24}$ W.

Next we apply Stephan-Boltzmann law to get the radius: $L = 4\,\pi\,R^2\sigma\,T^4$, the radius of Sirius A is $R_A = 1.39 \times 10^9$m.

Whereas the radius of Sirius B is: $R_B = 2.25 \times 10^6$ m

B) Let us convert the angular separation to radians: $\theta_1 = \frac{7}{206265} = 3.39 \times 10^{-5}$ rad. Based on the distance found in the previous question, the separation is
$s_1 = 3.39 \times 10^{-5} \times 2.86 = 9.71 \times 10^{-5}$ pc which converted into meters is: 3.00×10^{12}m.

C) We can evaluate the total mass from the relation: $M_{tot} = a^3/P^2$ where a is the separation in Astronomical Units, P is the period in years and the total mass is given solar masses. Their separation converted to AU is: 20 AU, thus the total mass is:
$M_{tot} = \frac{20.0^3}{50^2} = 3.20\,M_\odot$.

D) The mass of Sirius B is $\frac{1}{3}3.20 = 1.07M_\odot$. To evaluate the mean density we need to find its radius. As we have already evaluated the luminosity, we now need to evaluate the radius Using part A of this question, the radius is: $R = 2.25 \times 10^6$ m. The mean density is: $\bar{\rho} = 4.4 \times 10^{10} kg/m^3$.

E) Based on the above results (mass radius and density), Sirius B has a radius comparable to the radius of Earth, while its mass is close to the mass of the Sun. Therefore, it is a white dwarf, closer to 1.

Solution 6.6 A Cepheid period-luminosity relation

2024 | Kathmandu, Nepal | Theory | T4 | 15 points

Problem p. 81

T4

4. (a) (2 points) $L \propto P^{2.5} \rightarrow L = Kp^{2.5}$, taking log on both side then $\log L = \log K + 2.5 \log P$

   $\log L = \log F + \log 4 + \log \pi + 2 \log r$

   (1 points) $m_1 - m_2 = -2.5 log \left(\frac{F_1}{F_2} \right)$

   For ease of calculation we can use Vega as reference star and m be apparent magnitude of cepheid variable then

   (1 point) $\frac{-m}{2.5} = \log F_1 - \log F_o$

   $m = M + 5 log \left(\frac{r}{10pc} \right)$

   Solving these equation we get,

   (3 points)

   $$M = -6.25 LogP + C$$

   ,where $C = -2.5(LogK - Log4 - LogF_o - Log\pi) + 5Log10$

   (b) (1 point) So for above equation of magnitude luminosity relation C=-1.5,

   (1 point) we suppose, m is the average magnitude for luminosity, $\frac{1}{2}(L_{\max} + L_{\min})$.

   (1 point) Now $m - m_{ref} = -2.5 \log \left(\frac{\frac{1}{2}(L_1 + L_2)}{L_{ref}} \right)$

   $m - m_{ref} = 2.5 \log 2 - 2.5 \log \left(\frac{L_1 + L_2}{L_{ref}} \right)$

   $m = m_{ref} + 2.5 \log 2 - 2.5 \log \left(\frac{L_1 + L_2}{L_{ref}} \right)$

(1 point) $\frac{L_{tot}}{L_{ref}} = \left(\frac{L_1+L_2}{L_{ref}}\right) = \left(\frac{L_1}{L_{ref}} + \frac{L_2}{L_{ref}}\right)$

$= \left(10^{-0.4m_1} + 10^{-0.4m_2}\right) 10^{0.4m_{ref}}$

(1 point) Now substituting the values,
$m = 2.5 \cdot \log 2 - 2.5 \cdot \log(10^{-\frac{2}{5}m_1} + 10^{-\frac{2}{5}m_2})$

(1 point) From graph $m_{max} = 28$ and $m_{min} = 22$ thus m=22.75 and period is 20 days.

(1 point) So, using the Absolute magnitude relation with peiod we obtained above we can write M=-9.63.

(1 point) Substitute the value in this equation $m = M + 5log\left(\frac{r}{10pc}\right)$ we get, r=29.92 Mpc.

Solution 6.7 A spectroscopic binary system

2024 | Kathmandu, Nepal | Theory | T10 | 15 points

Problem p. 82

T10

1. (1 point) We know $V_1 = \omega a_1$ and $V_2 = \omega a_2$.

   (2 points) From Kepler's Third law: $P^2 = \frac{4\pi^2}{GM}a^3$, where $M = m_1 + m_2$ and $a = a_1 + a_2$.

   (1 point) We know that $P = \frac{2\pi}{\omega}$, $a_1 = \frac{m_2}{(m_1+m_2)}a$ and $a_2 = \frac{m_1}{(m_1+m_2)}a$

   (2 points) Replacing above value in Kepler's equation we get,

   $G(m_1 + m_2) = 4\pi^2 \frac{\omega^2}{4\pi^2} \frac{(m_1+m_2)^3}{m_2^3} a_1^3$

   $\frac{m_2^3}{(m_1+m_2)^2} = (\omega a_1)^3 \frac{P}{2\pi G}$

   (2 points) When we observe the star we can't measure the real orbital velocity but the component of it as the plane of orbit and astronomer line of sight have angle θ,thus

(1 point) $\frac{m_2^3}{(m_1+m_2)^2}(sin^3\theta) = (v_{1obs})^3\frac{P}{2\pi G}$, which is required mass function, where v_{1obs} is maximum observed orbital velocity

(1 point) similarly we can write another mass function, $\frac{m_1^3}{(m_1+m_2)^2}(\sin^3\theta) = (v_{2obs})^3\frac{P}{2\pi G}$

2. (1 point) We know $\frac{v_1}{v_2} = \frac{m_2}{m_1}$

   (2 points) From graph we can see that $\frac{v_1}{v_2} = 1.5$ thus $m_2 = 1.5m_1$, period is 40 days and v_{1obs} is 75 km/s,and for minimum value we consider $\sin\theta$ as 1,if we put these value in mass function, we get

   (1 point) $0.54m_1 = 1.74M_\odot$

   $m_1 = 3.2M_\odot$,similarly

   (1 point) $m_2 = 4.9M_\odot$

Solution 6.8 A radiating interstellar dust cloud

2024 | Kathmandu, Nepal | Theory | T12 | 15 points

Problem p. 84

T12

1. (2 points)
$$\lambda_{max} T = 2.898 \times 10^{-3} mK$$
(1 point) Thus, $\lambda_{max} = 57.96 \mu m$.

2. (2 points) Use the Stefan-Boltzmann Law and compare to the Sun to get:
$$R_{dust} = R_{\odot} \sqrt{\frac{L_{dust}}{L_{\odot}} \frac{T_{\odot}^4}{T_{dust}^4}}$$
(2 points) Substituting values, you get:
$$R_{dust} = 120.4 \text{pc}$$

3. (3 points)
$$d_{dust} = \frac{cz}{H_0} = \frac{0.01 \times 2.998 \times 10^5}{70} Mpc = 42.83 \text{ Mpc}$$
(2 points)
$$\theta = \frac{R_{dust}}{d_{dust}} rad = 0.580”$$

4. (3 points)
$$D > 1.22 \frac{\lambda'_{max}}{\theta} = 25.4 \text{m}$$
λ_{max} needs to have redshift accounted, i.e. $\lambda'_{max} = (1+z)\lambda_{max}$ (or some equivalent). Only 1 point gained if λ_{max} is used for correct formula.

The question says that the dust re-emits UV light, although its stated 50 K temperature and the official 57.96 micrometre peak correspond to infrared emission. The source wording is retained.

Solution 6.9 Binary System

2025 | Piatra Neamț, Romania | Theory | Th.1 | 10 points

Problem p. 85

Marking Scheme

a) From Wien's law, we have: $b = T \cdot \lambda$, where b is the Wien's constant, T is the surface temperature of the star, and λ is the wavelength corresponding to the maximum peak of radiation. Hence,

$$b = T_A \cdot \lambda_A = T_B \cdot \lambda_B \Rightarrow \frac{T_A}{T_B} = \frac{\lambda_B}{\lambda_A} \qquad \textbf{2pt}$$

The luminosity of a star is given by the Stefan-Boltzmann formula:

$$\begin{aligned} L &= 4\pi R^2 \sigma T^4 \Rightarrow \\ \frac{L_A}{L_B} &= \frac{4\pi R_A^2 \sigma T_A^4}{4\pi R_B^2 \sigma T_B^4} \\ &= \left(\frac{R_A}{R_B}\right)^2 \left(\frac{T_A}{T_B}\right)^4 = \left(\frac{R_A}{R_B}\right)^2 \left(\frac{\lambda_B}{\lambda_A}\right)^4 \\ &= \left(\frac{2R_B}{R_B}\right)^2 \left(\frac{400}{650}\right)^4 \\ &\approx 0.57 \end{aligned} \qquad \textbf{2pt}$$

Now, we infer the luminosity-absolute magnitude relationship:

$$\begin{aligned} \frac{L_A}{L_B} &= 10^{0.4(M_B - M_A)} \Rightarrow \\ \Delta M = M_A - M_B &= -\frac{1}{0.4} \log_{10}\left(\frac{L_A}{L_B}\right) \\ &= -\frac{1}{0.4} \log_{10} 0.57 \\ &\approx 0.60 \end{aligned} \qquad \textbf{2pt}$$

Note: Also accept 0.61 (possible due to rounding).

b) Denote by $L = L_A + L_B$ the total luminosity of the binary system, and with $M = 0$ the absolute magnitude of the system. We will compare these with the Sun's luminosity $L_\odot$ and

its absolute magnitude $M_\odot$. Using the same formula as above, we get:

$$\begin{aligned}
\frac{L}{L_\odot} &= 10^{0.4(M_\odot - M)} \\
\frac{L}{L_\odot} &= \frac{L_A + L_B}{L_\odot} = \frac{L_A}{L_\odot} + \frac{L_B}{L_\odot} \\
&= 10^{0.4(M_\odot - M_A)} + 10^{0.4(M_\odot - M_B)} && \textbf{1pt} \\
10^{0.4(M_\odot - M)} &= 10^{0.4(M_\odot - M_A)} + 10^{0.4(M_\odot - M_B)} \\
10^{-0.4M} &= 10^{-0.4M_A} + 10^{-0.4M_B} = \\
&= 10^{-0.4(M_B + \Delta M)} + 10^{-0.4M_B} && \textbf{1pt} \\
10^{-0.4M} &= 10^{-0.4M_B}(10^{-0.4\Delta M} + 1) \\
-0.4M &= -0.4M_B + \log_{10}(10^{-0.4\Delta M} + 1) \\
M_B &= M + \frac{1}{0.4}\log_{10}(10^{-0.4\Delta M} + 1) \\
&= 0 + \frac{1}{0.4}\log_{10}(10^{-0.4\cdot 0.61} + 1) \approx 0.49 && \textbf{1pt} \\
M_A &= M_B + \Delta M = 0.49 + 0.61 = 1.10 && \textbf{1pt}
\end{aligned}$$

Alternative Solution b)
Denote by $L = L_A + L_B$ the total luminosity of the binary system, and with $M = 0$ the absolute magnitude of the system.

$$\begin{aligned}
M - M_A &= -2.5\log\left(\frac{L_A + L_B}{L_A}\right) = -2.5\log\left(1 + \frac{L_B}{L_A}\right) && \textbf{2pt} \\
0 - M_A &= -2.5\log\left(1 + \frac{1}{0.57}\right) \\
M_A &= 2.5\log(2.74) = 1.10 && \textbf{1pt}
\end{aligned}$$

We already calculated the difference in magnitude, which means we can easily calculate M_B now:

$$\begin{aligned}
\Delta M = M_A - M_B \Rightarrow M_B &= M_A - \Delta M \\
M_B &= 1.1 - 0.60 = 0.50 && \textbf{1pt}
\end{aligned}$$

Solution 6.10 Lightcurves

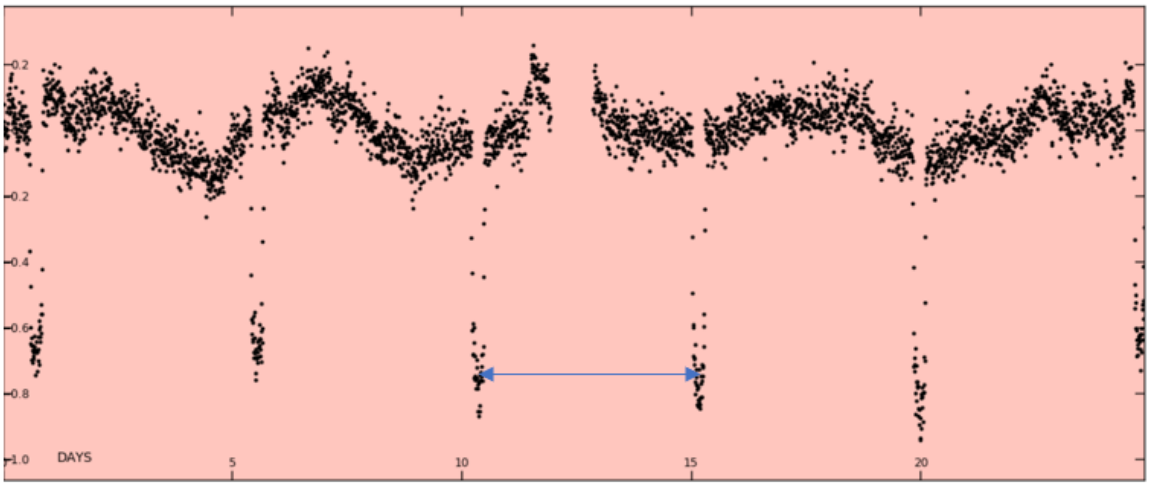
2025 | Piatra Neamț, Romania | Theory | Th.9 | 40 points

Problem p. 86

Marking Scheme

(A) Transiting Exoplanet (15 pt)

To calculate the period, we can proceed in different ways. The most straightforward way is to measure the distance between two consecutive minima in cm in the figure. Then, by measuring the distance between the 5 and 10 day marks, we can compute the scale of the figure on the OX axis. We then use this scale to calculate the period of the planet.



5 days ... 2.6 cm.

The scale will be 1.92 days/cm. The distance between two consecutive minima is 2.5 cm.

The period will then be:

$$P = 2.5 \times \frac{5}{2.6} \approx 4.81 \; days$$ **5pt**

Note for graders: The calculated scale might be different due to printing/zoom figure changes from the initial problem writing to the final version. However, the value of the period will not be affected by this. Answers within ± 0.1 days will receive full points

For a lower error in determining the period, we can measure longer distances on the paper:

- To determine the scale, we measure between the 5 and 20 day marks: 15 days ... 7.8 cm
- To determine the period, we measure the distance between the first and the one before last minima (the last one is not fully visible) and then we divide the distance by the number of periods: 10 cm divided by 4

$$P = \frac{10}{4} \times \frac{15}{7.8} \approx 4.81 \; days$$

Note for graders: Both methods will receive the maximum point score.

To calculate the ratio of the radii, we start by noting the minimum value in percent as $n\%$. From the description of the OY axis, we can write:

$$100 \times \frac{L_{transit} - L_{average}}{L_{average}} = n$$

This is just the formula given in the problem, written explicitly for a transiting moment of time.

$$L_{transit} - L_{average} = \frac{n}{100} \times L_{average}$$

where $L_{average}$ is the average luminosity of the system (the luminosity of the star without transit), and $L_{transit}$ is the luminosity of the system at the minimum caused by the transit. $L_{transit}$ can be written as:

$$L_{transit} = \frac{L_{average}}{\pi R^2} \left(\pi R^2 - \pi r^2\right)$$

Here R is the radius of the start and r is the radius of the planet. The calculation of n then becomes:

$$\frac{L_{average}}{\pi R^2} \left(\pi R^2 - \pi r^2\right) - L_{average} = \frac{n}{100} \times L_{average}$$

$$\left(\frac{r}{R}\right)^2 = -\frac{n}{100} \Rightarrow \frac{r}{R} = \sqrt{\frac{|n|}{100}} \quad \textbf{5pt}$$

Now we need to determine the scale of the figure on the OY axis. Measuring the distance between -0.2% and -0.8%, we get 2.2 cm. Thus:

$0.6\% \ldots$ 2.2 cm.

The depth of a minimum varies between 2.8 and 3.2 cm, depending on which minimum is chosen for the analysis. For the purpose of this problem, any value in the 2.8 – 3.1 interval will receive the maximum point score, as well as measuring all minima and taking an average of the depth of the transit (a more accurate method). The numerical value here will be calculated for 2.9 cm.

$$|n| = 2.9 \times \frac{0.6}{2.2} \approx 0.79\% \quad \textbf{3pt}$$

$$\frac{r}{R} = \frac{Planet\ Radius}{Star\ Radius} \approx 0.09 \quad \textbf{2pt}$$

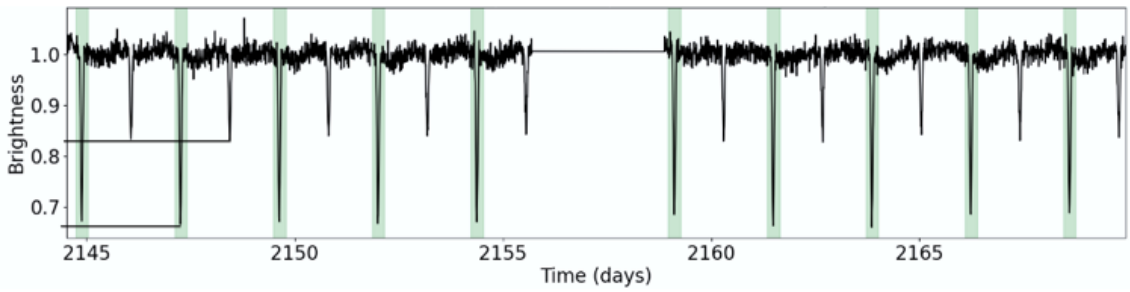
For the extremes values we would get:

$$\frac{r}{R} = \frac{Planet\ Radius}{Star\ Radius} \approx 0.087 \approx 0.09$$

$$\frac{r}{R} = \frac{Planet\ Radius}{Star\ Radius} \approx 0.093 \approx 0.09$$

By rounding the answer to the correct number of significant digits, we obtain the same result.

(B) Eclipsing Binary Star System (25 pt)



The scale of this figure on the OX axis is: 20 days ... 9.1 cm The measured value corresponding to the period is 1.1 cm.

We then get the period:

$$P = 1.1 \times \frac{20}{9.1} \approx 2.42\ days \quad \textbf{6pt}$$

Note for graders: The calculated scale might be different due to printing/zoom figure changes from initial problem writing to final version. However, the value of the period will not be affected by this. Answers within ± 0.1 days will receive full points

The average luminosity for a binary star system is:

$$L_{average} = L_R + L_r$$

where L_R is the luminosity of the large, cooler star of radius R and L_r is the luminosity of the smaller, hotter star of radius r. During the secondary minima (smaller, hotter star transiting the bigger, cooler star), the luminosity of the system will be:

$$L_{Secondary} = L_r + \frac{L_R}{\pi R^2}\left(\pi R^2 - \pi r^2\right)$$

During the principal minima (smaller, hotter star completely eclipsed by the bigger, cooler star), the luminosity of the system will be:

$$L_{Primary} = L_R$$ **6pt**

The corresponding luminosity ratios are:

$$n_{Primary} = \frac{L_{Primary}}{L_{average}} = \frac{L_R}{L_R + L_r}$$

$$n_{Secondary} = \frac{L_{Secondary}}{L_{average}} = \frac{L_r + \frac{L_R}{\pi R^2}\left(\pi R^2 - \pi r^2\right)}{L_R + L_r}$$

$$n_{Secondary} = \frac{L_R + L_r - L_R \times \frac{r^2}{R^2}}{L_R + L_r} = 1 - \frac{L_R}{L_R + L_r} \times \frac{r^2}{R^2}$$

$$n_{Secondary} = 1 - n_{Primary} \times \frac{r^2}{R^2}$$

$$\frac{r}{R} = \sqrt{\frac{1 - n_{Secondary}}{n_{Primary}}}$$ **5pt**

From the figure, we measure the Y-scale: 0.2 ... 1.1 cm Then, $1 - n_{Primary}$ will be, in cm: 1.8 cm.

$$n_{Primary} = 1 - 1.8 \times \frac{0.2}{1.1} = 1 - 0.327 \approx 0.67$$ **3pt**

$1 - n_{Secondary}$ will be, in cm: 0.9 cm.

$$n_{Secondary} = 1 - 0.9 \times \frac{0.2}{1.1} = 1 - 0.16 \approx 0.84$$ **3pt**

$$\frac{r}{R} = \sqrt{\frac{1 - 0.84}{0.67}} \approx 0.49$$ **2pt**

± 0.02 values accepted for full marks

7. Stellar Evolution

Theory problems

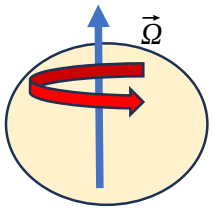
Problem 7.1 Rapidly spinning neutron star

2023 | Volos, Greece | Theory | Q4 | 10 points

Solution p. 103

Question 4: Rapidly spinning neutron star.

A neutron star is a rapidly spinning compact star. A particular neutron star has a mass of 1.40 $M_\odot$ and its radius is 10.0 km. What is the shortest possible spin period for it, before it starts shedding its mass away?



Universal gravitational constant	G	=	6.674×10^{-11} N m^2 kg^{-2}

Object	Mean radius (km)	Mass (kg)	Orbital Radius (au)
Sun	695 500	1.988×10^{30}	---

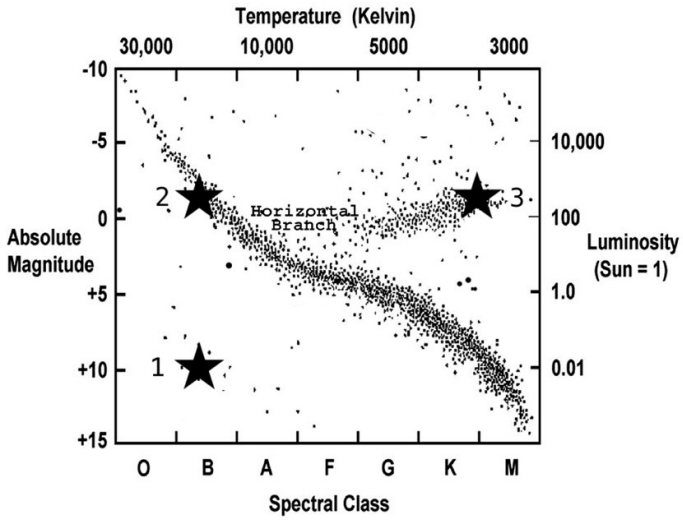
Problem 7.2 H-R diagram

2023 | Volos, Greece | Theory | Q10 | 10 points

Solution p. 103

Question 10: H-R diagram.

The image below shows the H-R diagram and three stars: 1, 2 and 3. For each statement below write "TRUE" if the statement is definitely true. Write "FALSE" if the statement is definitely false. Write "MAYBE" if the information is insufficient to draw a conclusion.



A) Star 1 is a main sequence star.

B) The radius of star 2 is larger than that of star 3.

C) Star 2 is younger than the current age of the Sun.

D) Star 3 is more massive than the Sun.

E) Star 3 is colder than the other two stars.

Solutions to Stellar Evolution

Solution 7.1 Rapidly spinning neutron star

2023 | Volos, Greece | Theory | Q4 | 10 points

Problem p. 101

The limit of mass shedding occurs when the centripetal/centrifugal acceleration equals the force of gravitational acceleration. This corresponds to:

$$\frac{GM}{r^2} = \frac{v^2}{r}$$

[4 points]

We find that the velocity, substituting the values for a neutron star, we find that the corresponding linear velocity on the equator due to rotation is 1.36×10^8m/s. We now need to evaluate the period: $\frac{2\pi}{P} r = v$, thus $P = \frac{2\pi r}{v} = 0.462$ ms **[6 points]**

Solution 7.2 H-R diagram

2023 | Volos, Greece | Theory | Q10 | 10 points

Problem p. 102

A) Star 1 is a main sequence star.	FALSE	**[2 points]**
B) The radius of star 2 is larger than that of star 3.	FALSE	**[2 points]**
C) Star 2 is younger than the current age of the Sun.	TRUE	**[2 points]**
D) Star 3 is more massive than the Sun.	MAYBE	**[2 points]**
E) Star 3 is colder than the other two stars.	TRUE	**[2 points]**

8. Galaxies and Cosmology

Theory problems

Problem 8.1 Andromeda Galaxy Rotation

2022 | Câmpulung Moldovenesc, Romania | Theory | Q5

Solution p. 107

I. Multiple choice problems – 100 points

For these problems, you have to choose only one correct answer. Please write your answer on answer-sheet. **Don't forget to write the number of the problem, i.e Grid 1. (c).**

Grid 5. Andromeda Galaxy Rotation

The following are known: the Earth – Andromeda galaxy distance, $R = 1.42 \cdot 10^{11} R_0$, where R_0 is the radius of the Earth's circular orbit around the Sun; the mass of our Galaxy, $M_G = 2.5 \cdot 10^{11} M_0$, where M_0 is the mass of the Sun; $T_0 =$ 1 terrestrial year, the period of rotation of the Earth around the Sun; the mass of the Andromeda galaxy, $M_A = 3.6 \cdot 10^{11} M_0$.
The period of the rotational motions of our galaxy and the Andromeda galaxy, around their common centre of mass, is:

a) $5.1 \cdot 10^{10}$ years b) $6.9 \cdot 10^{10}$ years c) $7.9 \cdot 10^{10}$ years d) $8.2 \cdot 10^{10}$ years

Problem 8.2 Cosmology

2023 | Volos, Greece | Theory | Q9 | 10 points

Solution p. 107

Question 9: Cosmology

Arrange the following events in the order in which they occurred (from earliest to the most recent).

A) Formation of the first stars.
B) Inflationary era of the universe
C) Formation of helium nuclei
D) Formation of neutral hydrogen
E) Formation of oxygen atoms

Problem 8.3 The Cosmic Horseshoe

2024 | Kathmandu, Nepal | Theory | T7 | 15 points

Solution p. 108

T7 (15 points)

The Cosmic Horseshoe is a gravitationally lensed galaxy at $z = 2.379$. It has a magnification factor ($\mu = 30$). The magnification that occurs due to gravitational lensing is analogous to that provided by an optical telescope, but is due to the gravitational effects of massive bodies. Answer some questions about this galaxy to help a researcher. [Note: AB magnitudes are converted analogously to the magnitude scale that you are familiar with. The average flux density of an object is the Flux (F) of that object divided by the bandwidth $\Delta\nu$ of the detector.]

1. At what wavelength (λ_{obs}) would you expect to detect the H_α emission line emitted by this galaxy? [Note: $\lambda_{rest} = 656.3$nm.]
2. Source plane reconstruction is the technique by which the lensing effects on the galaxy are estimated, in order to get an understanding of the galaxy's morphology. We use the technique and now need to determine the object's physical properties. Using the *Hubble Space Telescope (HST)*, the F606W band AB magnitude is found to be 20.5. What is the object's magnitude after correcting for lensing?
3. One can convert from AB magnitudes to flux density by using the fact that a source at 0 AB magnitude has 3631 Jy of flux density. What is (a) the apparent flux density of the Cosmic Horseshoe and (b) what is the magnification corrected flux density?
4. The Cosmic Horseshoe is observed to have its F606W UV continuum emission spread across an area of $13\mathrm{arcsec}^2$. What is its apparent UV continuum emission surface density, in units of mag arcsec^{-2}?
5. If the angular scale at the Horseshoe's redshift is 8 kpc arcsec^{-1}, what is the radius of the galaxy, assuming that its a face-on disk?

Problem 8.4 Critical density and tennis balls

2024 | Kathmandu, Nepal | Theory | T11 | 15 points

Solution p. 109

T11 (15 points)

In a classical approximation to get a feel for expanding universe, one may assume that the universe is uniformly filled cloud with small particles, and that the critical density (ρ_c) is the density of the cloud at which the expansion of the universe (the Hubble-Lemaitre law) stops at infinite time.

1. Derive a classical expression for (ρ_c) of the universe.
2. Find the value of critical density in SI units.
3. If the universe was filled with tennis balls of mass 0.145 kg each distributed uniformly (no other mass) and the mass density of this universe equal to the critical density then how many tennis balls would be included in a sphere of size of the Sun?

Universal Gravitational constant	$G = 6.674 \times 10^{-11}\ \mathrm{N\ m^2\ kg^{-2}}$
Hubble constant	$H_0 = 70\ \mathrm{km\ s^{-1}\ Mpc^{-1}}$
1 parsec	$1\ \mathrm{pc} = 3.086 \times 10^{16}\ \mathrm{m} = 206\,265\ \mathrm{AU} = 3.262\ \mathrm{ly}$

Object	Mean radius (km)	Mass (kg)	Semi-major axis (AU)	Eccentricity	Albedo
Sun	$695\,500$	1.988×10^{30}	---	---	---

Solutions to Galaxies and Cosmology

Solution 8.1 Andromeda Galaxy Rotation

2022 | Câmpulung Moldovenesc, Romania | Theory | Q5

Problem p. 104

Solution

From the study of light diffraction it is known that the angular distance between two objects for which they can be observed separately is approximately. λ / D. It means that the change in the position of the constellation Andromeda can be observed if its angular displacement is:

$$\varphi_0 = 1.22 \cdot \frac{\lambda}{D} = 1.22 \cdot \frac{5 \cdot 10^{-7}}{6} = 10.16 \cdot 10^{-8} \text{ radians.}$$

If T is the period of rotation of our Galaxy and the Andromeda Constellation, respectively, around their common center of mass, then the time required for the angular displacement φ_0 of the Andromeda constellation is:

$$\tau = \frac{\varphi_0}{\omega} = \frac{\varphi_0}{2\pi} T.$$

For two binary systems, using the generalized form of Kepler's third law, we can write that:

$$\left(\frac{T_1}{T_2}\right)^2 \left(\frac{M_1 + m_1}{M_2 + m_2}\right) = \left(\frac{a_1}{a_2}\right)^3,$$

out of which, for the galaxy - Andromeda and Sun - Earth systems, results in:

$$\left(\frac{T}{T_0}\right)^2 \left(\frac{M_G + M_A}{M_0 + M_P}\right) = \left(\frac{R}{R_0}\right)^3; \; M_P << M_0; \; T_0 = 1 \text{ terrestrial year;}$$

$$\left(\frac{T}{T_0}\right)^2 \left(\frac{M_G + M_A}{M_0}\right) = \left(\frac{R}{R_0}\right)^3;$$

$$T^2 = T_0^2 \cdot \left(\frac{R}{R_0}\right)^3 \cdot \left(\frac{M_0}{M_G + M_A}\right); \; \tau = \frac{\varphi_0}{2\pi} T; \; \varphi_0 = 1.22 \cdot \frac{\lambda}{D};$$

$$\tau = 1.22 \cdot \frac{\lambda}{2\pi D} T_0 \left(\frac{R}{R_0}\right)^{3/2} \left(\frac{M_0}{M_G + M_A}\right)^{1/2} \approx 10^3 \text{ years.}$$

The supplied S5 file solves a different version asking how long a 6 m telescope at wavelength 5 × 10^-7 m needs to detect Andromeda's orbital motion. The official reasoning provides an orbital-period formula, but its final value of about 10^3 years is the detection time, not the period asked for in Q5.

Solution 8.2 Cosmology

2023 | Volos, Greece | Theory | Q9 | 10 points

Problem p. 104

The correct answer is: B, C, D, A, E

Solution 8.3 The Cosmic Horseshoe

2024 | Kathmandu, Nepal | Theory | T7 | 15 points

Problem p. 105

T7

1. (1 point)
$$\lambda_{obs} = (1+z) \times \lambda_{rest} = 3.379 \times 656.3\text{nm}$$
(1 point)
$$= 2217.6\text{nm}$$

2. (2 points) Lensing is multiplicative on the flux. So:
$$m_{corr} = m_{obs} + 2.5\ \log(30) = 24.19$$

3. (2 points)
$$f_{obs} = 3631 \times 10^{-\frac{20.5}{2.5}}\,\text{Jy} = 2.29 \times 10^{-5}\text{Jy}$$
(2 points)
$$f_{corr} = \frac{f_{obs}}{\mu} = 7.64 \times 10^{-7}\text{Jy}$$

4. (2 points) $\Sigma_{UV} = f_{obs}/13$.

(2 points) In magnitudes, $\Sigma'_{UV} = -2.5\log(\Sigma_{UV}/3631) = 23.3\ \text{mag arcsec}^{-2}$

5. (3 points)
$$r = \sqrt{13 \times 8^2/\pi\mu} = 3\ \text{kpc}$$
[1 point for correct placement of magnification factor, 1 for squaring scale, 1 for correct answer.]

Solution 8.4 Critical density and tennis balls

2024 | Kathmandu, Nepal | Theory | T11 | 15 points

Problem p. 106

T11

1. (1 point) Use the escape velocity definition:

$$v_{esc} = \sqrt{\frac{2GM}{R}}$$

(1 point) together with Hubble's law:

$$v = H_0 R$$

And:

$$M = \frac{4}{3}\pi R^3 \rho_c$$

(4 points) Substitute Hubble's Law and Mass in the escape velocity definition to obtain:

$$\rho_c = \frac{3H_0^2}{8\pi G}$$

2. (2 units)

$$H_0 = 2.2685 \times 10^{-18}\ \mathrm{s}^{-1}$$

(2 points) Get: $\rho_c = 9.20 \times 10^{-27}\mathrm{kg\ m}^{-3}$

3. (1 point)

$$M = \frac{4}{3}\pi \rho_c R_\odot^3$$

(1 point)

$$N = \frac{M}{M_{tennis}}$$

(3 points) Substitute values to get:

$$N = 89.45(\text{either 89 or 90}$$

(both acceptable).

9. Observational Problems

Sky maps and charts

Problem 9.1 Blind Sky Map

2022 | Câmpulung Moldovenesc, Romania | Sky map | Q13

Solution p. 145

You have 50 minutes to finish this part.

Please use only the pencil to make the drawings and mark-ups.

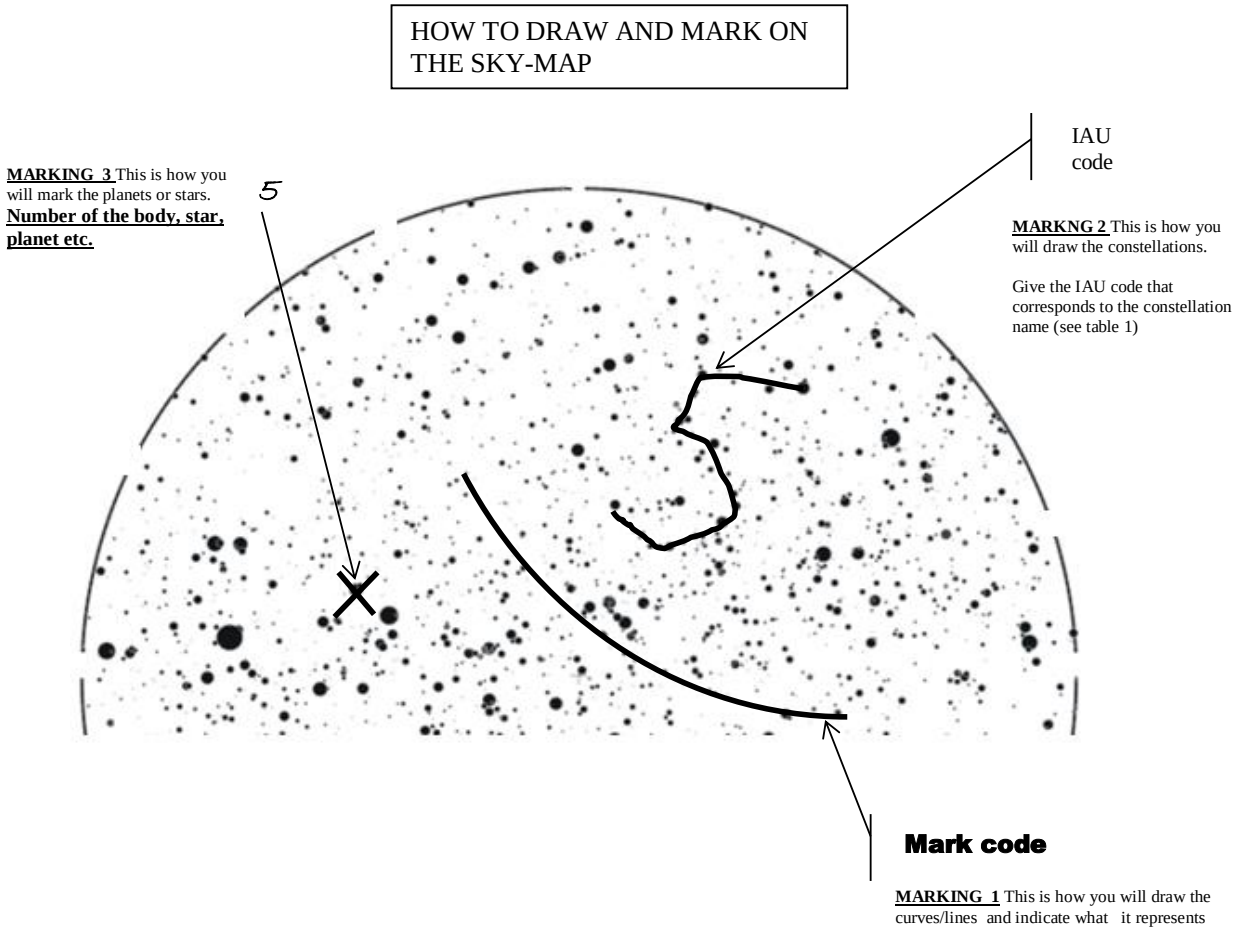
After you finish the work fill your student ID on the answer sheet as well on the sky map.

Put the 2 papers in the folder, leave on the table the compass, the ruler and the pencil.

Thank you!

In this part you will use the sky-map found in the envelope. Please use the pencil for marking and drawing lines on the sky-map. Use the indication 1, 2 and 3 to draw lines and mark objects on the map, as seen in the figure bellow.

The map is in azimuthal projection (distance from the centre is in proportion to zenith distance) .



The map represents the sky in somewhere in Romania. The Zenith point is in the centre of the chart.

1. Find out the latitude of place where map have been done.

Latitude of place:

. Draw the greatest circumpolar circle **(Mark A) (on map)**.

2. Draw on the map with continuous lines the celestial equator (**Mark B**), the ecliptic(**Mark C**), the galactic equator (**Mark G**), and the local meridian (**Mark M**),.
3. Mark on the map the cardinal points (**N,S,E,W**) and show 3 planets (**P1, P2, P3**) from the Solar System.
4. Identify and mark on the map the 5 brightest stars (magnitude less +1,00). Number the star starting from **1** – the brightest, and continue with the fainter ones till number **5** for the faintest. Fill in the following table the Bayer name of the five identified stars

Marking on the map			
	1	Name of the star	
	2	Name of the star	
	3	Name of the star	
	4	Name of the star	
	5	Name of the star	

5. Draw on the map 10 constellations you can identify. Each constellation you found, it will be identified on the map after the letters (IAU code) corresponding to its name in table 1.
6. Mark on the map the positions of the following objects:
   a. The following Messier objects: M31, M27, M15;
   b. β Cet, δ UMi

Marking on the map	**6**	**M31**
	7	**M27**
	8	**M15**
	9	**β Cet,**
	10	**δ UMi**

7.

Estimate the sidereal time of the map; write the value in the box	

8.

Estimate the equatorial coordinates of the star Altair (α Aquilae). Write your answer in the box	

Problem 9.2 Navigating the night sky I

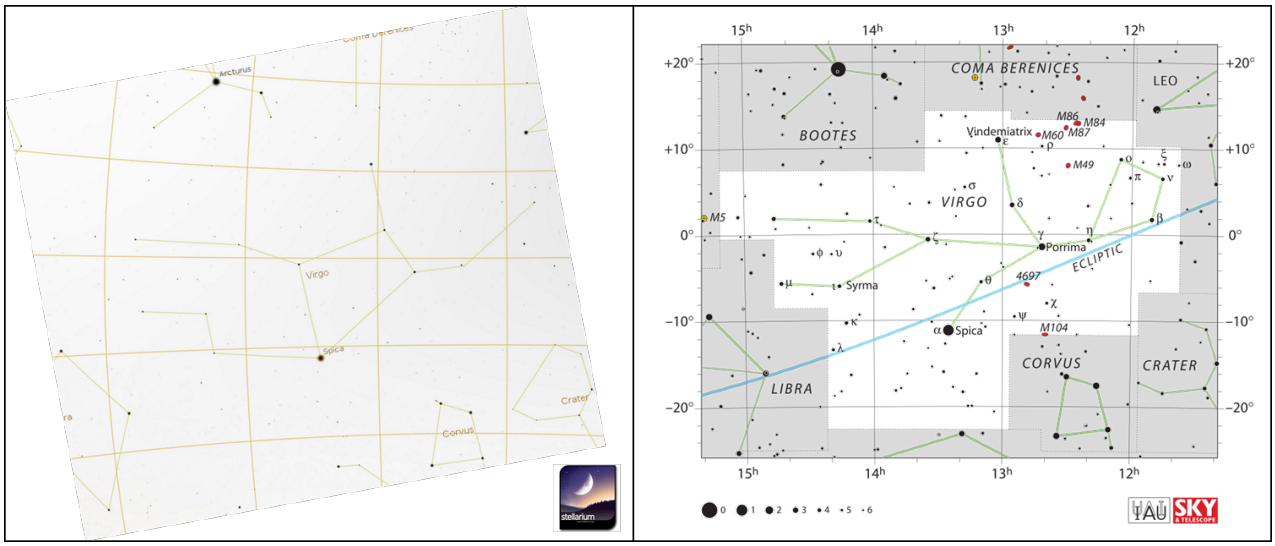
2023 | Volos, Greece | Sky map | Q13 | 45 points

Solution p. 147

Classroom Test part 1: Navigating the night sky

Instructions:

1. You have 40 minutes in total to solve this part.
2. In this exercise, you are given two wide-field maps (around 100° wide) of the night sky, labeled as "**Map 1**" and "**Map 2**". Map 1 is for an observer of the southern hemisphere, while Map 2 is for an observer of the northern hemisphere. The Milky Way is not depicted on the maps.
3. Write your answers on the maps themselves.
4. You will be asked questions about both of these maps. You can spend the available time as you wish between the two maps.
5. For all of the following questions, if you provide more than the requested answers, following rules will apply:
   A. For questions 1a, 1b, 2a and 2b, there will be no penalty for one extra answer, i.e. if you are asked to mark **N** objects in any sub-question and you mark **N+1** objects.
   B. For **each extra object** after that you will lose partial points (negative marking).
   C. You will not receive negative marking for any wrong answers though.
6. Whenever you are asked to draw constellations on the map, you can use any common stick figure drawing depicted in atlases or planetarium softwares (such as Stellarium). We will accept all reasonable drawings, but you are advised to keep your drawing as simple and recognisable as possible. For example, it would be preferable to depict Andromeda as the two back legs of Pegasus rather than a stick figure (a person) as in the IAU maps. Below you are given an example depicting two acceptable drawings of Virgo.



Map 1 – Southern sky:

a) Draw stick figures for any 10 constellations on the map and write their IAU code next to them. **(10 points)**

Notes:

- In case you do not know how to draw the constellation stick figure, but do know the area in which it resides, you can still provide the IAU code of the constellation in that area. You can refer to the table given for a list of IAU codes.
- You should draw and name only those constellations that are (almost) completely visible in the map. The constellations that are partially cut-off from the map will not be accepted as valid answers.

b) Identify any 20 stars on the map and provide either their proper name or their Bayer designation in English. **(10 points)**

Notes:

- Both proper name and Bayer name are acceptable, but you do not get extra points for knowing both.
- It is not compulsory to only choose the stars from the constellations you picked in the previous question.

c) A very bright star is missing from the map. Mark the position of the missing star with an "x" on the map. **(2.5 points)**

Map 2 – Northern sky:

a) Draw stick figures of any 10 constellations on the map and write their IAU code next to them (same instructions as question 1a). **(10 points)**

b) Identify on the map 20 stars and provide either their proper name or their Bayer abbreviation in English. (same instructions as 1b). **(10 points)**

c) A Supernova is present in the sky map. Identify and circle the same. **(2.5 points)**

IAU Constellations List

No	Constellation	IAU code		No	Constellation	IAU code
1	Andromeda	And		45	Lacerta	Lac
2	Antlia	Ant		46	Leo	Leo
3	Apus	Aps		47	Leo Minor	LMi
4	Aquarius	Aqr		48	Lepus	Lep
5	Aquila	Aql		49	Libra	Lib
6	Ara	Ara		50	Lupus	Lup
7	Aries	Ari		51	Lynx	Lyn
8	Auriga	Aur		52	Lyra	Lyr
9	Boötes	Boo		53	Mensa	Men
10	Caelum	Cae		54	Microscopium	Mic
11	Camelopardalis	Cam		55	Monoceros	Mon
12	Cancer	Cnc		56	Musca	Mus
13	Canes Venatici	CVn		57	Norma	Nor
14	Canis Major	CMa		58	Octans	Oct
15	Canis Minor	CMi		59	Ophiuchus	Oph
16	Capricornus	Cap		60	Orion	Ori
17	Carina	Car		61	Pavo	Pav
18	Cassiopeia	Cas		62	Pegasus	Peg
19	Centaurus	Cen		63	Perseus	Per
20	Cepheus	Cep		64	Phoenix	Phe
21	Cetus	Cet		65	Pictor	Pic
22	Chamaeleon	Cha		66	Pisces	Psc
23	Circinus	Cir		67	Piscis Austrinus	PsA
24	Columba	Col		68	Puppis	Pup
25	Coma Berenices	Com		69	Pyxis	Pyx
26	Corona Australis	CrA		70	Reticulum	Ret
27	Corona Borealis	CrB		71	Sagitta	Sge
28	Corvus	Crv		72	Sagittarius	Sgr
29	Crater	Crt		73	Scorpius	Sco
30	Crux	Cru		74	Sculptor	Scl
31	Cygnus	Cyg		75	Scutum	Sct
32	Delphinus	Del		76	Serpens	Ser
33	Dorado	Dor		77	Sextans	Sex
34	Draco	Dra		78	Taurus	Tau
35	Equuleus	Equ		79	Telescopium	Tel
36	Eridanus	Eri		80	Triangulum	Tri
37	Fornax	For		81	Triangulum Australe	TrA
38	Gemini	Gem		82	Tucana	Tuc
39	Grus	Gru		83	Ursa Major	UMa
40	Hercules	Her		84	Ursa Minor	UMi
41	Horologium	Hor		85	Vela	Vel
42	Hydra	Hya		86	Virgo	Vir
43	Hydrus	Hyi		87	Volans	Vol
44	Indus	Ind		88	Vulpecula	Vul


Map 1


Map 2

Problem 9.3 Navigating the night sky II

2023 | Volos, Greece | Sky map | Q14 | 45 points

Solution p. 147

Classroom Test part 2: Navigating the night sky

Instructions:

1. You have 40 minutes in total to solve this part.
2. In this exercise, you are given three tables in the answer sheet. Table 2.1ab and Table 2.1c corresponds to part 2.1 below, titled “Messier Objects”. Table 2.2 corresponds to part 2.2 below titled “Bright stars”.
3. Mark your answers in these tables.
4. You can spend the available time as you wish between the two tables.
5. For both parts (2.1 and 2.2), if you provide more than the requested answers, only the first 10 answers in each column will be evaluated. You will however not be negatively marked for any wrong answers.

Part 2.1: Messier Objects

The table 2.1ab provides a list of the 110 Messier objects. The first row of the table is filled in as an example.

a) Write the letter code of the correct type for any 10 Messier objects in this table at the corresponding row of the “Type” column. **(5 points)**

Note: The types to choose from are: “Galaxy” (**GL**), “Nebula” (**NB**), “Globular Cluster” (**GC**), “Open Cluster” (**OC**) and “other” (**OT**). A few Messier objects do not belong to any of the first four categories, and as such, are categorised as “other”.

b) Write the IAU code of the constellation of any 10 Messier objects in this table (not necessarily the same as the ones chosen in the previous question) at the corresponding row of the “Constellation” column. **(5 points)**

c) You will find below a list of 25 objects in Table 2.1c in the answer sheet, which are also found in the Messier list. For any 10 Messier objects from this list, write the correct Messier number in front of them. **(10 points)**

Part 2.2: Bright Stars

The Table 2.2 provides a list of the 92 brightest stars (excluding the Sun) up to 2.50 mag. The table provides the proper name of the stars, their Bayer designation, their apparent visual magnitude (m_v), and their distance from the Sun (in light-years). The stars are given in alphabetical order.

d) Identify any 10 stars on the list that are either a binary or part of a multiple star system. To do so, put a tick mark ($\surd$) at the corresponding row of the "Binary or Multiple" column.

(5 points)

Note: For the case of binary or multiple star systems, the visual magnitude given is that of the entire system (combined magnitude), as observed by the naked eye.

e) Identify any 10 stars on the list that are variable stars (of any kind). To do so, put a tick mark ($\surd$) at the corresponding row of the "Variable" column.

(10 points)

Note: For the case of variable stars, the visual magnitude given is the approximate time averaged visual magnitude as observed by the naked eye.

f) Most stars are currently classified under the Morgan–Keenan (MK) system using the letters O, B, A, F, G, K, and M, a sequence from the hottest (O type) to the coolest (M type). For the sake of this exercise, we will refer to stars of spectral type G, K, or M as "cool" stars. Identify any 10 stars on the list that are "cool". To do so, put a tick mark ($\surd$) at the corresponding row of the "Cool" column. **(10 points)**

Note: For the case of binary or multiple star systems of the list, their members are either of the same type (e.g., all of them are "cool") or their brightest component is the main contributor to the observed colour. For example, in a binary system with both a "hot" and a "cool" star, the "cool" one is far brighter and makes gives the impression of a red or orange star when observed with the naked eye.

Table 2.1ab

M#	Type	Const.		M#	Type	Const.		M#	Type	Const.
M1	**NB**	**TAU**		M38				M75		
M2				M39				M76		
M3				M40				M77		
M4				M41				M78		
M5				M42				M79		
M6				M43				M80		
M7				M44				M81		
M8				M45				M82		
M9				M46				M83		
M10				M47				M84		
M11				M48				M85		
M12				M49				M86		
M13				M50				M87		
M14				M51				M88		
M15				M52				M89		
M16				M53				M90		
M17				M54				M91		
M18				M55				M92		
M19				M56				M93		
M20				M57				M94		
M21				M58				M95		
M22				M59				M96		
M23				M60				M97		
M24				M61				M98		
M25				M62				M99		
M26				M63				M100		

M27				M64				M101		
M28				M65				M102		
M29				M66				M103		
M30				M67				M104		
M31				M68				M105		
M32				M69				M106		
M33				M70				M107		
M34				M71				M108		
M35				M72				M109		
M36				M73				M110		
M37				M74						

Table 2.1c

Name	M#
Andromeda Galaxy	
Beehive Cluster	
Black Eye Galaxy	
Bode's Galaxy	
Butterfly Cluster	
Cat's Eye Galaxy	
Cigar Galaxy	
Crab Nebula	
Dumbbell Nebula	

Name	M#
Eagle Nebula	
Lagoon Nebula	
Le Gentil	
Omega Nebula	
Orion Nebula	
Owl Nebula	
Pinwheel Galaxy	
Pleiades	

Name	M#
Ptolemy Cluster	
Ring Nebula	
Sombrero Galaxy	
Spindle Galaxy	
Triangulum Galaxy	
Trifid Nebula	
Whirlpool Galaxy	
Wild Duck Cluster	

Table 2.2

Rank	Proper name (IAU)	Bayer designation	m_v	Distance (ly)	Binary or Multiple	Variable	Cool
2	—	α Cen	-0.27	4.34			
75	—	α Lup	2.30	550			
76	—	ε Cen	2.30	380			
79	—	η Cen	2.35	310			
82	—	κ Sco	2.39	460			
89	—	γ Cas	2.47	610			
30	—	γ Vel	1.72	840			
9	Achernar	α Eri	0.46	139			
92	Acrab	β Sco	2.50	404			
13	Acrux	α Cru	0.76	320			
22	Adhara	ε CMa	1.50	430			
14	Aldebaran	α Tau	0.86	65			
57	Algieba	γ Leo	2.08	130			
61	Algol	β Per	2.12	93			
43	Alhena	γ Gem	1.92	100			
33	Alioth	ε UMa	1.77	83			
91	Aljanah	ε Cyg	2.48	72			
40	Alkaid	η UMa	1.86	100			
72	Almach	γ And	2.26	350			
31	Alnair	α Gru	1.74	100			
29	Alnilam	ε Ori	1.69	2,000			
32	Alnitak	ζ Ori	1.77	1,300			
48	Alphard	α Hya	2.00	180			
66	Alphecca	α CrB	2.23	75			
55	Alpheratz	α And	2.06	97			
45	Alsephina	δ Vel	1.96	80			
12	Altair	α Aql	0.76	17			
87	Aludra	η CMa	2.45	2,000			
81	Ankaa	α Phe	2.38	77			
15	Antares	α Sco	0.96	550			

3	Arcturus	α Boo	-0.05	37			
64	Aspidiske	ι Car	2.21	690			
42	Atria	α TrA	1.91	390			
39	Avior	ε Car	1.86	600			
26	Bellatrix	γ Ori	1.64	250			
10	Betelgeuse	α Ori	0.50	550			
4	Canopus	α Car	0.00	310			
6	Capella	α Aur	0.08	43			
73	Caph	β Cas	2.28	54			
24	Castor	α Gem	1.62	51			
19	Deneb	α Cyg	1.25	2,600			
60	Denebola	β Leo	2.11	36			
50	Diphda	β Cet	2.02	96			
77	Dschubba	δ Sco	2.31	400			
34	Dubhe	α UMa	1.79	120			
27	Elnath	β Tau	1.65	130			
69	Eltanin	γ Dra	2.23	150			
83	Enif	ε Peg	2.40	670			
18	Fomalhaut	α PsA	1.16	25			
25	Gacrux	γ Cru	1.64	89			
11	Hadar	β Cen	0.61	390			
49	Hamal	α Ari	2.00	66			
74	Izar	ε Boo	2.29	202			
38	Kaus Australis	ε Sgr	1.85	140			
58	Kochab	β UMi	2.08	130			
78	Larawag	ε Sco	2.31	65			
90	Markab	α Peg	2.48	140			
88	Markeb	κ Vel	2.46	540			
41	Menkalinan	β Aur	1.90	80			
54	Menkent	θ Cen	2.06	59			
80	Merak	β UMa	2.37	79			
28	Miaplacidus	β Car	1.69	110			

20	Mimosa	β Cru	1.25	280			
67	Mintaka	δ Ori	2.23	900			
53	Mirach	β And	2.05	200			
35	Mirfak	α Per	1.80	510			
46	Mirzam	β CMa	1.98	500			
51	Mizar	ζ UMa	2.04	83			
63	Muhlifain	γ Cen	2.17	130			
71	Naos	ζ Pup	2.25	1,080			
52	Nunki	σ Sgr	2.05	230			
44	Peacock	α Pav	1.94	180			
86	Phecda	γ UMa	2.44	84			
47	Polaris	α UMi	1.98	430			
17	Pollux	β Gem	1.14	34			
8	Procyon	α CMi	0.34	11			
56	Rasalhague	α Oph	2.07	47			
21	Regulus	α Leo	1.39	79			
7	Rigel	β Ori	0.13	860			
85	Sabik	η Oph	2.43	88			
68	Sadr	γ Cyg	2.23	1,500			
59	Saiph	κ Ori	2.09	720			
37	Sargas	θ Sco	1.84	330			
84	Scheat	β Peg	2.42	200			
70	Schedar	α Cas	2.24	230			
23	Shaula	λ Sco	1.62	570			
1	Sirius	α CMa	-1.46	8.6			
16	Spica	α Vir	0.97	250			
65	Suhail	λ Vel	2.21	570			
62	Tiaki	β Gru	2.15	170			
5	Vega	α Lyr	0.03	25			
36	Wezen	δ CMa	1.82	1,800			

Problem 9.4 A sky chart from Nepal

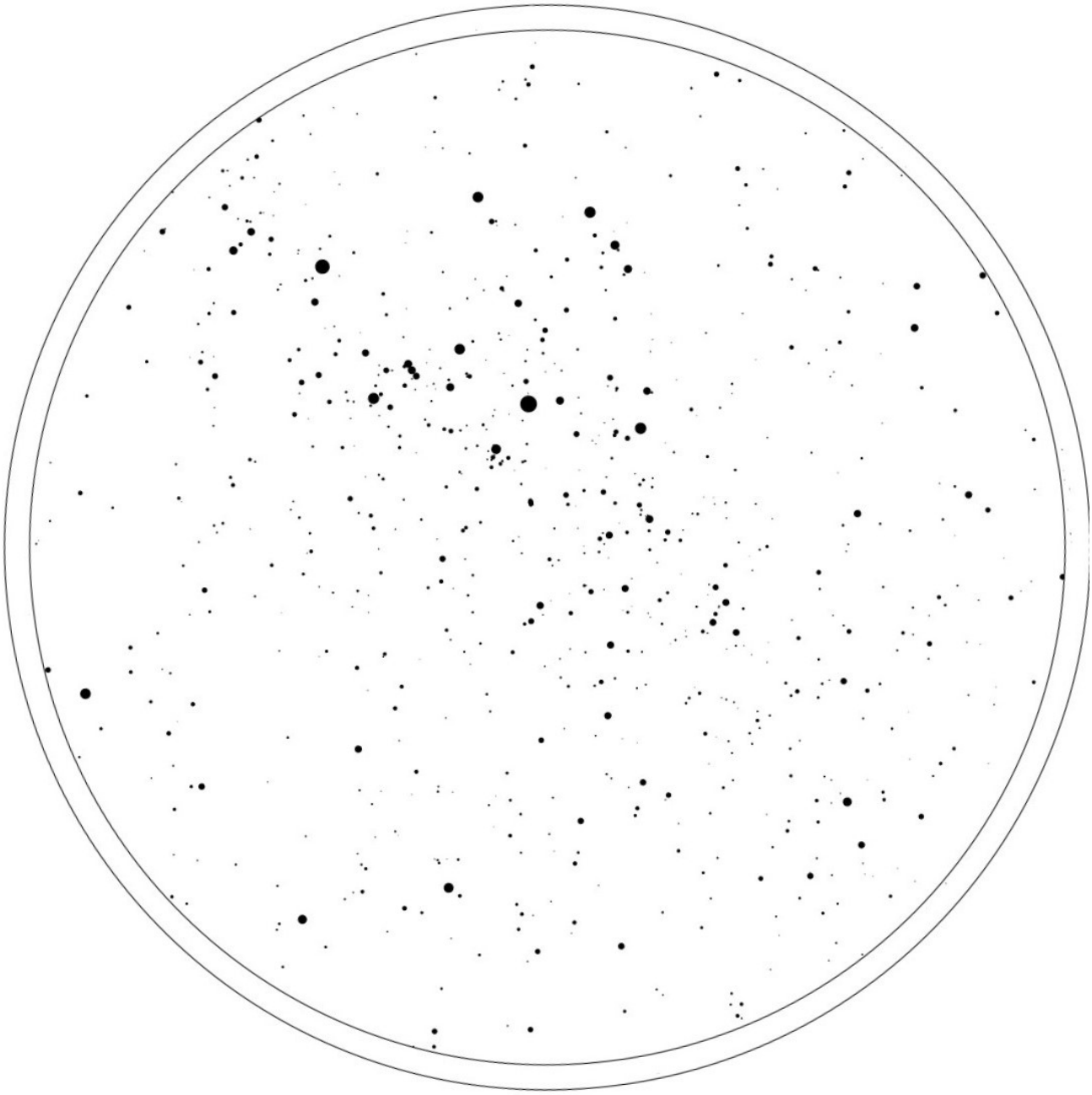
2024 | Kathmandu, Nepal | Sky map | OC1 | 45 points

Solution p. 148

OC1 (45 points)

The map in the answersheet represents the sky in azimuthal equidistant projection. The Zenith is at the center of the chart.

1. Estimate the latitude of the location from where this sky map is captured. [4 points]
2. Label the cardinal points(N, E, S, W) on the map and mark the three planets from the Solar System in the sky map with a small square around each of them. [7 points]
3. Draw on the map with continuous lines the celestial equator (EQ), the ecliptic (EC), and the local meridian (ME) and label them. [4.5 points]
4. Trace 10 constellations in the sky map and label them as C1, C2,....,C10 and write the corresponding latin name of the constellations or IAU code in the table below. [10 points]
5. Identify and circle the 4 brightest stars in the given sky map. Label the star from S1 – the brightest, and continue with the fainter ones till number S4 for the faintest. Fill in the following table the Bayer designation or the official IAU name of the four identified stars. [6 points]
6. Mark (with a small circle) positions of the following objects and label them. [10.5 points]
    - M42
    - M81
    - M41
    - M31
    - γ Ori
    - α Per
    - α Cyg
7. Estimate the equatorial coordinates of the star Aldebaran (α Tau). [3 points]



<table>
<tr><td>1.</td><td>$\phi_{location} =$</td></tr>
<tr><td>2.</td><td>On the chart.</td></tr>
<tr><td>3.</td><td>On the chart.</td></tr>
<tr><td>4.</td><td>
<table>
<tr><th>Number</th><th>Constellation Name</th></tr>
<tr><td>C1</td><td></td></tr>
<tr><td>C2</td><td></td></tr>
<tr><td>C3</td><td></td></tr>
<tr><td>C4</td><td></td></tr>
<tr><td>C5</td><td></td></tr>
<tr><td>C6</td><td></td></tr>
<tr><td>C7</td><td></td></tr>
<tr><td>C8</td><td></td></tr>
<tr><td>C9</td><td></td></tr>
<tr><td>C10</td><td></td></tr>
</table>
</td></tr>
<tr><td>5.</td><td>
<table>
<tr><th>Code</th><th>Star</th></tr>
<tr><td>S1</td><td></td></tr>
<tr><td>S2</td><td></td></tr>
<tr><td>S3</td><td></td></tr>
<tr><td>S4</td><td></td></tr>
</table>
</td></tr>
<tr><td>6.</td><td>On the chart.</td></tr>
<tr><td>7.</td><td>$\alpha_{Aldebaran} =$
$\delta_{Aldebaran} =$</td></tr>
</table>

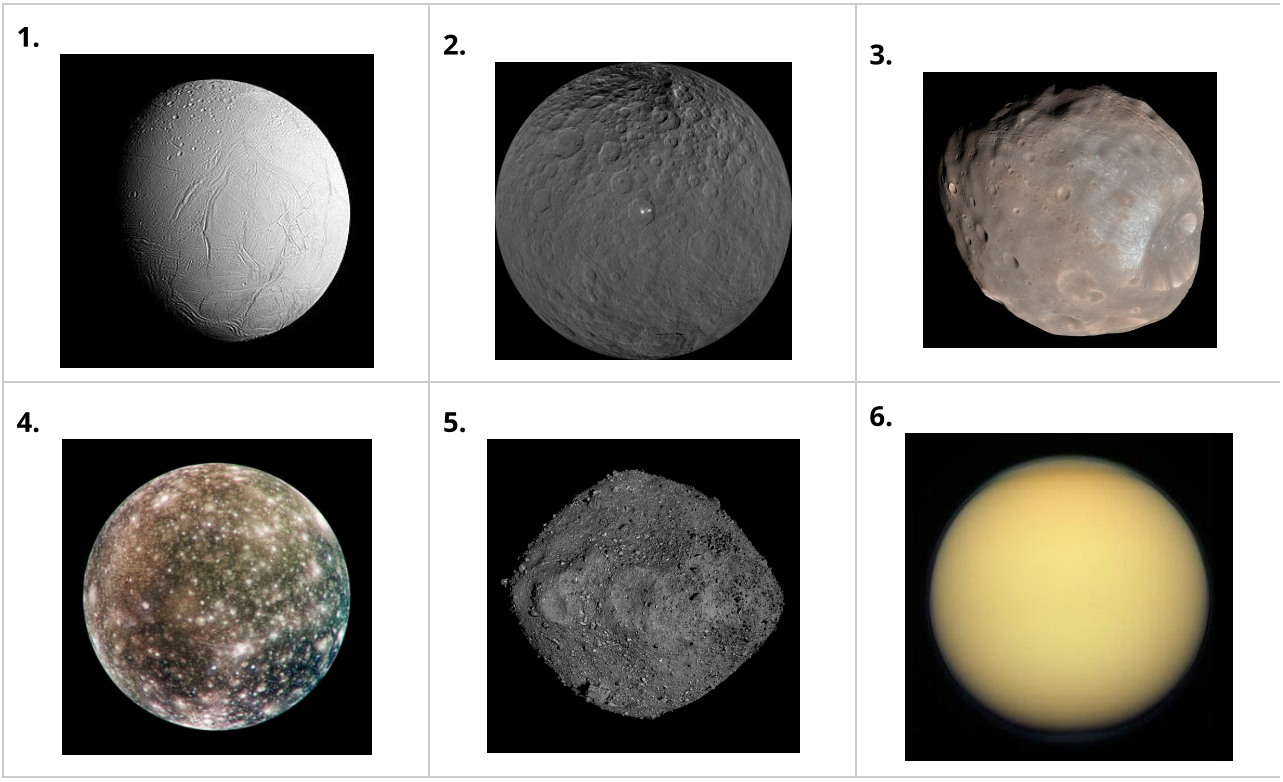
Problem 9.5 Classifying Solar System objects

2024 | Kathmandu, Nepal | Sky map | OC2 | 9 points

Solution p. 153

OC2 (9 points)

In the table given in the answersheet, fill in the numbers of the pictures as per their category.



OC2

Asteroids	Dwarf Planets	Satellites

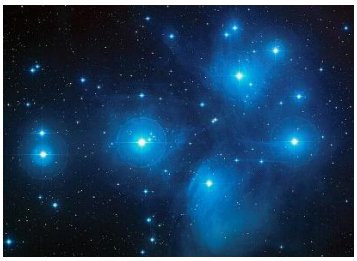
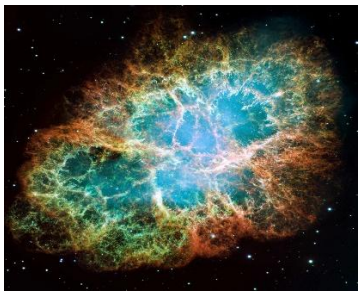
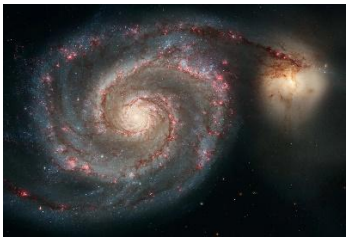
Problem 9.6 Identifying deep-sky objects

2024 | Kathmandu, Nepal | Sky map | OC3 | 18 points

Solution p. 154

OC3 (18 points)

Complete the table given in the answersheet regarding the objects below:

Object	Image	Object	Image
O1		O4	
O2		O5	
O3		O6	

OC3

Object	Messier Number (e.g. M32)	Constellation (e.g. ORI or Orion)
O1		
O2		
O3		
O4		
O5		
O6		

Problem 9.7 Know Your Sky

2025 | Piatra Neamț, Romania | Sky map | OM1 | 40 points

Solution p. 156

OM 1 – Know your sky (40 Points)

Use the sky map ”Map-OM1” to answer the questions below. The projection type of the map is stereographic and planet/moon are present on the map.

OM 1.1 – Marking Stars on the Map (10 p)

Mark all the stars listed in Table 1 with a square (□) drawn around each star on the provided Map-OM1. Label every marked star with its corresponding object number.

Star no.	Star name	Star no.	Star name
1	α Ari (Hamal)	6	β CMi (Gomeisa)
2	δ Cyg (Fawaris)	7	α Lep (Arneb)
3	β Dra (Rastaban)	8	δ UMa (Megrez)
4	α Cet (Menkar)	9	β Tau (Elnath)
5	δ Eri (Rana)	10	α And (Alpheratz)

Table 1: List of stars with their names

OM 1.2 – Messier Objects (8 p)

Using the provided plus signs (+) on the map and label them corresponding to the following 8 Messier objects: M15, M31, M34, M35, M39, M42, M44, M45 from the Messier Catalog.

OM 1.3 – Meridian and constellations (10 p)

The given map is centered at Zenith. Draw the meridian and then draw with star-line patterns the constellations along/near the meridian and note the IAU abbreviations on the sheet.

OM 1.4 – Celestial equator (2 p)

Draw the celestial equator on Map-OM1. Mark it as "EQ".

OM 1.5 – Cardinal Points (2 p)

Mark the cardinal points as N, S, W, and E on Map-OM1.

OM 1.6 – Ecliptic plane (2 p)

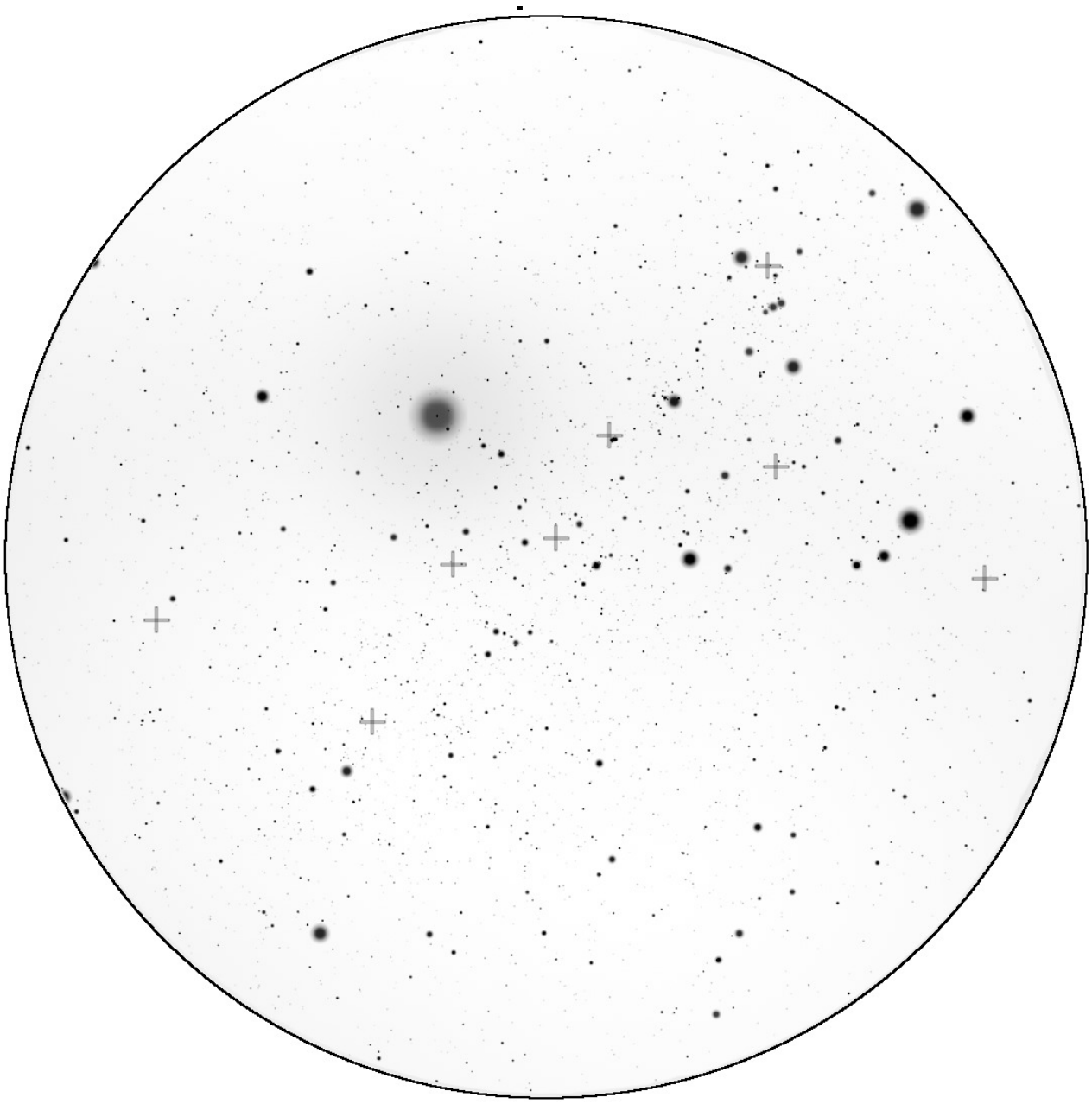
Draw the ecliptic on the Map-OM1 and label it with “EC”.

OM 1.7 – Galactic plane (4 p)

Draw the galactic equator on the Map-OM1 and label it as “GE”. Mark either the galactic center (GC) or anticenter (GA), whichever is visible on the map.

OM 1.8 – Circumpolar Circle (2 p)

Draw the circumpolar circle on the Map-OM1 and label it as “CC”.



Problem 9.8 Chart and Time

2025 | Piatra Neamț, Romania | Sky map | OM2 | 10 points

Solution p. 157

OM 2 – Chart and time (10 Points)

Use the OM2 – Chart and Time page to answer the following questions. The given map is centered on the Zenith and shows two constellations, C1 and C2, together with the horizon and a coordinate grid. The corresponding answer sheet is the Blind Sky Map – OM2 Chart and Time.

OM 2.1 – Constellation identi cation (2p)

Identify the C1 and C2 constellations and write their latin names or IAU abbreviation in the table below, map "OM2 - Chart and time".

OM 2.2 – Mark the stars (2p)

Draw an arrow each indicating the α stars of the C1 and C2 constellations.

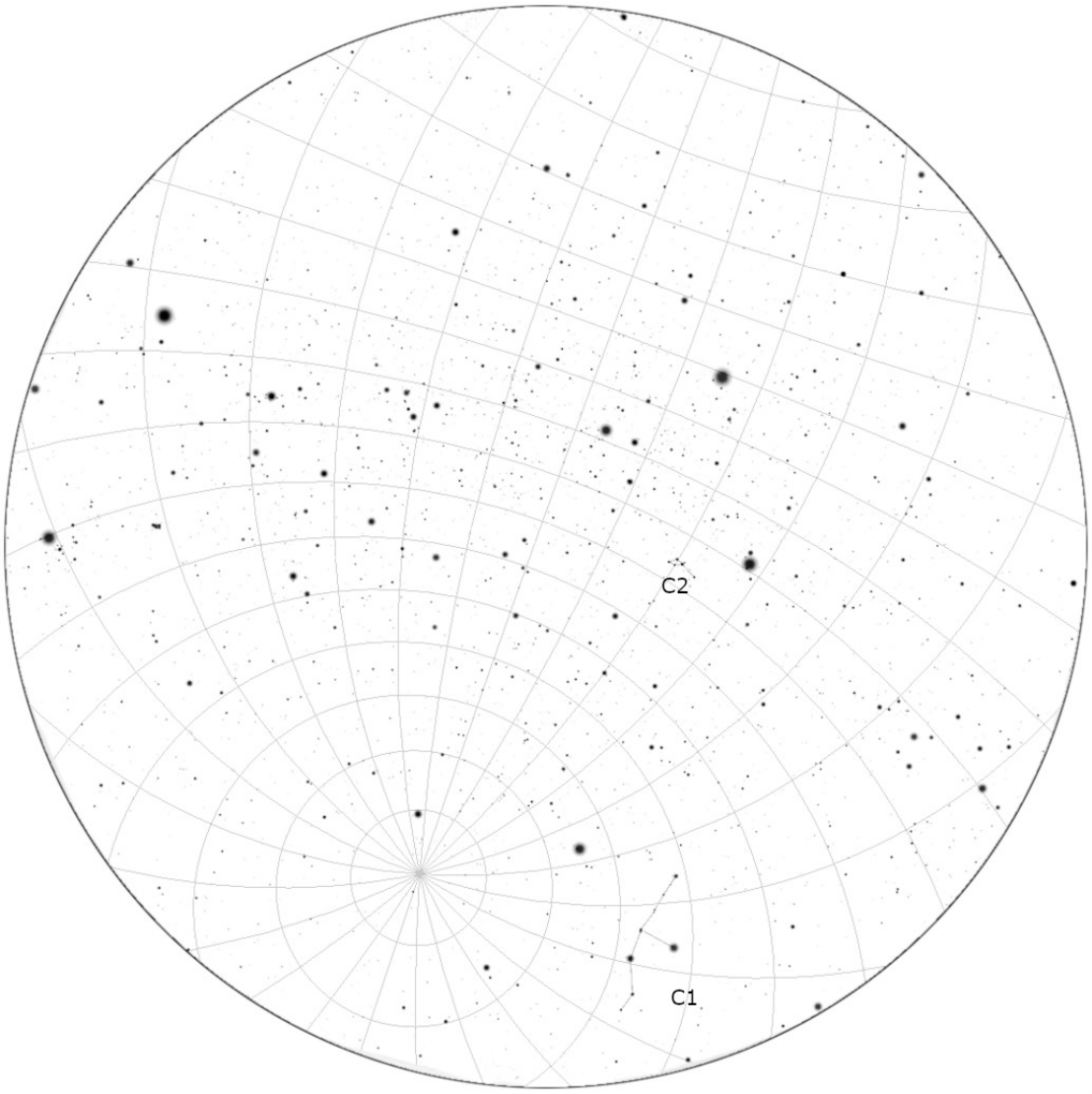
OM 2.3 – Autumnal and vernal equinox (2 p)

One of the equinox points, either the vernal equinox (VE) or autumnal equinox (AE), is present on the map. Mark it with an X and label it with VE or AE.

OM 2.4 – Grid identi cation (4 p)

Identify which coordinate grid is present on the map. Place an X mark in the table below map - OM2 - Chart and time.

OM2 - Chart and time



2.1	Name of Constellation (Latin name or IAU abbreviation)
C1	
C2	

2.4	ecliptic	azimuthal	galactic	equatorial
Coordinate grid				

Problem 9.9 Missing Stars

2025 | Piatra Neamț, Romania | Sky map | OM3 | 10 points

Solution p. 158

OM 3 - Missing stars (10 Points)

Use the OM3 – Missing Stars map to answer the following questions. There are four stars missing (with visual magnitudes ≤ 2.5).

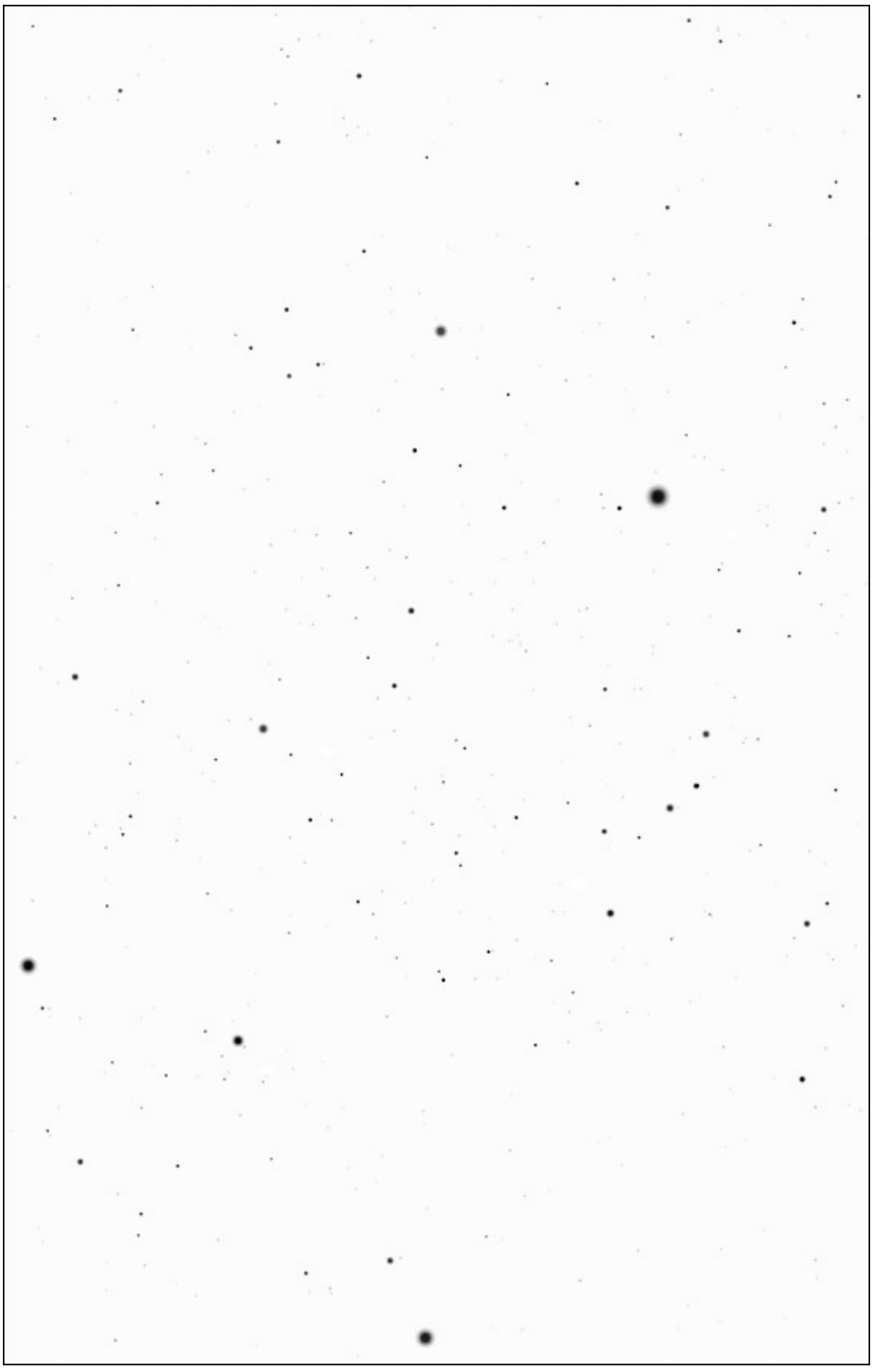
OM 3.1 (4 p)

Find which stars are missing and mark them on the map using a plus sign (+) and label them S1, S2, S3, S4.

OM 3.2 (6 p)

For each missing star, write on the answer sheet: • The IAU star name or Bayer designation. • Whether the object is a double or multiple star: write Y if it is, or N if it is not. Record your answers in the table below Map OM3 - Missing stars.

OM3 - Missing stars



Star 1	
Name	
Binary	
Star 2	
Name	
Binary	
Star 3	
Name	
Binary	
Star 4	
Name	
Binary	

Telescope

Problem 9.10 Sun observations

2023 | Volos, Greece | Telescope | Q15 | 45 points

Solution p. 159

Day Observation

Instructions:

1. You have 20 minutes in total to solve the questions.
2. **Avoid any direct viewing of the Sun.** The telescope you are going to use will be already equipped with a solar filter and the finder scope will have been removed or covered.
3. Write your answers on the separate answer sheet provided to you. Answers / calculations on the question paper will not be evaluated.

Sun Observations (45 points)

a) Carefully point the telescope towards the Sun and centre it in the field of view. Then, ask your examiner to confirm you targeted the Sun successfully and give them your answer-sheet to take the corresponding note. If you cannot target the Sun, the examiner will target it for you, but you will lose all points of this question. **(5 points)**

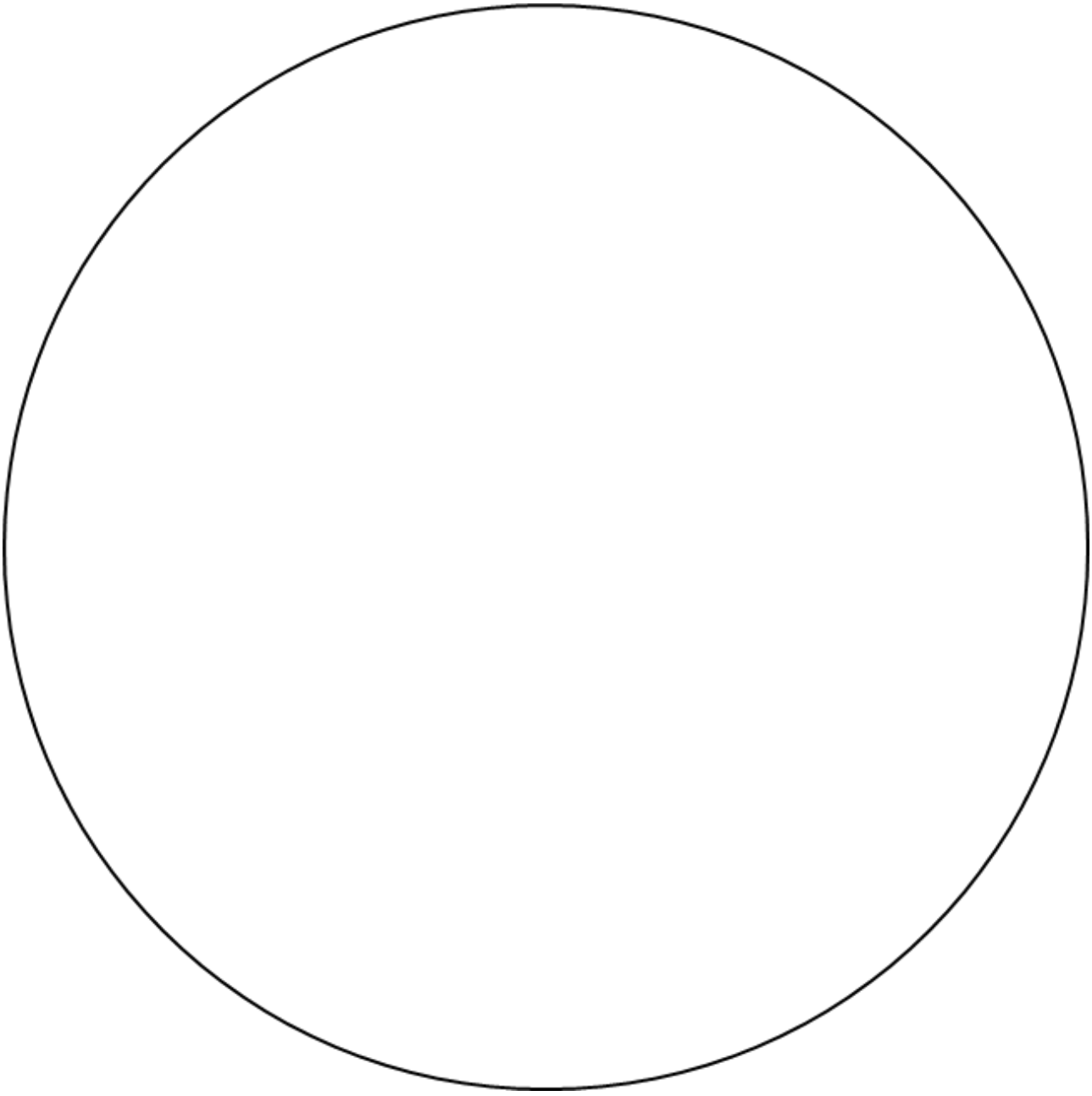
b) Within the disk provided on the answer sheet, draw all the sunspots you can see through the telescope. **(10 points)**

c) Measure the declination of the Sun using the telescope. **(5 points)**

d) Estimate the right ascension of the Sun (in hours and minutes). **(10 points)**

e) What is going to be the approximate declination of the Sun in 3 months? **(5 points)**

f) Call the examiner and demonstrate to them how you can “scan” with your telescope the celestial equator (make the telescope pass through all the equatorial constellations). Once you are done, give the examiner your answer-sheet to take the corresponding note. **(10 points)**



Problem 9.11 Cloudy sky observations

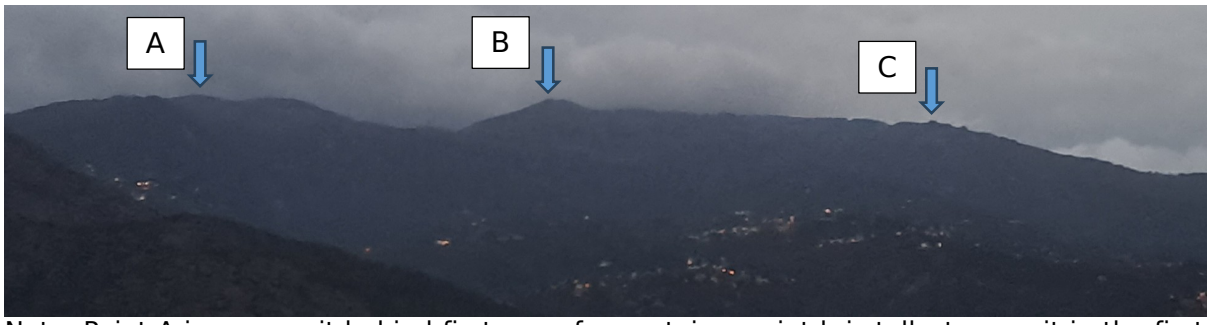
2023 | Volos, Greece | Telescope | Q16 | 45 points

Solution p. 159

Cloudy Sky Observation

Instructions:

1. You have a total of 30 minutes to solve the questions.
2. The telescope and finder tube are already aligned. Do not try to change the alignment. Both RA and declination dials are already calibrated.
3. Write your answers on the separate answer sheet provided to you. Answers / calculations on the question paper will not be evaluated.
4. The north-south line is parallel to the balcony railing, north is on your left.



Note: Point A is a summit behind first row of mountains, point b is tallest summit in the first row of mountains and point C is a square building at the right end of mountain.

A) Point the telescope to the points that are indicated with arrows in the image above. Use the dials on the telescope to determine the equatorial coordinates of these three points. **(15 points)**

B) Find the angular distance between points A and C. **(10 points)**

C) Call the examiner and demonstrate to them how you can “scan” with your telescope the celestial equator (make the telescope pass through all the equatorial constellations). Once you are done, give the examiner your answer-sheet to take the corresponding note. **(10 points)**

D) Point your telescope to the direction with equatorial coordinates 13h 20min and 2° S. Then, ask your examiner to confirm you targeted the right coordinates. **(10 points)**

Problem 9.12 The Moon

2024 | Kathmandu, Nepal | Telescope | OT1 | 32 points

Solution p. 160

OT1 (The Moon) (32 points)

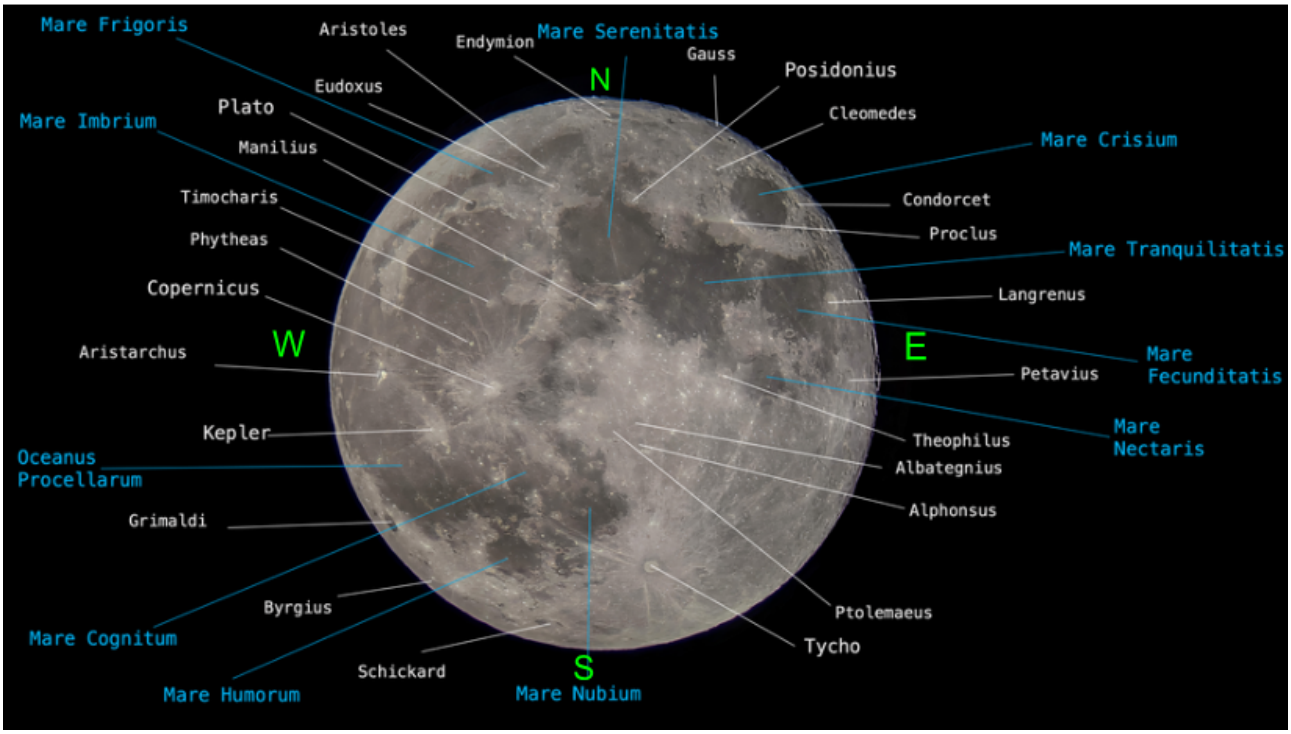
You have 2 minutes 30 seconds to point and focus the telescope to the Moon. If you are not able to point it within 2 minutes 30 seconds ask the supervisor to point it for you. (6 points)

A zoomed-in view of the Moon will be shown for 90 seconds and a zoomed-out view of the Moon will be shown for next 60 seconds.

On your answer sheet:

1. Draw the Moon, including the lunar terminator. (4 points)
2. Label the cardinal points on the Moon drawn. (4 points)
3. With the help of the moon chart provided on the next page, identify three visible craters and three maria and mark their positions on your drawing. (12 points)

Based on your observations, you have to complete a table of information about the moon. (6 points)



Age of the Moon (No. of days since new moon)	
Hemisphere from where the Moon is being watched	
Official IAU name of the constellation in which the Moon is located	

Problem 9.13 Jupiter and its moons

2024 | Kathmandu, Nepal | Telescope | OT2 | 16 points

Solution p. 161

OT2 (16 points)

An animation of a planet and its moons will be displayed on the screen. The animation corresponds to a single location over a 3-day period.

1. Identify the planet shown in the images by ticking the correct option.
2. Draw initial and final positions of the moons on your answer sheet and label the moons as A, B, C, and D. E.g. if one moon is labeled as A in first drawing, the same should be labeled as A in second drawing.
3. Arrange the labels (i.e. A, B, C, D) of the observed moons according to their size from largest to the smallest.

OT2

1.	☐ Mercury ☐ Venus ☐ Mars ☐ Jupiter ☐ Saturn ☐ Uranus ☐ Neptune	
2.	**Drawing 1**	**Drawing 2**
3.		

Problem 9.14 Telescope Adjustment Knobs

2025 | Piatra Neamț, Romania | Telescope | OT1 | 4 points

Solution p. 162

Instructions:

You will get 5 minutes before the start of the examination to read the instructions and questions.

You will be informed regarding the start of the round by the supervisors.

The total time that you will get with the telescope is 20 minutes.

Supervisors can only assist you in locking/unlocking the telescope axis - at request.

All telescopes are equipped with a 25 mm eyepiece.

The telescope scales are calibrated so that the hour angle axis represents right ascension.

Please be careful when using the telescope. Handle it with care, do not push it or stumble upon the tripod legs.

After that, you will be taken out of the observation hall.

OT 1 (4 points)

On your telescope, the adjustment knobs are marked with 1 and 2. Mark with an X on your answer sheet, the answer to the following questions:

a. Which coordinate does knob No. 1 refer to?
b. Which coordinate does knob No. 2 refer to?

Problem 9.15 Coordinate Identification

2025 | Piatra Neamț, Romania | Telescope | OT2 | 9 points

Solution p. 163

OT 2 (9 points)

On the wall opposite to your position, you will find a sign numbered **1**. Point the telescope towards the specified sign and you will see a representation. For the star marked with the symbol *****, answer the following (mark with X on the answer sheet):

a. What is the name of the coordinate marked with letter **A**?
b. What is the name of the coordinate marked with letter **B**?
c. What is the name of the coordinate marked with letter **C**?

Problem 9.16 Constellation through the Eyepiece

2025 | Piatra Neamț, Romania | Telescope | OT3 | 20 points

Solution p. 163

OT 3 (20 points)

On the wall opposite to your position, you will find a sign named **O**. Point the telescope towards the specified sign, where you should see a constellation. Answer the following:

a. Draw the constellation seen through the eyepiece on your answer sheet;
b. Specify the IAU name or IAU abbreviation of the constellation;
c. Mark the Alpha star with a circle on your drawing;
d. Write the IAU name of the Alpha star;
e. Between stars Gamma (γ) and Beta (β) of this constellation, there is a Messier object. Specify its IAU name OR the Messier catalogue number;
f. Specify the type of the object (Mark with X on your answer sheet);
g. On your drawing, mark the position of this Messier object, with a triangle (Δ).

Problem 9.17 Objects and Telescope Coordinates

2025 | Piatra Neamț, Romania | Telescope | OT4 | 27 points

Solution p. 165

OT 4 (27 points)

On the wall opposite to you, you will find signs named **R**, **N** and **Y**. Point the telescope towards each specific sign and, for each sign, answer the following:

a. Write the IAU Name or Messier Catalogue Number (if applicable) of the celestial Object found on the sign;
b. Right Ascension, read from the telescope for the sign;
c. Declination, read from the telescope for the sign.

Night observation

Problem 9.18 Observations with the Naked Eye

2022 | Câmpulung Moldovenesc, Romania | Night observation | Q14

Solution p. 168

Working time: 30 minutes of which 5 minutes accommodation with the subjects.

NIGHT OBSERVATION ROUND

You have at your disposal: Newtonian telescope with EQ5 mount, with the diameter of the main mirror of 200 mm, focal length of 1000 mm; 25 mm eyepiece and 10 mm eyepiece.

A. Observations with the naked eye.

1. Estimate the sidereal time at the start of the test, with an error of ±10 minutes.

Answer	

2. Find the angular distance with an error of ± 3 degrees between:
(αPer_ Mirfak and βAnd_ Mirach)
(βCas_Caph and γCyg_Sadar)

Answer(αPer_ Mirfak and βAnd_ Mirach)	
Answer (βCas_Caph and γCyg_Sadar)	

3. Specify two constellations located at the meridian, specify the type of culmination and the alpha star.

	Constellation at meridian	Type of culmination	Alpha star
Answer			
Answer			

4. Estimate with an error of ±10 minutes the right ascension of the planet Jupiter.

Answer	

5. Estimate, for the star βARI_Sheratan, the apparent magnitude (error of ±0.4), declination (error of ±3 degrees) and right ascension (±10 minutes)

	apparent magnitude (±0.4 error)	declination (±3 degree error)	right ascension (±10 minutes)
Answer			

Problem 9.19 Observations Using the Astronomical Telescope

2022 | Câmpulung Moldovenesc, Romania | Night observation | Q15

Solution p. 168

Working time: 30 minutes of which 5 minutes accommodation with the subjects.

NIGHT OBSERVATION ROUND

You have at your disposal: Newtonian telescope with EQ5 mount, with the diameter of the main mirror of 200 mm, focal length of 1000 mm; 25 mm eyepiece and 10 mm eyepiece.

B. Observations using the astronomical telescope.

1. Point the telescope towards the beta star βCyg. Specify the type of star.

	Correct positioning		Star type
Answer	Yes	Not	

2. Point the telescope towards Messier M45 (which constellation it is in and what it is).

	Correct positioning		The constellation in which it is located	Type
Answer	Yes	Not		

3. Point the telescope towards the planet Jupiter, using the eyepiece with the focal length 25 mm. Make a drawing of what you observe through the telescope, and identify the Galilean satellites that can be seen through the eyepiece with a focal length 10 mm.

Answer Correct positioning	Yes	Not
Answer Drawing and identification Galilean satellites		

Solutions to Observational Problems

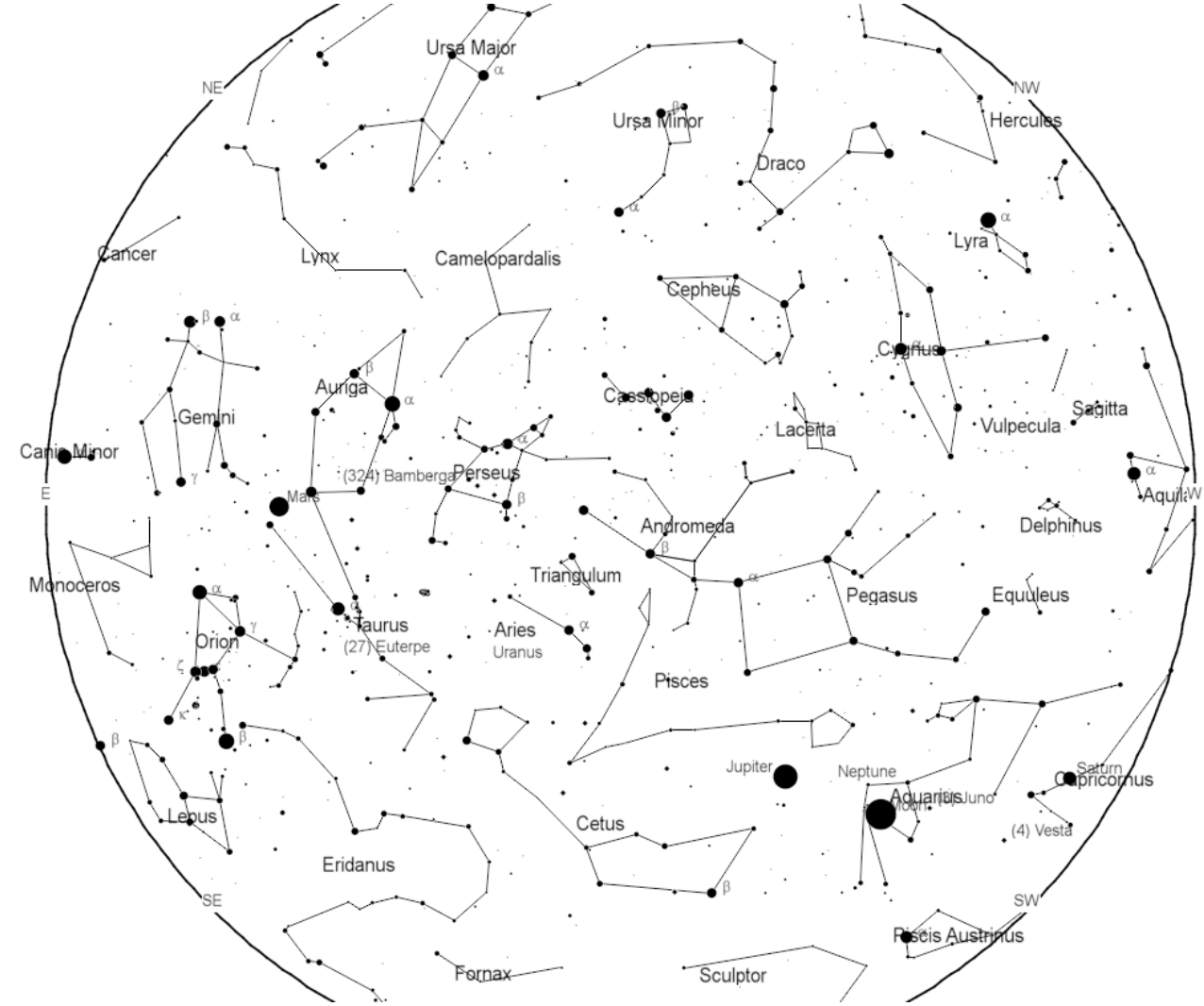
Solution 9.1 Blind Sky Map

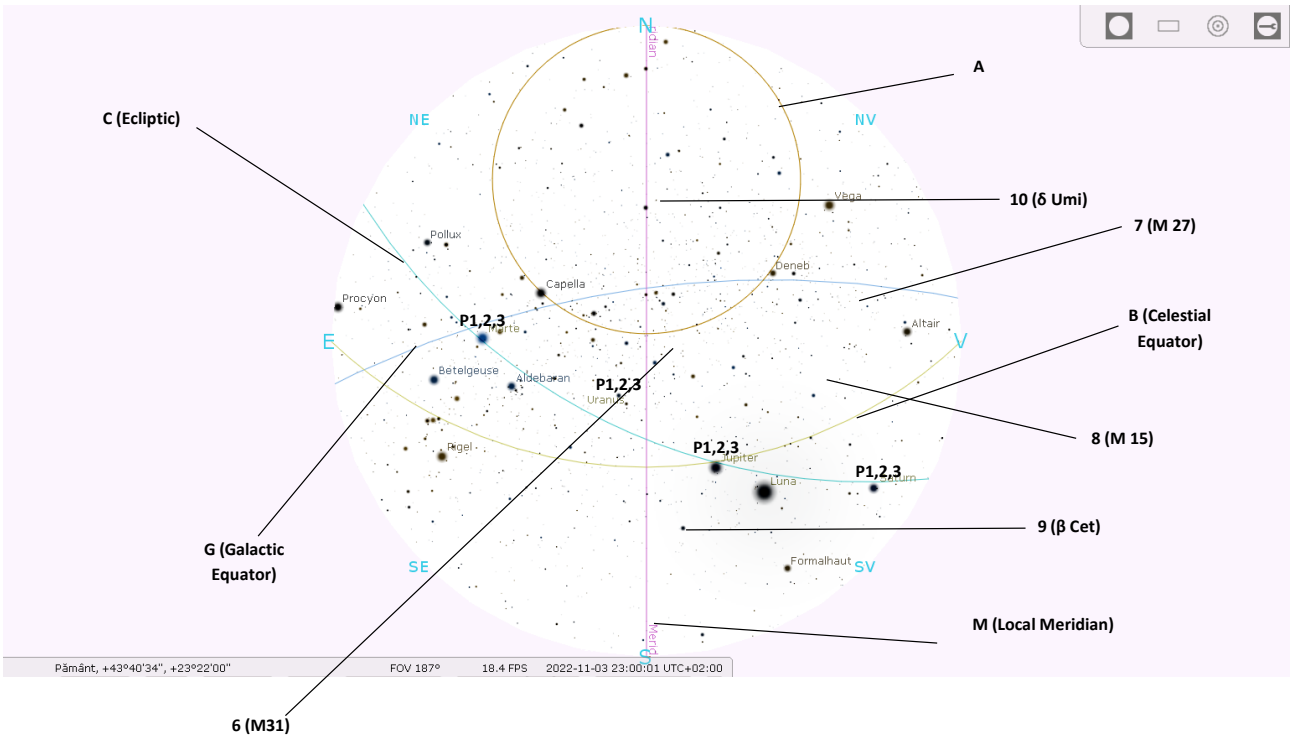
2022 | Câmpulung Moldovenesc, Romania | Sky map | Q13

Problem p. 110

Blind map solution

1. $\varphi = 43^0 40'$
2. See the map below
3. See the map below, any three of the visible planets, in any order
4. See the map below
5. See the map below
6. See the map below
7. s = 1^h 34^m
8. α = $19^h 51^m 53^s$ **or** H=$05^h 41^m 59^s$
   δ= + $8^0 56'$





The separately issued blind sky map and the referenced IAU-code Table 1 are absent from the supplied question PDF. Two annotated charts survive in the official solution and are reproduced there.

Solution 9.2 Navigating the night sky I

2023 | Volos, Greece | Sky map | Q13 | 45 points

Problem p. 112

No official solution is included in the supplied 2023 observation papers. The surviving file contains the question and blank answer sheets.

Solution 9.3 Navigating the night sky II

2023 | Volos, Greece | Sky map | Q14 | 45 points

Problem p. 117

No official solution is included in the supplied 2023 observation papers. The surviving file contains the question and blank answer sheets.

Solution 9.4 A sky chart from Nepal

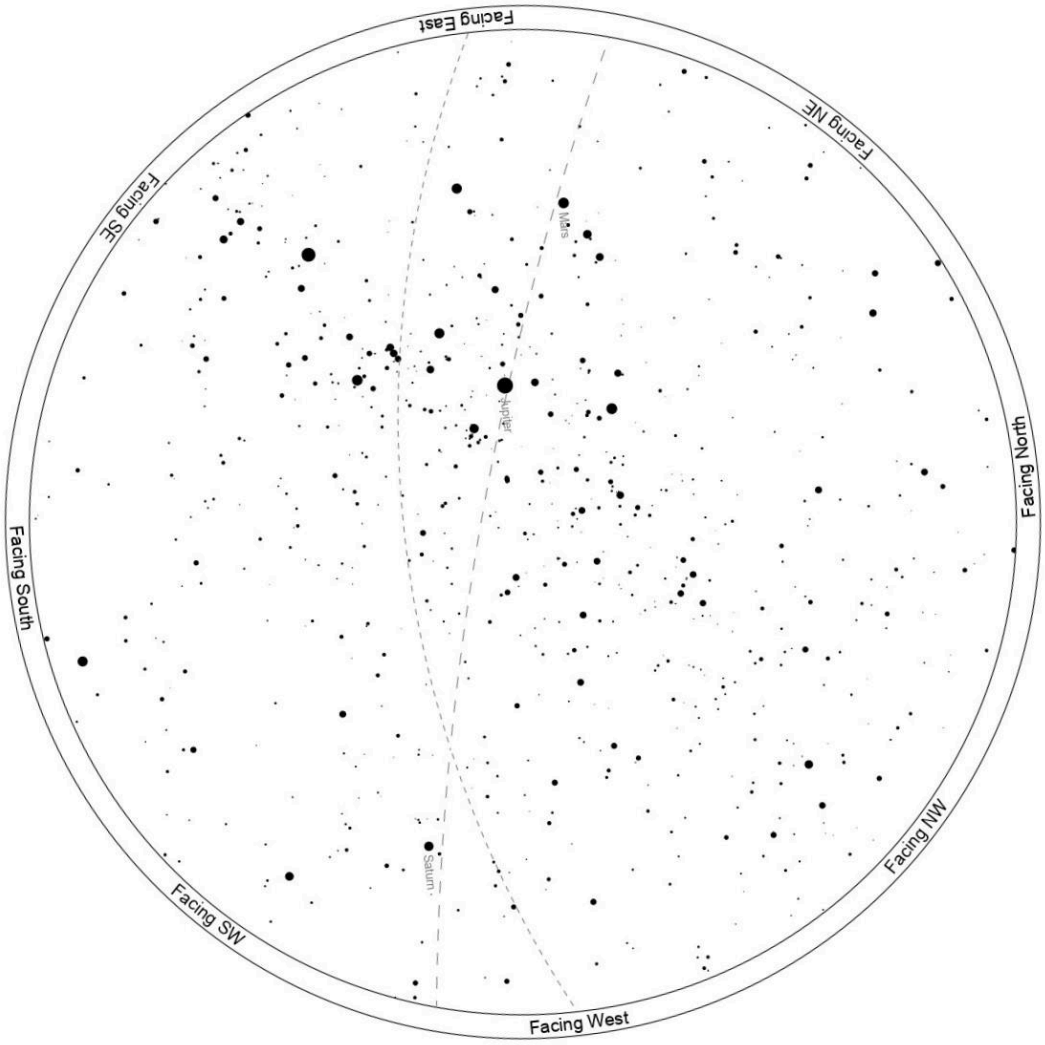
2024 | Kathmandu, Nepal | Sky map | OC1 | 45 points

Problem p. 124

OC1

1. **Answer: 27°7’ N**
   [Answer between 27-28° gains all 4 points, with each degree of inaccuracy 0.5 points are lost. e.g. if a student answer 26° or 29°, they would receive 3.5 points]

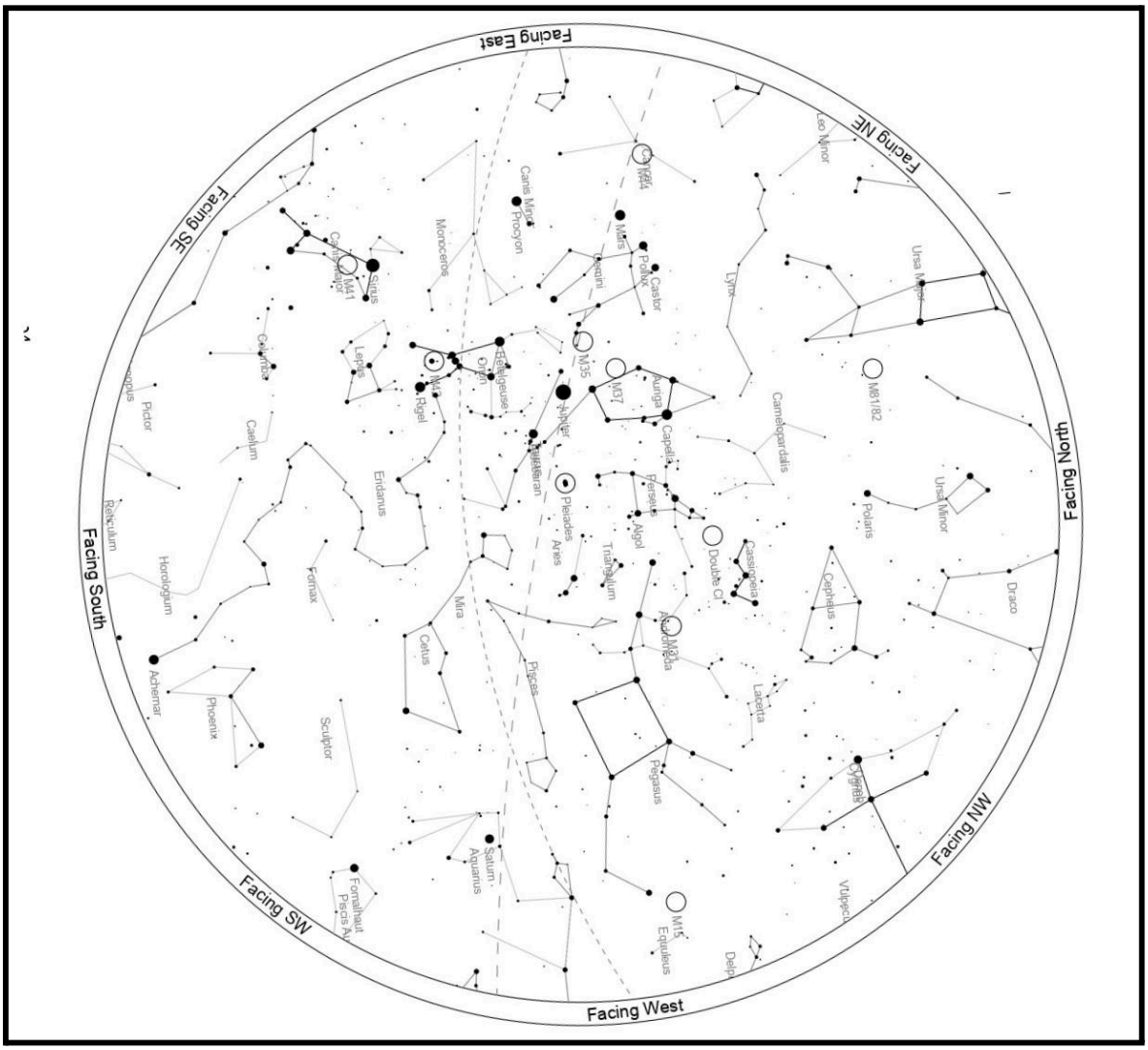
2. Answer:



[With 10° of inaccuracy, 0.5 points is lost for cardinal points, the planets have to be circled correctly to gain full points. Naming of the planet is not needed]

3. Same map as above except for the local meridian to go straightly from North to South.
   [10 degree of inaccuracy will lead to a 0.5 points loss]

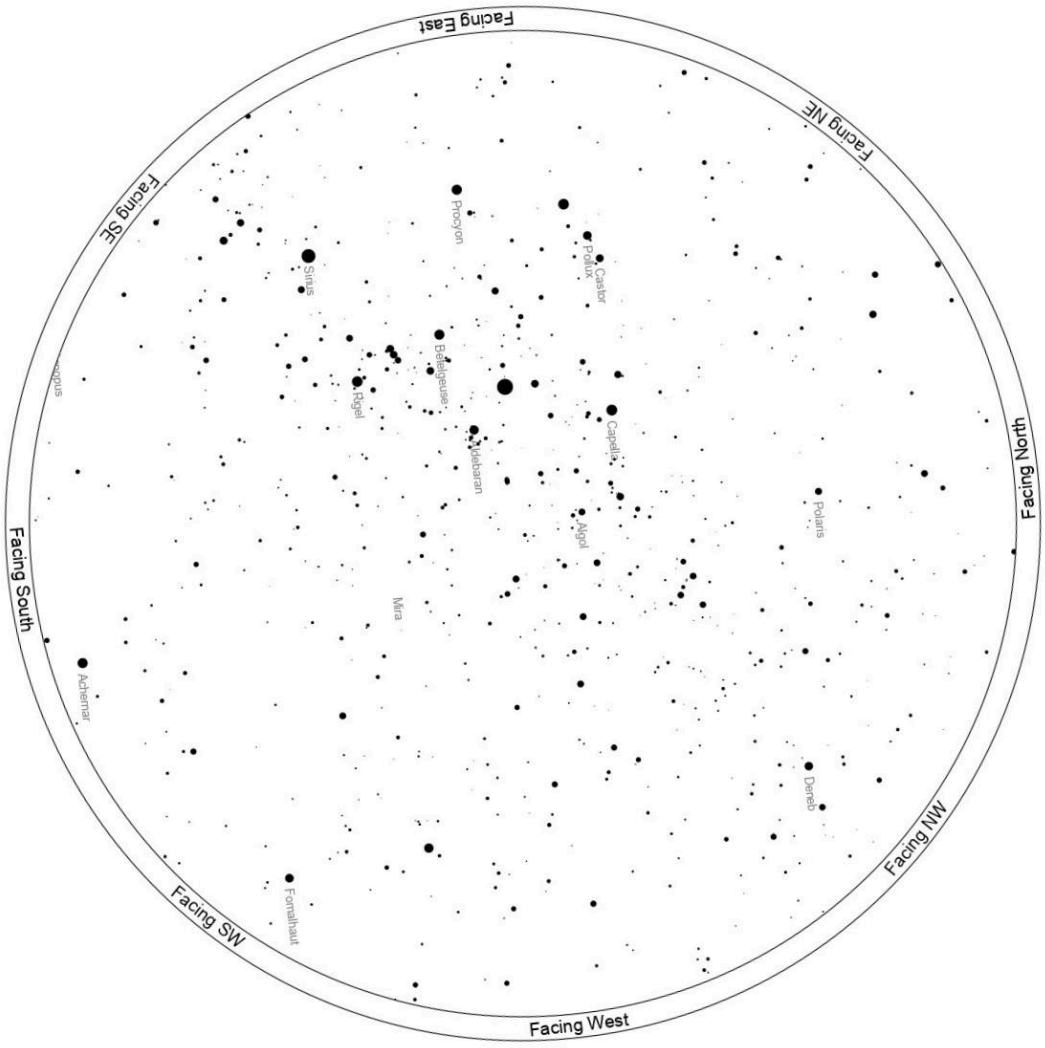
4. Can be any constellations shown in the sky map below.
   [If the tracing composes of at least the brightest stars then the student will earn full points, missing one or more prominent star will lead to a loss of 0.5 point]



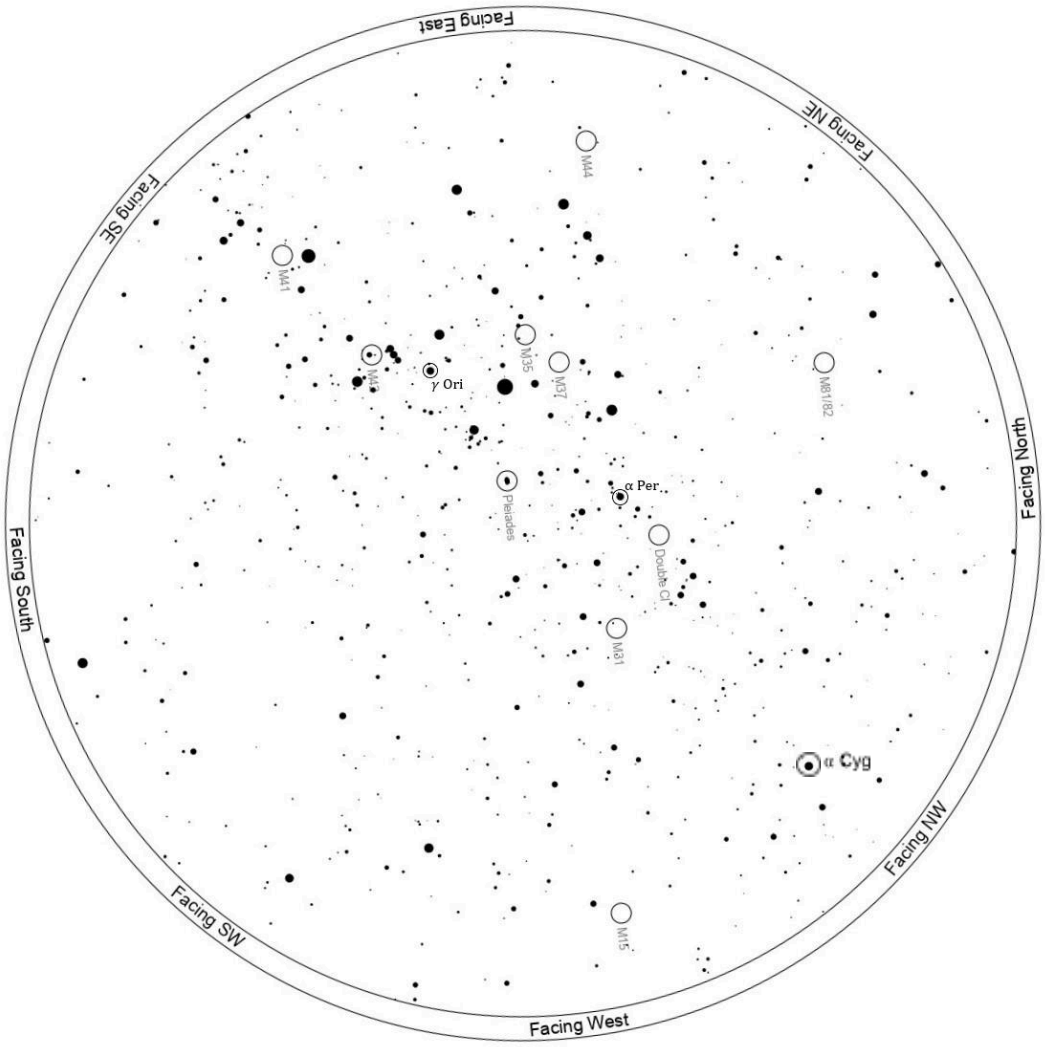
5. Answer:
    A. Alpha Canis Majoris (Sirius)
    B. Alpha Aurigae (Capella)
    C. Beta Orionis (Rigel)
    D. Alpha Canis Minoris (Procyon)

    [0.5 points for each star can be earned by writing the right Bayer designated name or abbreviation of the star, 0.5 points for each star can be earned if the star is in the right order in the list, 0.5 points for each star can be earned if the star is marked correctly in the sky map]

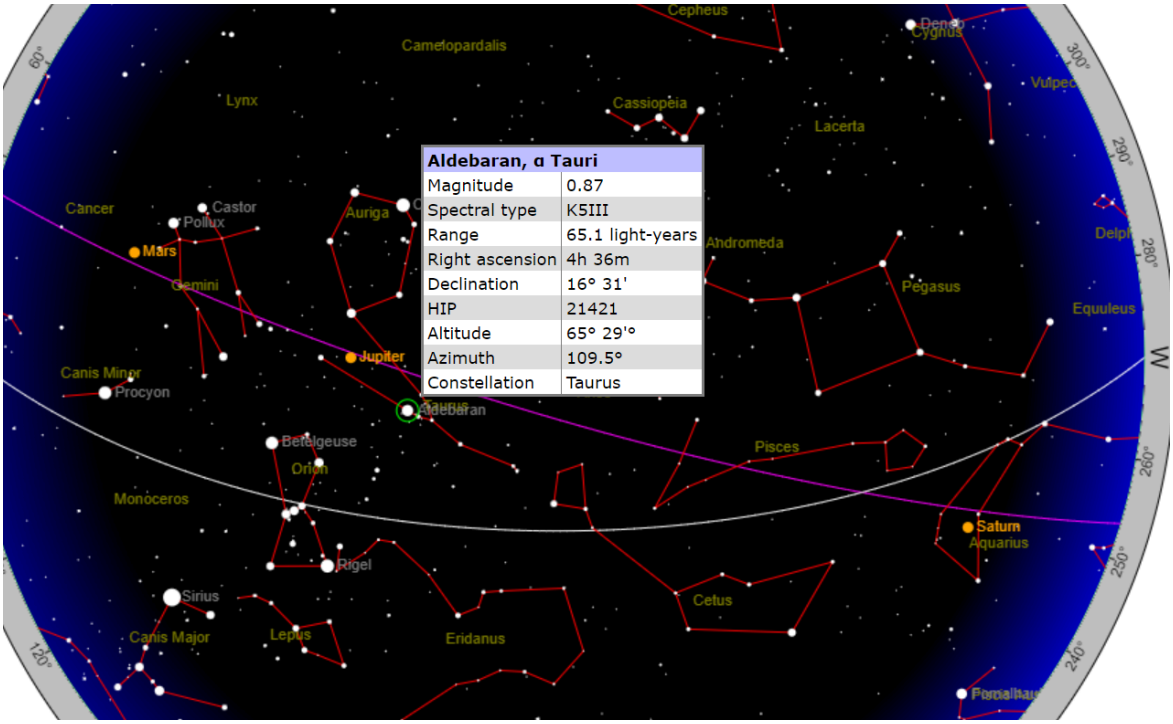
    Note: Sky map is attached below



6. Answer: Sky map is attached. Look for M42, M81, M41, M31, γ Orionis (Ori), α Persei (Per) and α Cyg.
   [If the object labeled is within the size of the circles below, then the student earns full credit. If the object is in the constellation its supposed to be but not in the right place, the student earns 0.5 points]



7. Answer: Equatorial coordinates of the star Aldebaran (α Tauri)
   Right Ascension: 4h 36m
   Declination: 16° 31'
   [If the RA and Dec is between
   3.5h to 5.5h and 15.5° to 17.5°, the student earns full credit (2.5 points).
   2.5h to 6.5h and 14.5° to 18.5° = 2 points,
   1.5h to 7.5h and 13.5° to 19.5° = 1.5 points,
   0.5h to 8.5h and 12.5° to 20.5° = 1.0 points,
   -0.5h to 9.5h and 11.5° to 21.5° = 0.5 points,
   Anything beyond this range will not earn any credit.]



The combined question paper and answer key use a 45-point OC1 within a 72-point observation round. The separate observation PDF instead gives OC1 43.5 points. The official key calls 2.5 points full credit for part 7 although the question allocates 3 points.

Solution 9.5 Classifying Solar System objects

2024 | Kathmandu, Nepal | Sky map | OC2 | 9 points

Problem p. 127

OC2(1.5x6)

Asteroids	Dwarf Planets	Satellites
5	2	1,3,4,6

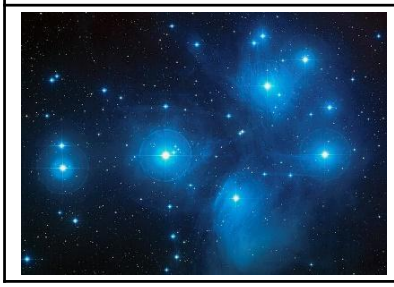
The combined question paper and official key use six pictures and 9 points. The separate observation PDF gives nine pictures and 13.5 points. This edition follows the matching six-picture version.

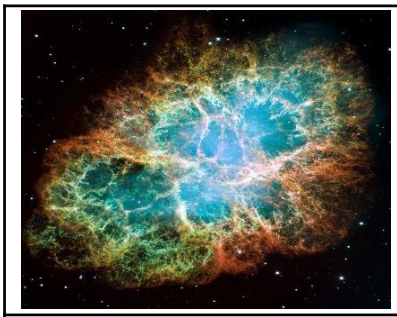
Solution 9.6 Identifying deep-sky objects

2024 | Kathmandu, Nepal | Sky map | OC3 | 18 points

Problem p. 128

OC3(1.5x12)

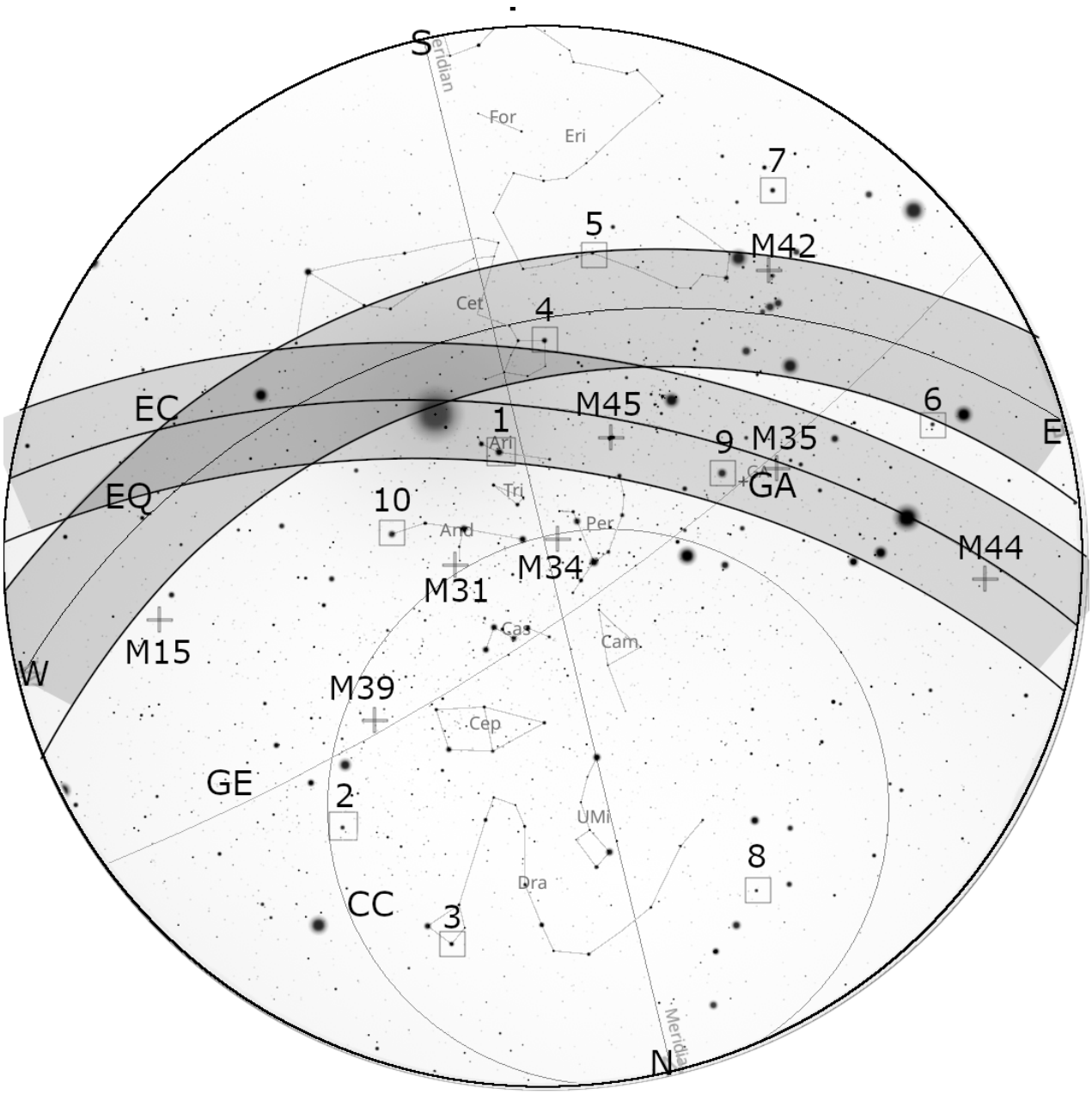
	M 97	Ursa Major
	M101	Ursa Major
	M51	Canes Venatici
	M45	Taurus

	M1	Taurus
	M42/M43 both will get full marks	Orion

Solution 9.7 Know Your Sky

2025 | Piatra Neamț, Romania | Sky map | OM1 | 40 points

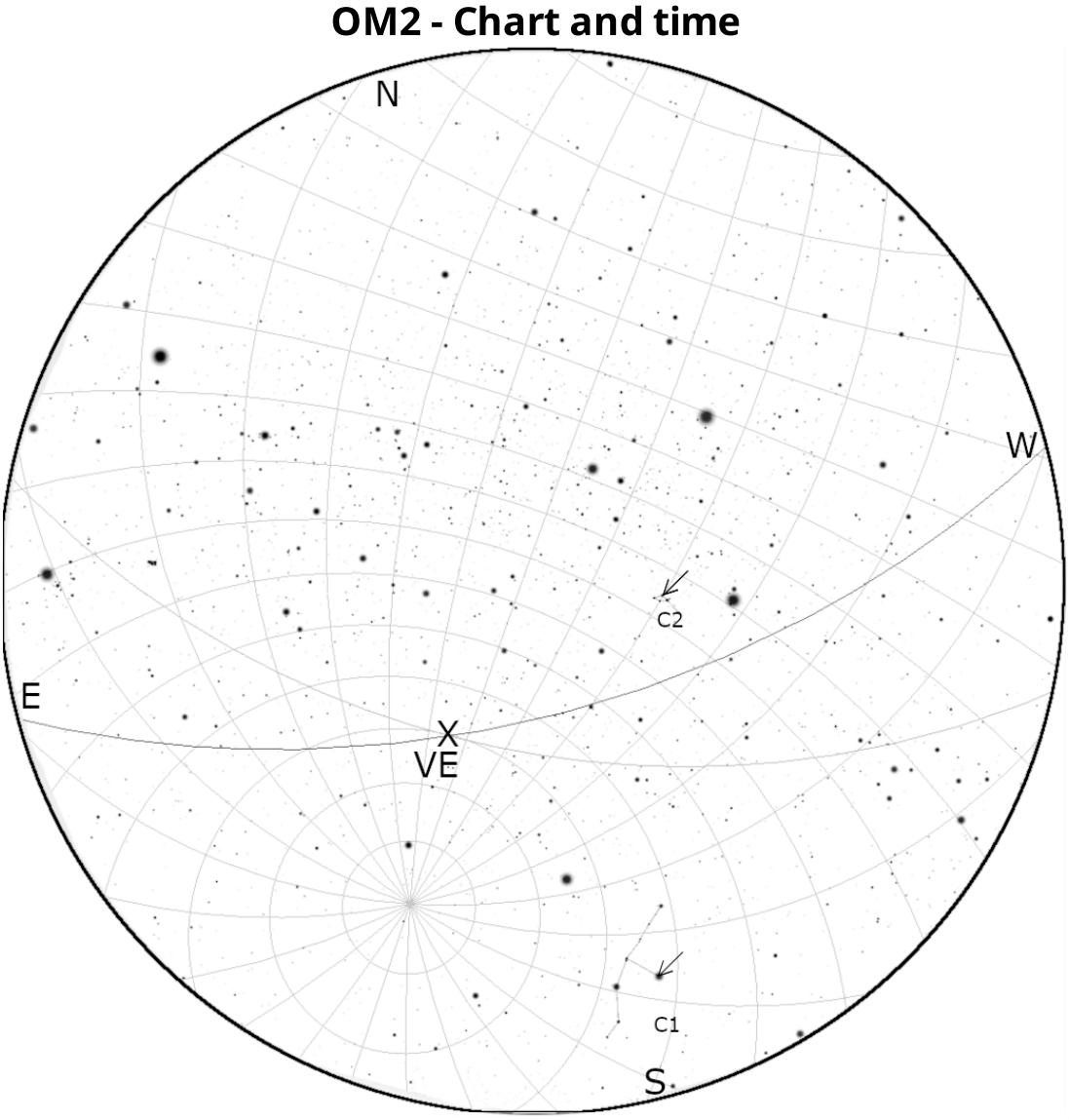
Problem p. 130



Solution 9.8 Chart and Time

2025 | Piatra Neamț, Romania | Sky map | OM2 | 10 points

Problem p. 132



2.1	Name of Constellation (Latin name or IAU abbreviation)
C1	Gru / Grus
C2	Del / Delphinus

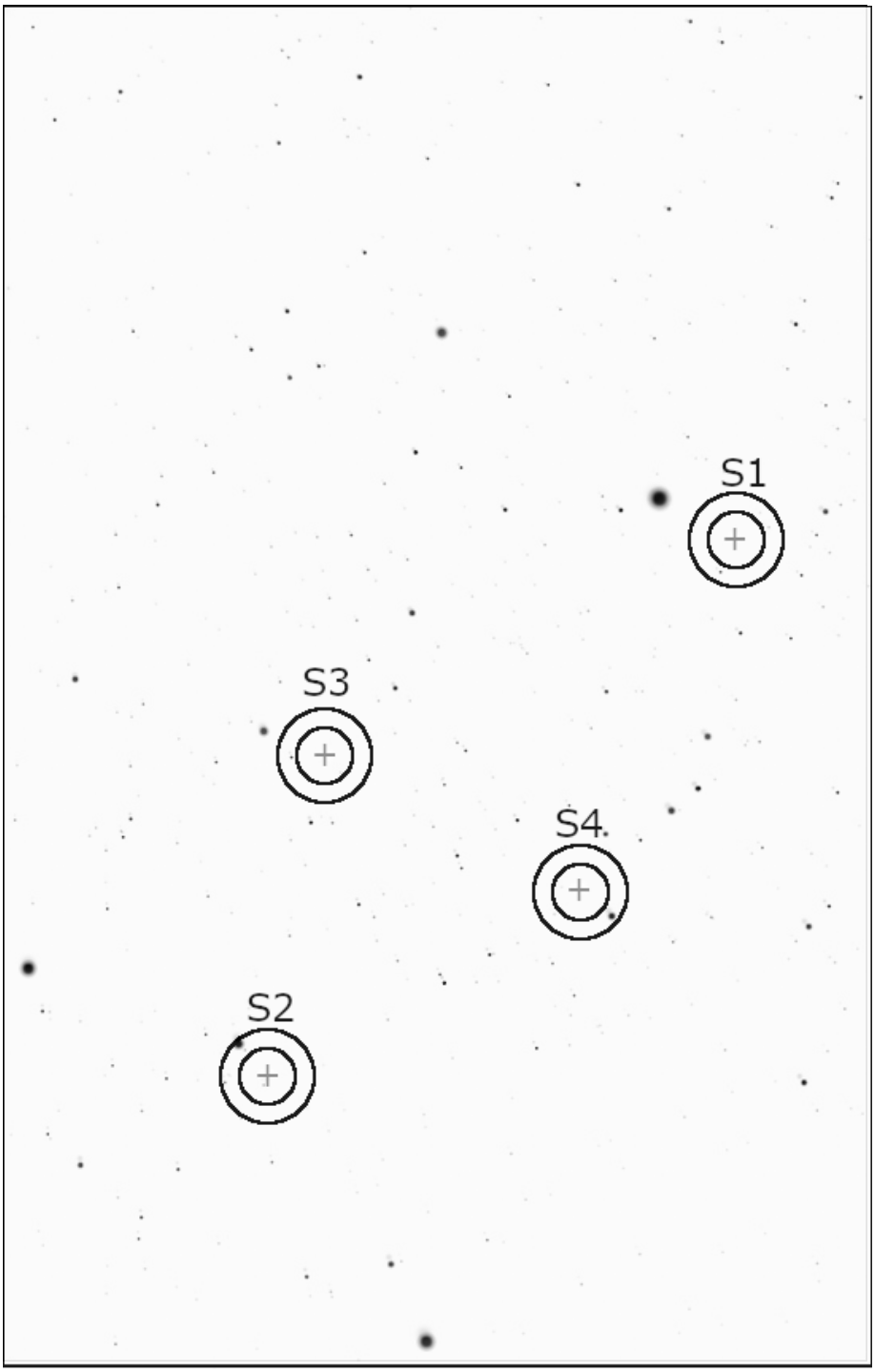
2.4	ecliptic	azimuthal	galactic	equatorial
Coordinate grid			X	

Solution 9.9 Missing Stars

2025 | Piatra Neamț, Romania | Sky map | OM3 | 10 points

Problem p. 134

OM3 - Missing stars



Star 1	
Name	Izar
Binary	Y/Double Star
Star 2	
Name	Castor
Binary	Y/Double Star
Star 3	
Name	Algieba
Binary	Y/Double Star
Star 4	
Name	Merak
Binary	N/NOT double

Solution 9.10 Sun observations

2023 | Volos, Greece | Telescope | Q15 | 45 points

Problem p. 136

No official solution is included in the supplied 2023 observation papers. The surviving file contains the question and blank answer sheets.

Solution 9.11 Cloudy sky observations

2023 | Volos, Greece | Telescope | Q16 | 45 points

Problem p. 138

No official solution is included in the supplied 2023 observation papers. The surviving file contains the question and blank answer sheets.

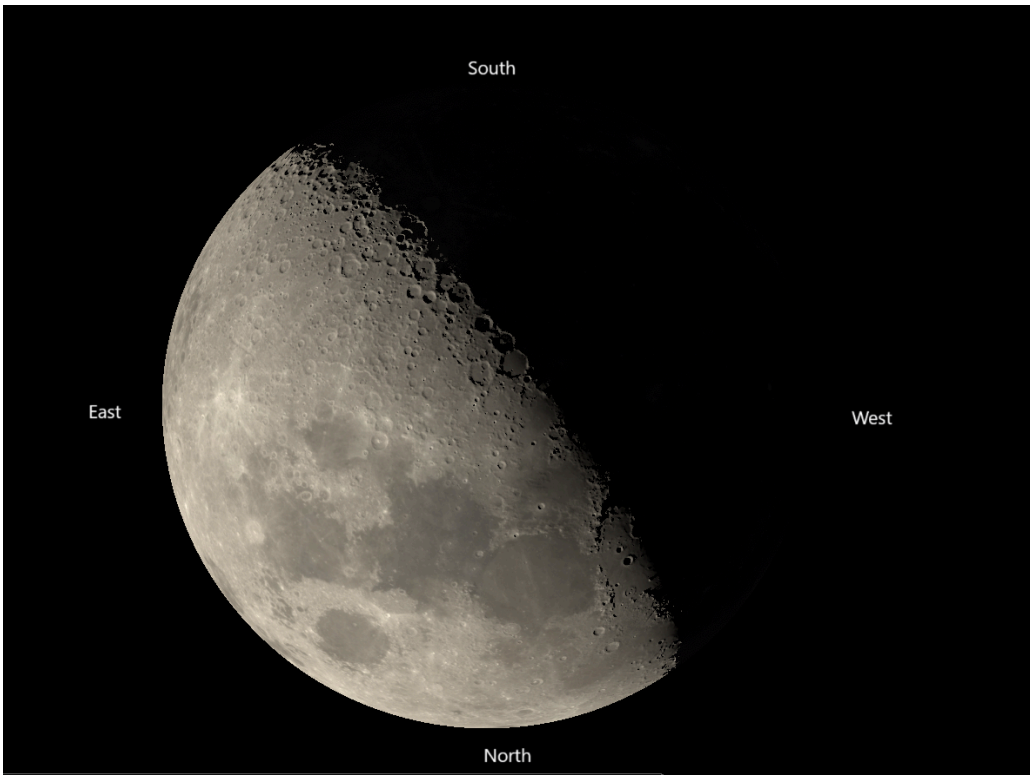
Solution 9.12 The Moon

2024 | Kathmandu, Nepal | Telescope | OT1 | 32 points

Problem p. 139

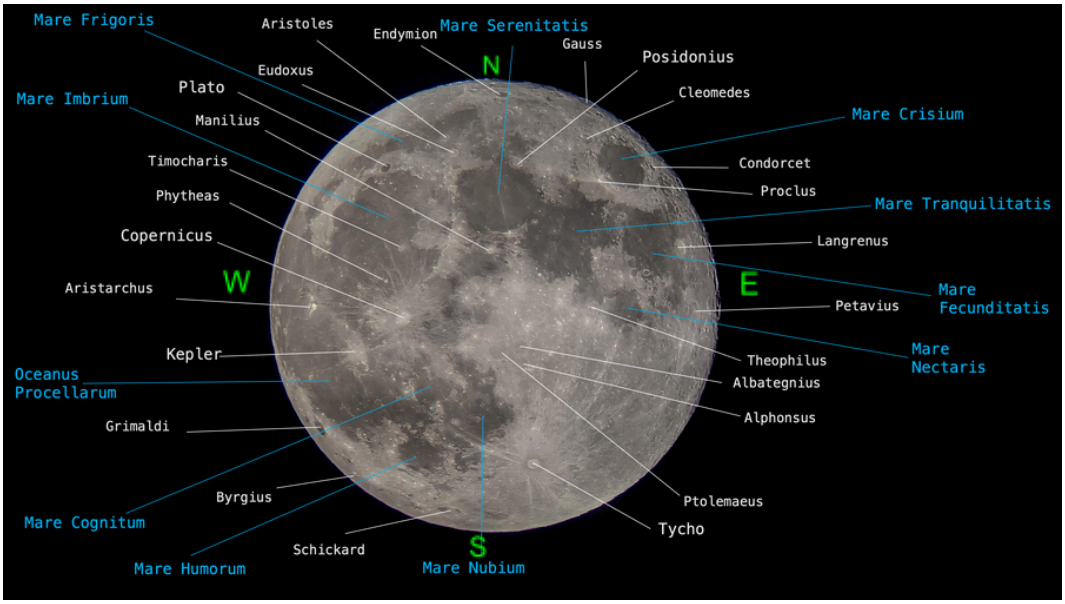
The Moon (32 points)

6 Points for pointing marked by the supervisor.



1. **2 points for drawing the moon**
   2 points for drawing the lunar terminator properly.
2. **4 points for cardinal points**

3. **(2x3 = 6 points for maria, 2x3 = 6 points for craters)**



4.

Age of the Moon (2 points)	8.3 days (8 - 8.5 days will gain full marks) (7.5 - 9 will gain half marks)
Hemisphere (Northern or Southern) from where the Moon is being watched (2 points)	Southern
Official IAU name of the constellation in which the Moon is located (2 points)	Taurus

Solution 9.13 Jupiter and its moons

2024 | Kathmandu, Nepal | Telescope | OT2 | 16 points

Problem p. 140

The Planet (16 points)

1. Jupiter (2 points)
2. 4 points for drawing initial and final positions. 4 points for labelling them correctly.
3. 1.5x4=6 for correct order.

Image 1 solution

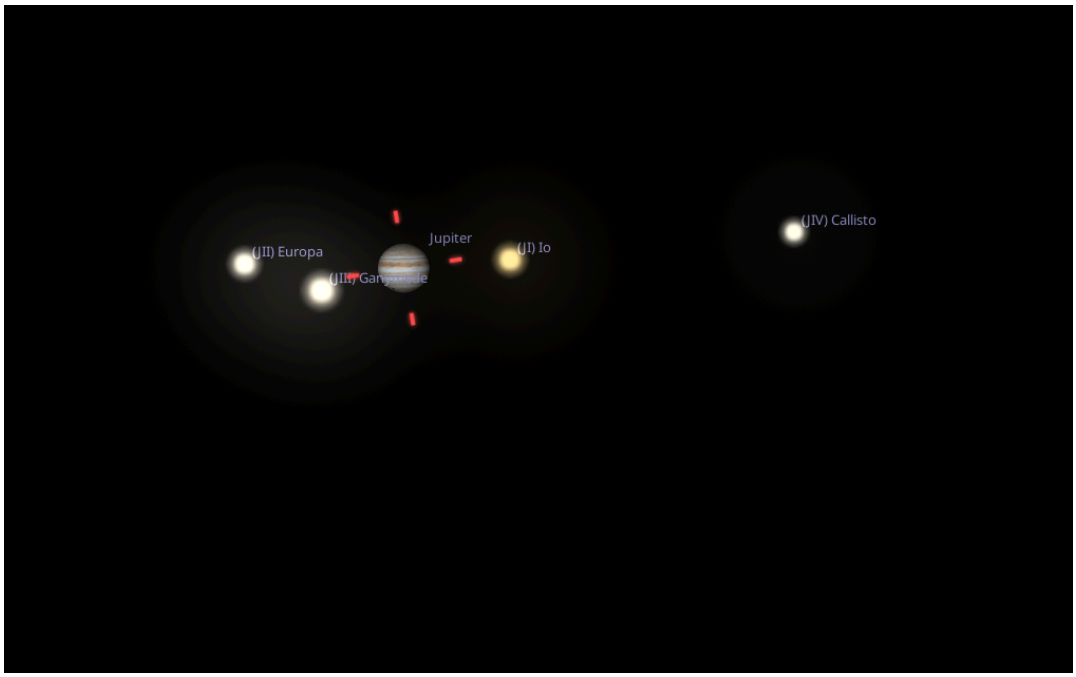
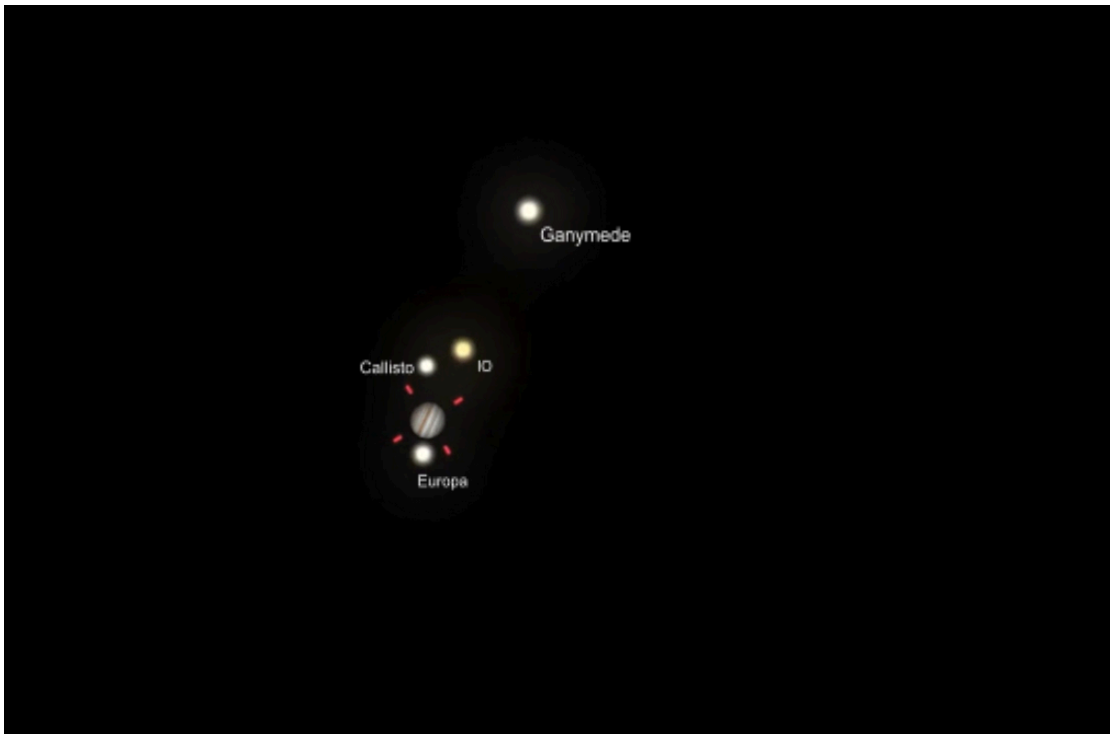


Image 2 solution



Note: Students won't be writing the names. They will label them as A,B,C,D. The moon must have the same label in both images.

3.
Order:
Ganymede
Callisto
IO
Europa

Solution 9.14 Telescope Adjustment Knobs

2025 | Piatra Neamţ, Romania | Telescope | OT1 | 4 points

Problem p. 141

ANSWERS

OT1 – 2 points for each correct answer

1.	Altitude	Right Ascension	Declination	Azimuth
a.		**X**		
b.			**X**	

Solution 9.15 Coordinate Identification

2025 | Piatra Neamț, Romania | Telescope | OT2 | 9 points

Problem p. 141

OT2 – 3 points for each correct answer

2.	Azimuth	Local Hour Angle	Sideral Hour Angle	Zenith Angle	Altitude	Declination	Meridian Angle
A					X		
B				X			
C	X						

Solution 9.16 Constellation through the Eyepiece

2025 | Piatra Neamț, Romania | Telescope | OT3 | 20 points

Problem p. 142

OT3

a,c,g. **Drawing of the constellation – 5p; Marking the star – 2p; Marking of M57 – 2 p.**



<table>
<tr><td>b.</td><td colspan="3">Lyra OR Lyr</td><td colspan="4">2 points</td></tr>
<tr><td>d.</td><td colspan="3">Vega</td><td colspan="4">3 points</td></tr>
<tr><td>e.</td><td colspan="3">Ring Nebula OR M57</td><td colspan="4">3 points</td></tr>
<tr><td rowspan="2">f.</td><td rowspan="2">3 points</td><td>Galaxy</td><td>Supernova</td><td>Star Cluster</td><td>Planetary Nebula</td><td>Dwarf planet</td><td>Reflection nebula</td></tr>
<tr><td></td><td></td><td></td><td>X</td><td></td><td></td></tr>
</table>

Solution 9.17 Objects and Telescope Coordinates

2025 | Piatra Neamț, Romania | Telescope | OT4 | 27 points

Problem p. 142

OT4 – 3 points for each correct answer

Sign	a. Latin name or Messier number (if applicable)	b. Right Ascension	c. Declination °
		Telescop 1	
R	**Orion Nebula OR M42**	22 h 10 m	32
N	**Andromeda Galaxy OR M31**	22 h 30 m	26
Y	**Double Star Albireo**	23 h 00 m	22
		Telescop 2	
R	**Orion Nebula OR M42**	21h 30 m	40
N	**Andromeda Galaxy OR M31**	22 h 00 m	32
Y	**Double Star Albireo**	22 h 30 m	30
		Telescop 3	
R	**Orion Nebula OR M42**	20 h 00 m	44
N	**Andromeda Galaxy OR M31**	20 h 50 m	40
Y	**Double Star Albireo**	21 h 30 m	38
		Telescop 4	
R	**Orion Nebula OR M42**	19 h 40 m	48
N	**Andromeda Galaxy OR M31**	20 h 20 m	44
Y	**Double Star Albireo**	21 h 00 m	42

		Telescop 5	
R	**Orion Nebula OR M42**	19 h 40 m	48
N	**Andromeda Galaxy OR M31**	20 h 20 m	44
Y	**Double Star Albireo**	21 h 20 m	42
		Telescop 6	
R	**Orion Nebula OR M42**	22 h 10 m	42
N	**Andromeda Galaxy OR M31**	22 h 30 m	35
Y	**Double Star Albireo**	23 h 10 m	30
		Telescop 7	
R	**Orion Nebula OR M42**	20 h 40 m	47
N	**Andromeda Galaxy OR M31**	21 h 20 m	40
Y	**Double Star Albireo**	22 h 10 m	36
		Telescop 8	
R	**Orion Nebula OR M42**	21 h 30 m	34
N	**Andromeda Galaxy OR M31**	22 h 00 m	28
Y	**Double Star Albireo**	22 h 40 m	24
		Telescop 9	
R	**Orion Nebula OR M42**	22 h 10 m	24
N	**Andromeda Galaxy OR M31**	22 h 40 m	18
Y	**Double Star Albireo**	23 h 20 m	16
		Telescop 10	
R	**Orion Nebula OR M42**	19 h 50 m	46
N	**Andromeda Galaxy OR M31**	20 h 30 m	48
Y	**Double Star Albireo**	19 h 00 m	48

		Telescop 11	
R	**Orion Nebula OR M42**	22 h 30 m	30
N	**Andromeda Galaxy OR M31**	23 h 00 m	28
Y	**Double Star Albireo**	21h 55 m	36
		Telescop 12	
R	**Orion Nebula OR M42**	20 h 00 m	42
N	**Andromeda Galaxy OR M31**	20h 30 m	40
Y	**Double Star Albireo**	19 h 10 m	44
		Telescop 13	
R	**Orion Nebula OR M42**	21 h 00 m	32
N	**Andromeda Galaxy OR M31**	21 h 30 m	32
Y	**Double Star Albireo**	20 h 20 m	36
		Telescop 14	
R	**Orion Nebula OR M42**	21 h 30 m	36
N	**Andromeda Galaxy OR M31**	22 h 00 m	34
Y	**Double Star Albireo**	21 h 00 m	42
		Telescop 15	
R	**Orion Nebula OR M42**	21 h 30 m	40
N	**Andromeda Galaxy OR M31**	22 h 00 m	38
Y	**Double Star Albireo**	20 h 50 m	44
		Telescop 16	
R	**Orion Nebula OR M42**	21 h 20 m	36
N	**Andromeda Galaxy OR M31**	21 h 50 m	36
Y	**Double Star Albireo**	20 h 40 m	40

		Telescop 17	
R	**Orion Nebula OR M42**	20 h 50 m	32
N	**Andromeda Galaxy OR M31**	21 h 20 m	32
Y	**Double Star Albireo**	20 h 10 m	36
		Telescop 18	
R	**Orion Nebula OR M42**	21 h 00 m	36
N	**Andromeda Galaxy OR M31**	21 h 30 m	36
Y	**Double Star Albireo**	20 h 20 m	40

Accepted error:

- ± 10 min for R.A
- ± 2° for Dec

Solution 9.18 Observations with the Naked Eye

2022 | Câmpulung Moldovenesc, Romania | Night observation | Q14

Problem p. 143

No official solution or marking key for the 2022 naked-eye night-observation tasks is included in the supplied archive.

Solution 9.19 Observations Using the Astronomical Telescope

2022 | Câmpulung Moldovenesc, Romania | Night observation | Q15

Problem p. 144

No official solution or marking key for the 2022 telescope night-observation tasks is included in the supplied archive.

Index by year

2022

5.1 Q1 The Stars Observed with the Telescope	prob. 67	sol. 71
1.1 Q2 The Sun on the Horizon	prob. 3	sol. 7
4.1 Q3 The Escape Velocity	prob. 27	sol. 40
5.2 Q4 The Terrestrial Satellite	prob. 67	sol. 72
8.1 Q5 Andromeda Galaxy Rotation	prob. 104	sol. 107
3.1 Q6 The Sun as Seen from Saturn	prob. 19	sol. 23
4.2 Q7 Balls Suspended inside a Terrestrial Satellite	prob. 27	sol. 42
4.3 Q8 Spaceship to the Sun	prob. 29	sol. 44
6.1 Q9 The Moon, Seen on the Way to the Moon	prob. 77	sol. 87
6.2 Q10 The Luminosity of a Star	prob. 77	sol. 89
4.4 Q11 Zodiac IOAA-J 2022	prob. 29	sol. 46
4.5 Q12 The Rocket that Destroys the Threatening Asteroid	prob. 35	sol. 52
9.1 Q13 Blind Sky Map	prob. 110	sol. 145
9.18 Q14 Observations with the Naked Eye	prob. 143	sol. 168
9.19 Q15 Observations Using the Astronomical Telescope	prob. 144	sol. 168

2023

1.2 Q1 Horizon crossing	prob. 4	sol. 9
4.6 Q2 Planet orbits	prob. 36	sol. 54
6.3 Q3 Unresolved binary star	prob. 78	sol. 90
7.1 Q4 Rapidly spinning neutron star	prob. 101	sol. 103
2.1 Q5 Moon phases	prob. 15	sol. 17
3.2 Q6 Solar eclipse	prob. 19	sol. 24
6.4 Q7 Parallax and proper motion	prob. 78	sol. 90
5.3 Q8 Telescope	prob. 68	sol. 74
8.2 Q9 Cosmology	prob. 104	sol. 107
7.2 Q10 H-R diagram	prob. 102	sol. 103
6.5 Q11 Sirius A and B	prob. 79	sol. 91
4.7 Q12 Satellites	prob. 36	sol. 55
9.2 Q13 Navigating the night sky I	prob. 112	sol. 147
9.3 Q14 Navigating the night sky II	prob. 117	sol. 147
9.10 Q15 Sun observations	prob. 136	sol. 159
9.11 Q16 Cloudy sky observations	prob. 138	sol. 159

2024

5.4 T1 Telescope limiting magnitude and interstellar extinction	prob. 69	sol. 74
4.8 T2 Energy and speed in an elliptical orbit	prob. 37	sol. 57
2.2 T3 Longitude and local time in Nepal	prob. 15	sol. 17
6.6 T4 A Cepheid period-luminosity relation	prob. 81	sol. 92
4.9 T5 Altitude of a geostationary satellite	prob. 37	sol. 58
2.3 T6 The lunar calendar and Dashain	prob. 16	sol. 18
8.3 T7 The Cosmic Horseshoe	prob. 105	sol. 108
1.3 T8 Rising Sagarmatha and angular separation	prob. 4	sol. 9
4.10 T9 Free fall inside a gas cloud	prob. 38	sol. 59
6.7 T10 A spectroscopic binary system	prob. 82	sol. 93
8.4 T11 Critical density and tennis balls	prob. 106	sol. 109
6.8 T12 A radiating interstellar dust cloud	prob. 84	sol. 95
9.4 OC1 A sky chart from Nepal	prob. 124	sol. 148
9.5 OC2 Classifying Solar System objects	prob. 127	sol. 153
9.6 OC3 Identifying deep-sky objects	prob. 128	sol. 154
9.12 OT1 The Moon	prob. 139	sol. 160
9.13 OT2 Jupiter and its moons	prob. 140	sol. 161

2025

6.9 Th.1 Binary System	prob. 85	sol. 96
4.11 Th.2 Speed of Asteroid	prob. 38	sol. 60
1.4 Th.3 Where is Theodor?	prob. 5	sol. 10
5.5 Th.4 Exploring NEAs	prob. 70	sol. 75
1.5 Th.5 Ceahlau Mountain	prob. 6	sol. 12
1.6 Th.6 The Flying Aquila	prob. 6	sol. 13
3.3 Th.7 Lunar Eclipse	prob. 20	sol. 24
4.12 Th.8 Roche Limit, Hill Sphere, Lagrange Points	prob. 39	sol. 61
6.10 Th.9 Lightcurves	prob. 86	sol. 97
9.7 OM1 Know Your Sky	prob. 130	sol. 156
9.8 OM2 Chart and Time	prob. 132	sol. 157
9.9 OM3 Missing Stars	prob. 134	sol. 158
9.14 OT1 Telescope Adjustment Knobs	prob. 141	sol. 162
9.15 OT2 Coordinate Identification	prob. 141	sol. 163
9.16 OT3 Constellation through the Eyepiece	prob. 142	sol. 163
9.17 OT4 Objects and Telescope Coordinates	prob. 142	sol. 165